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Nowell et al.

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(54) **FLUID END ASSEMBLY**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

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(65) **Prior Publication Data**

US 2024/0426294 A1 Dec. 26, 2024

Related U.S. Application Data

(63) Continuation of application No. 18/080,235, filed on
Dec. 13, 2022, now abandoned, which is a
(Continued)

(51) **Int. Cl.**
F04B 53/22 (2006.01)
F04B 1/0408 (2020.01)

(Continued)

(52) **U.S. Cl.**
CPC **F04B 53/22** (2013.01); **F04B 1/0408**
(2013.01); **F04B 1/0421** (2013.01); **F04B**
1/122 (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC F04B 53/22; F04B 53/16; F04B 1/0408;
F04B 1/0421; F04B 1/122; F04B 1/145;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,005,412 A 10/1961 Camp
3,053,500 A 9/1962 Atkinson
(Continued)

FOREIGN PATENT DOCUMENTS

WO WO2014144113 A2 9/2014

OTHER PUBLICATIONS

GD Energy Products, LLC (Petitioner) v. *Kerr Machine Co.* (Patent
Owner) Petition to Institute an Inter Partes Review for U.S. Pat. No.
12,152,582, Feb. 6, 2025, 165 pages.

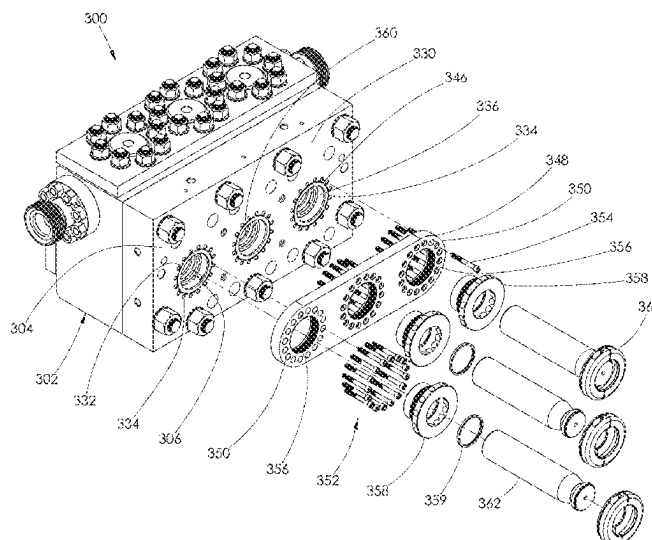
Primary Examiner — Patrick C Williams

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(57) **ABSTRACT**

A fluid end having its fluid flow bores sealed without
threading a retaining nut into the walls of each bore. The
fluid ends may be assembled using a plurality of different
kits, each kit comprising a fluid end body, a component, a
retainer element, and a fastening system. The retainer ele-
ment holds the component within each of the bores formed
in the fluid end body and the fastening system secures the
retainer element to the body. The fastening system com-
prises a plurality of externally threaded studs, washers and
nuts in some embodiments. In other embodiments, the
fastening system comprises a plurality of screws.

21 Claims, 39 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/215,155, filed on Mar. 29, 2021, now Pat. No. 11,536,267, which is a continuation-in-part of application No. 16/897,659, filed on Jun. 10, 2020, now Pat. No. 10,962,001, which is a continuation-in-part of application No. 16/876,414, filed on May 18, 2020, now Pat. No. 11,408,419, which is a continuation of application No. 16/035,126, filed on Jul. 13, 2018, now Pat. No. 10,670,013.

- (60) Provisional application No. 63/049,974, filed on Jul. 9, 2020, provisional application No. 62/859,256, filed on Jun. 10, 2019, provisional application No. 62/562,588, filed on Sep. 25, 2017, provisional application No. 62/536,297, filed on Jul. 24, 2017, provisional application No. 62/532,574, filed on Jul. 14, 2017.

(51) Int. Cl.

F04B 1/0421 (2020.01)
F04B 1/0448 (2020.01)
F04B 1/0461 (2020.01)
F04B 1/0538 (2020.01)
F04B 1/122 (2020.01)
F04B 1/145 (2020.01)
F04B 1/16 (2006.01)
F04B 1/18 (2020.01)
F04B 53/16 (2006.01)
F16B 35/00 (2006.01)
F16B 39/24 (2006.01)
F16B 43/00 (2006.01)

(52) U.S. Cl.

CPC **F04B 1/145** (2013.01); **F04B 1/16** (2013.01); **F04B 1/18** (2013.01); **F04B 53/16** (2013.01); **F16B 35/005** (2013.01); **F16B 39/24** (2013.01); **F16B 43/00** (2013.01); **F04B 1/0448** (2013.01); **F04B 1/0461** (2013.01); **F04B 1/0538** (2013.01); **F05B 2260/301** (2013.01)

(58) Field of Classification Search

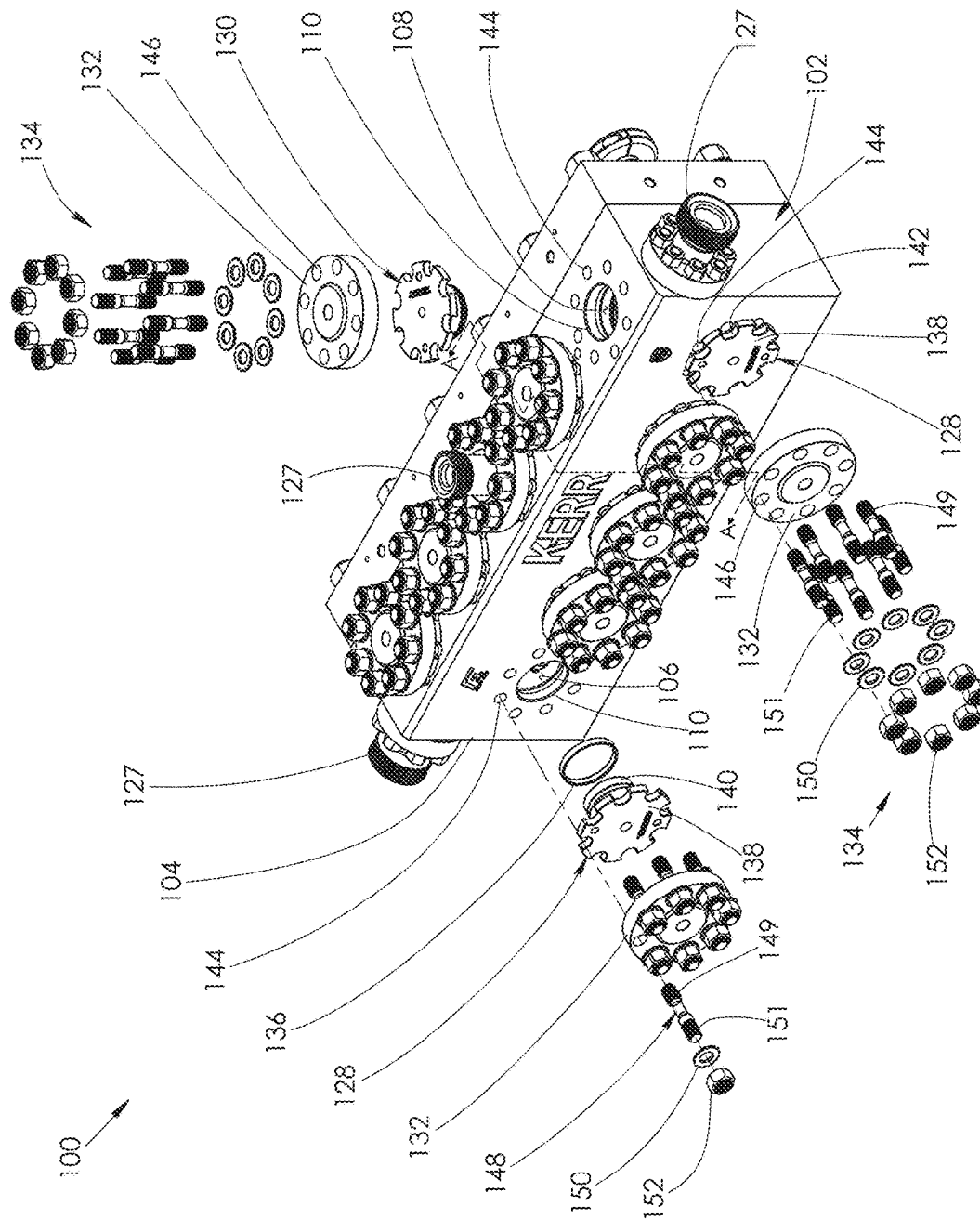
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 See application file for complete search history.

(56)**References Cited**

U.S. PATENT DOCUMENTS

3,146,724 A	9/1964	Cornelson	
3,427,988 A	2/1969	Redman et al.	
3,679,332 A	7/1972	Yohpe	
4,432,386 A	2/1984	Pacht	
4,520,837 A	6/1985	Cole et al.	
4,758,135 A	7/1988	Woodward et al.	
4,768,933 A	9/1988	Stachowiak	
4,771,801 A	9/1988	Crump et al.	
4,773,833 A	9/1988	Wilkinson et al.	
4,861,241 A	8/1989	Gamboa et al.	
5,073,096 A	12/1991	King et al.	
5,226,445 A	7/1993	Surjaatmadja	
5,362,215 A	11/1994	King	
6,167,959 B1	1/2001	Bassinger et al.	
7,290,560 B2	11/2007	Orr et al.	
7,296,591 B2	11/2007	Moe et al.	
7,335,002 B2	2/2008	Vicars	
7,506,574 B2	3/2009	Jensen et al.	
7,866,346 B1	1/2011	Walters	
8,100,407 B2 *	1/2012	Stanton	F16J 15/183 277/522
8,360,094 B2	1/2013	Steinbock et al.	
8,365,754 B2	2/2013	Riley et al.	
9,188,121 B1	11/2015	Dille	
10,221,847 B2	3/2019	Dyer	
10,760,567 B2	9/2020	Salih et al.	
2004/0234404 A1	11/2004	Vicars	
2006/0002806 A1	1/2006	Baxter	
2009/0123303 A1	5/2009	Ohnishi	
2011/0189040 A1	8/2011	Vicars	
2011/0206547 A1	8/2011	Kim	
2011/0236238 A1	9/2011	Cordes	
2012/0141308 A1	6/2012	Saini	
2013/0202458 A1	8/2013	Byrne	
2013/0319220 A1	12/2013	Luharuka	
2014/0348677 A1	11/2014	Moeller	
2015/0132157 A1	5/2015	Whaley	
2017/0218951 A1	8/2017	Graham	
2018/0363642 A1	12/2018	Salih et al.	
2019/0011051 A1	1/2019	Yeung	
2020/0232455 A1	7/2020	Blume	

* cited by examiner



 GLOBE

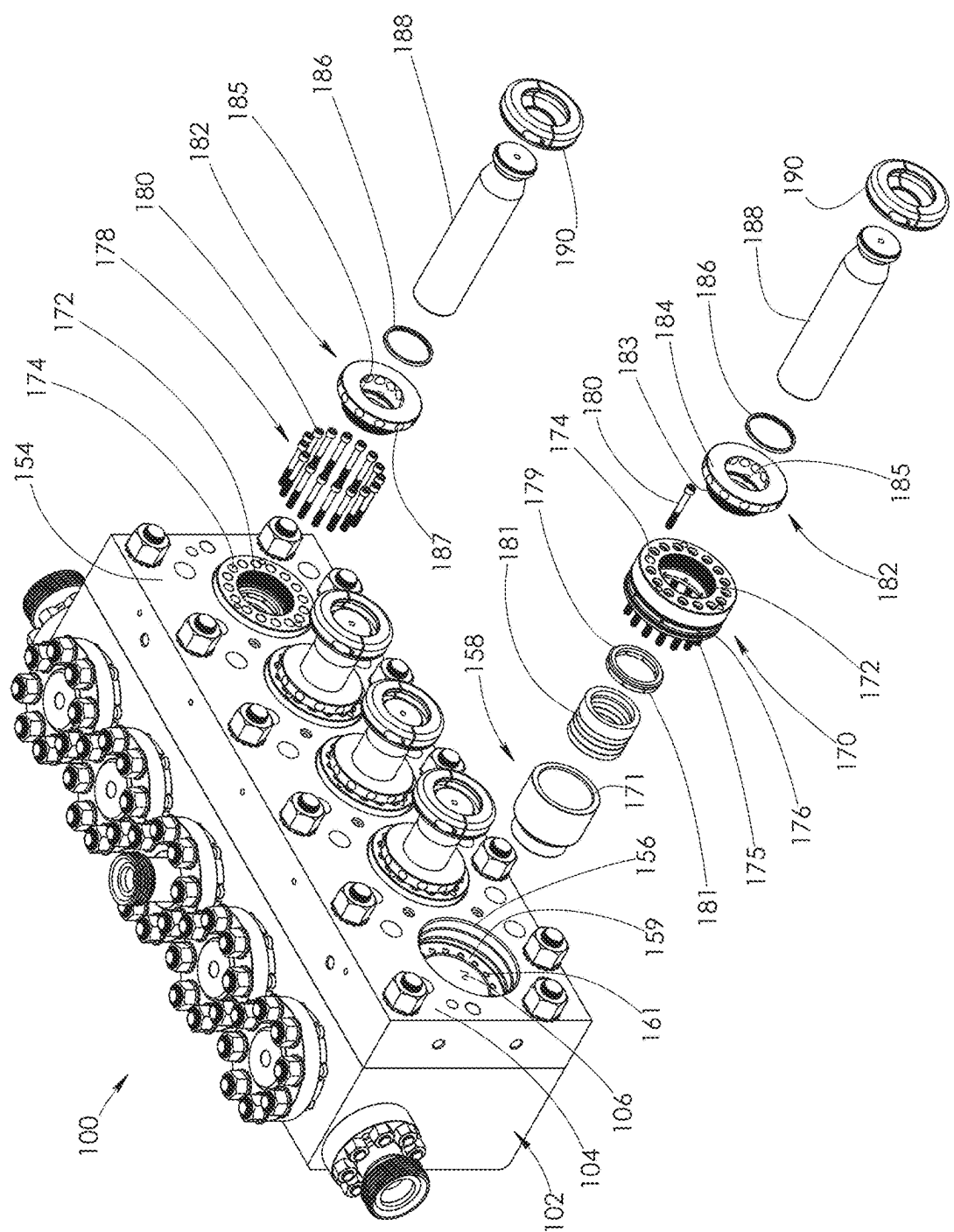


FIG. 2

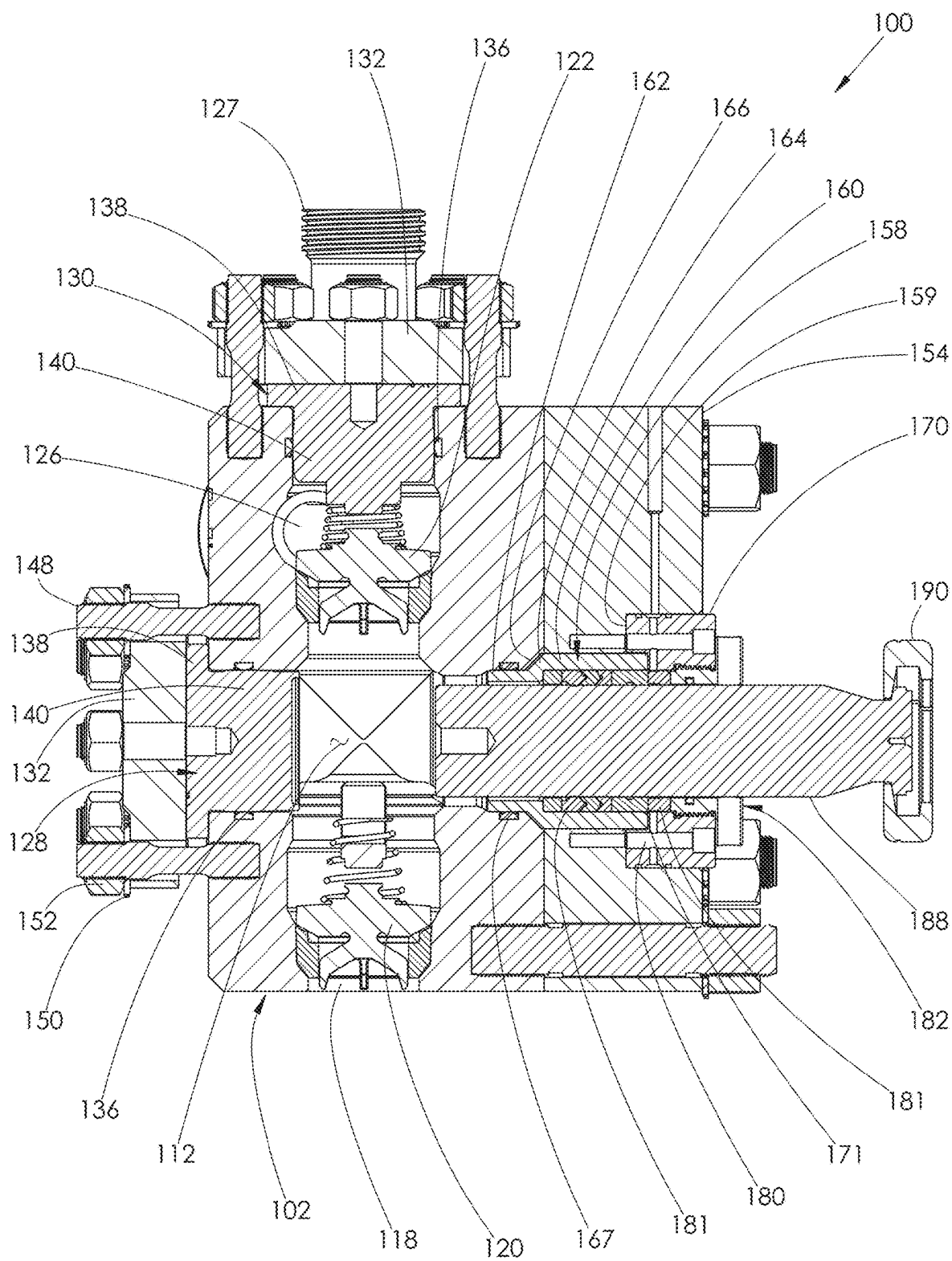


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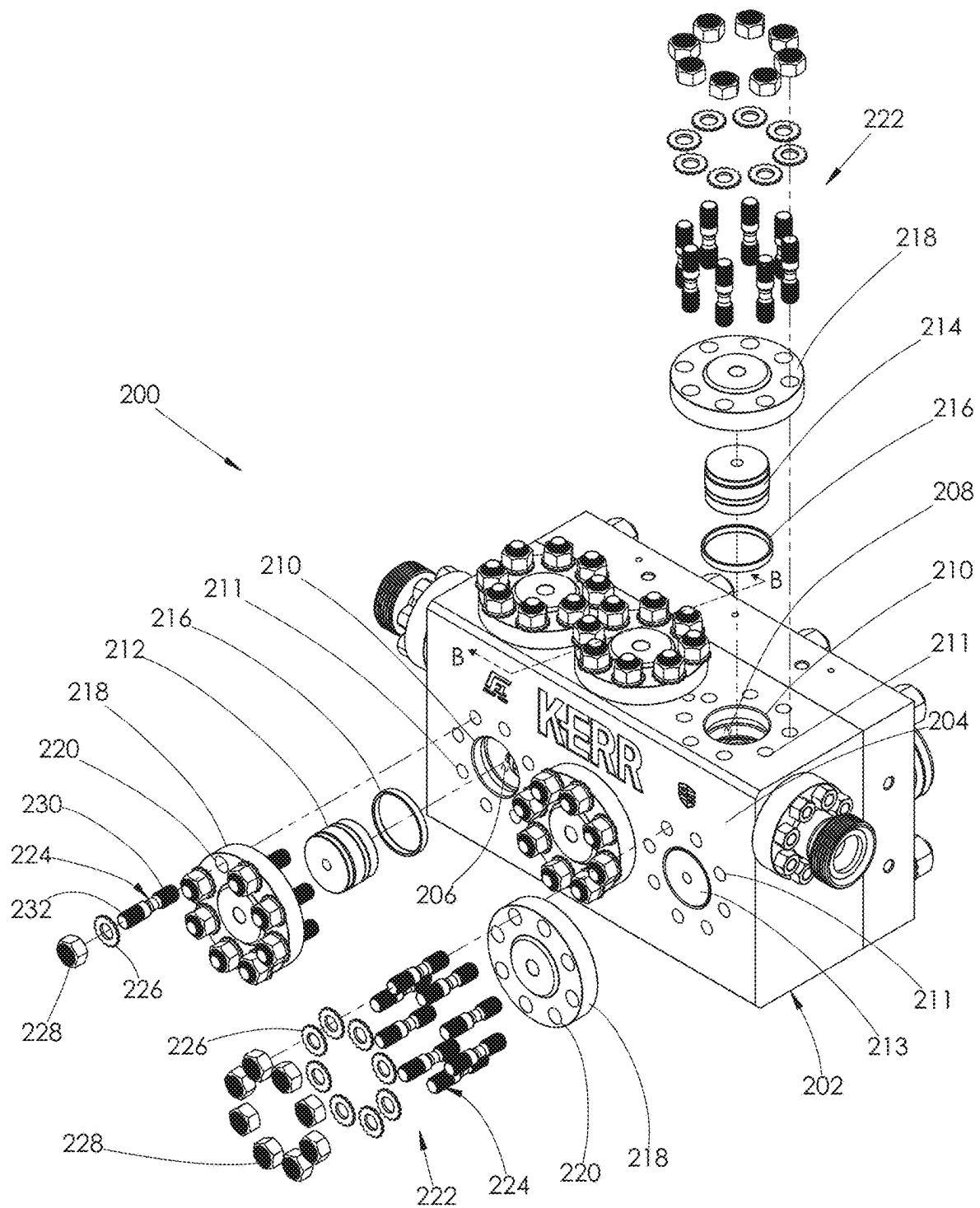


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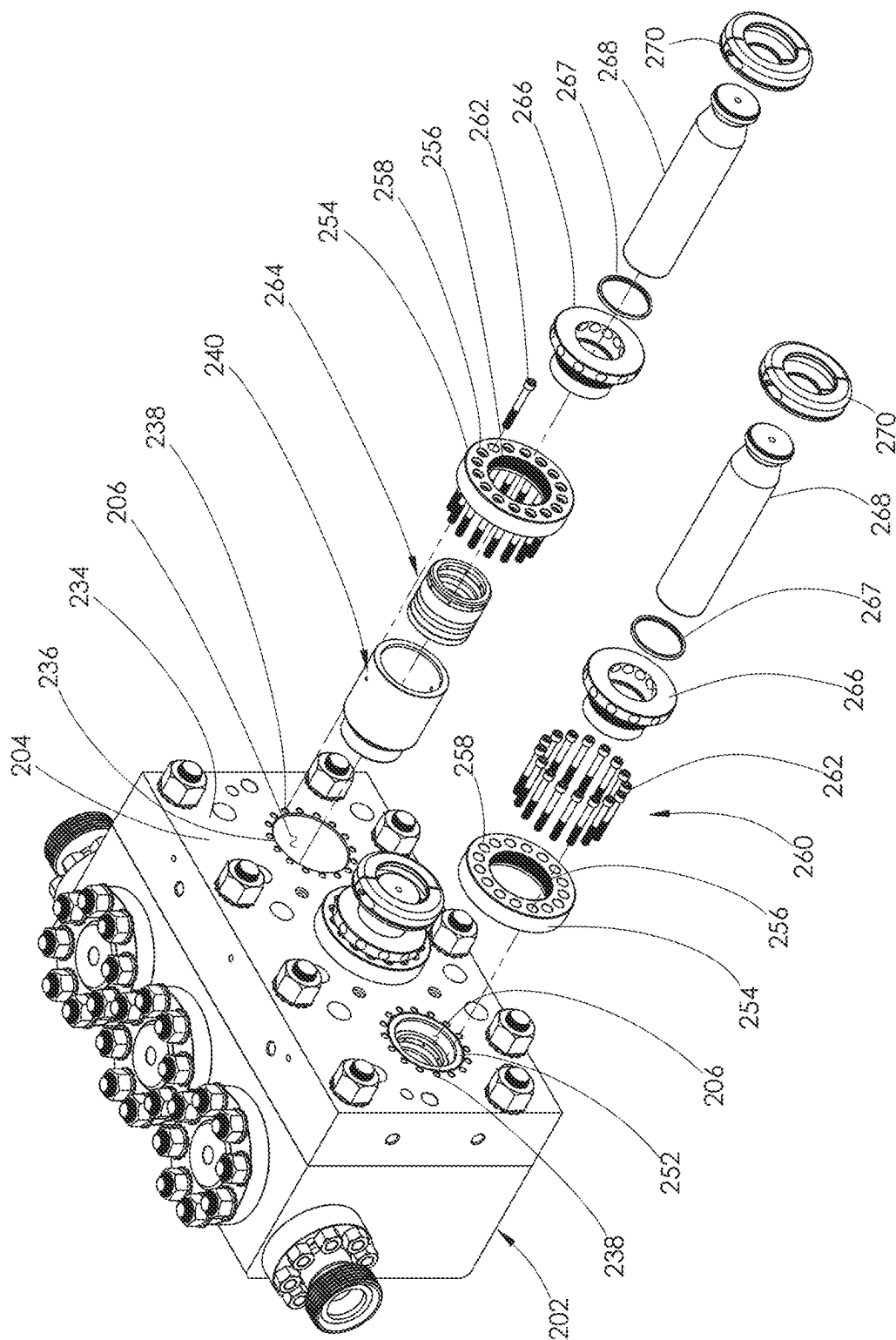


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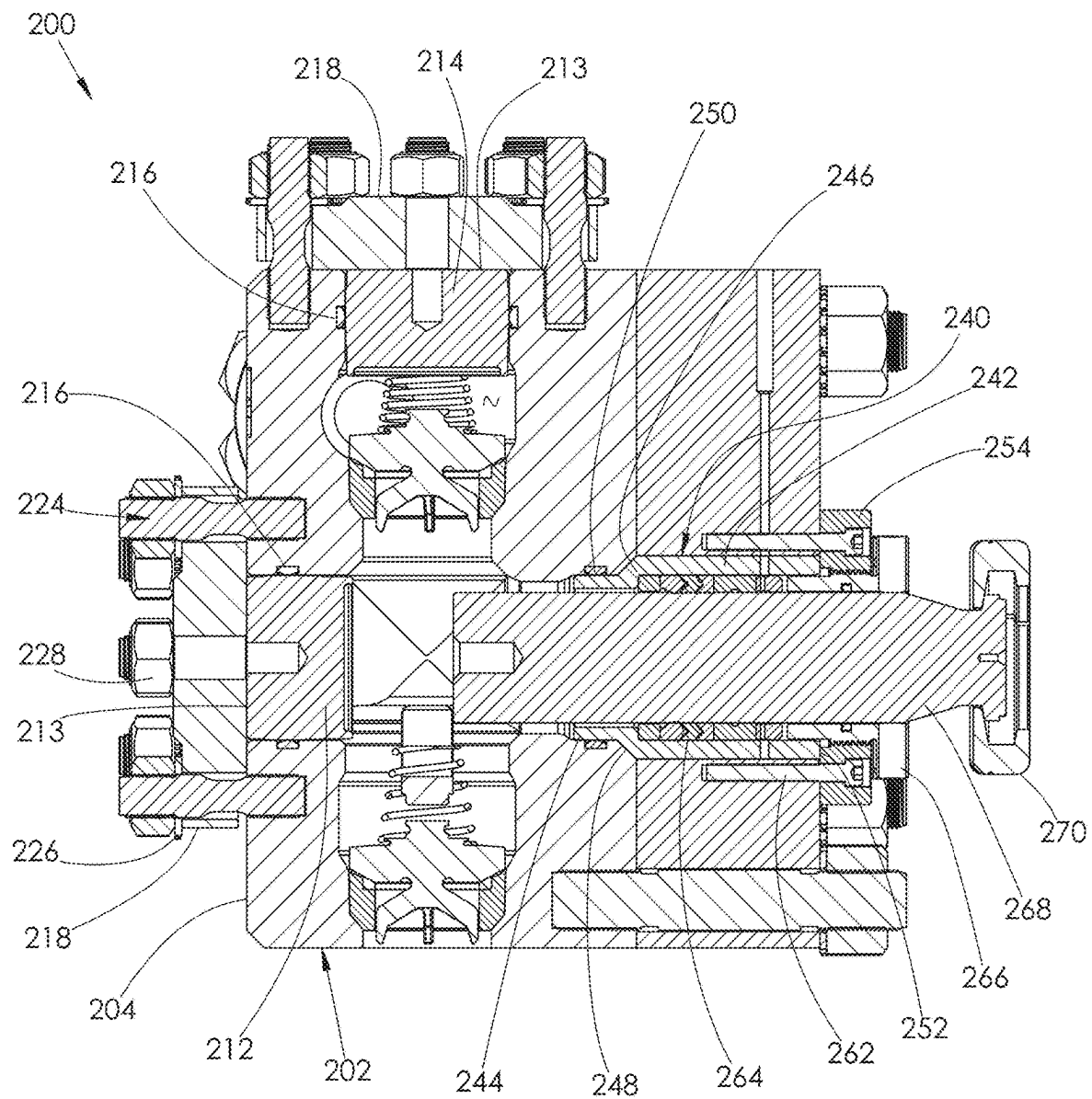


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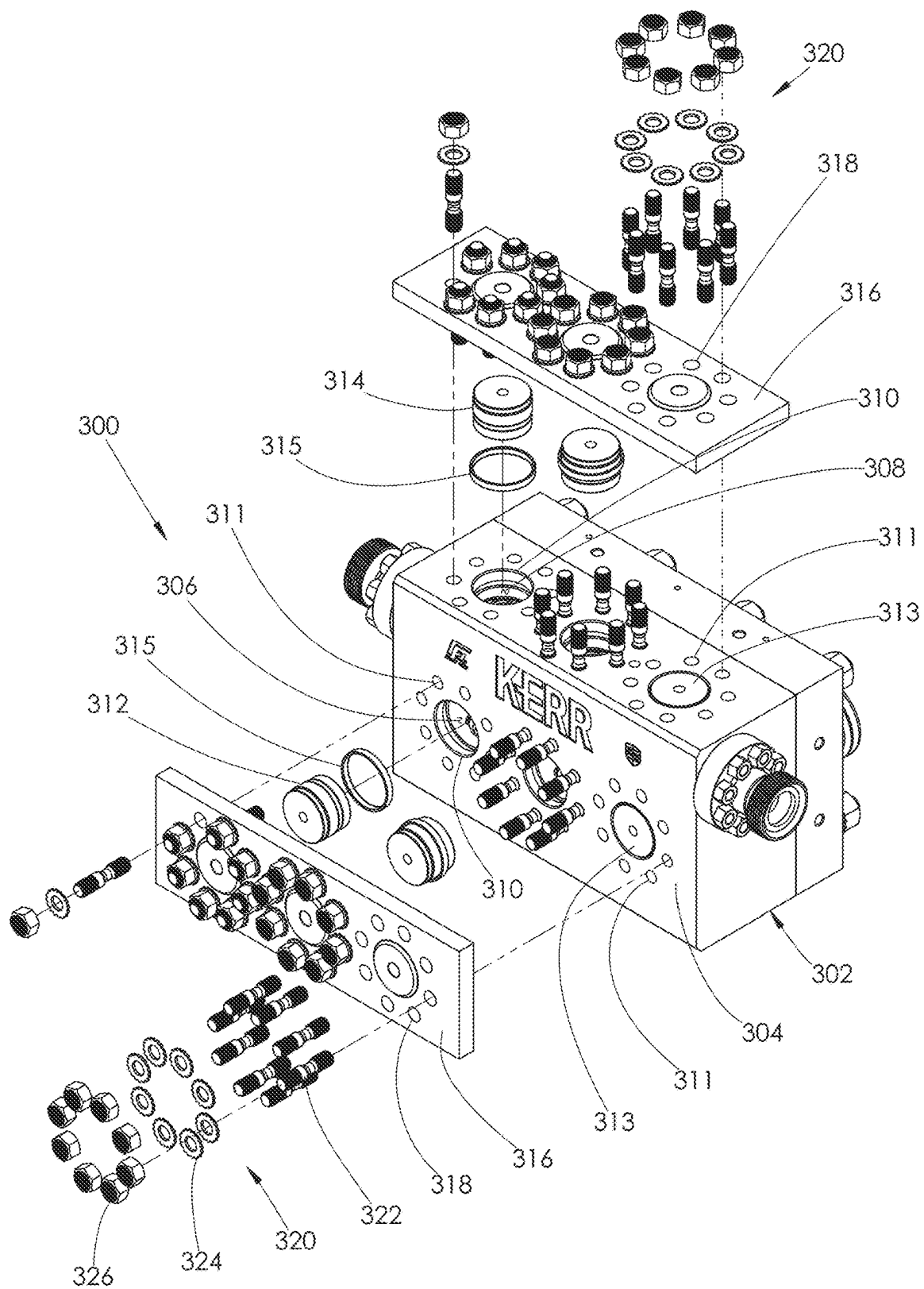


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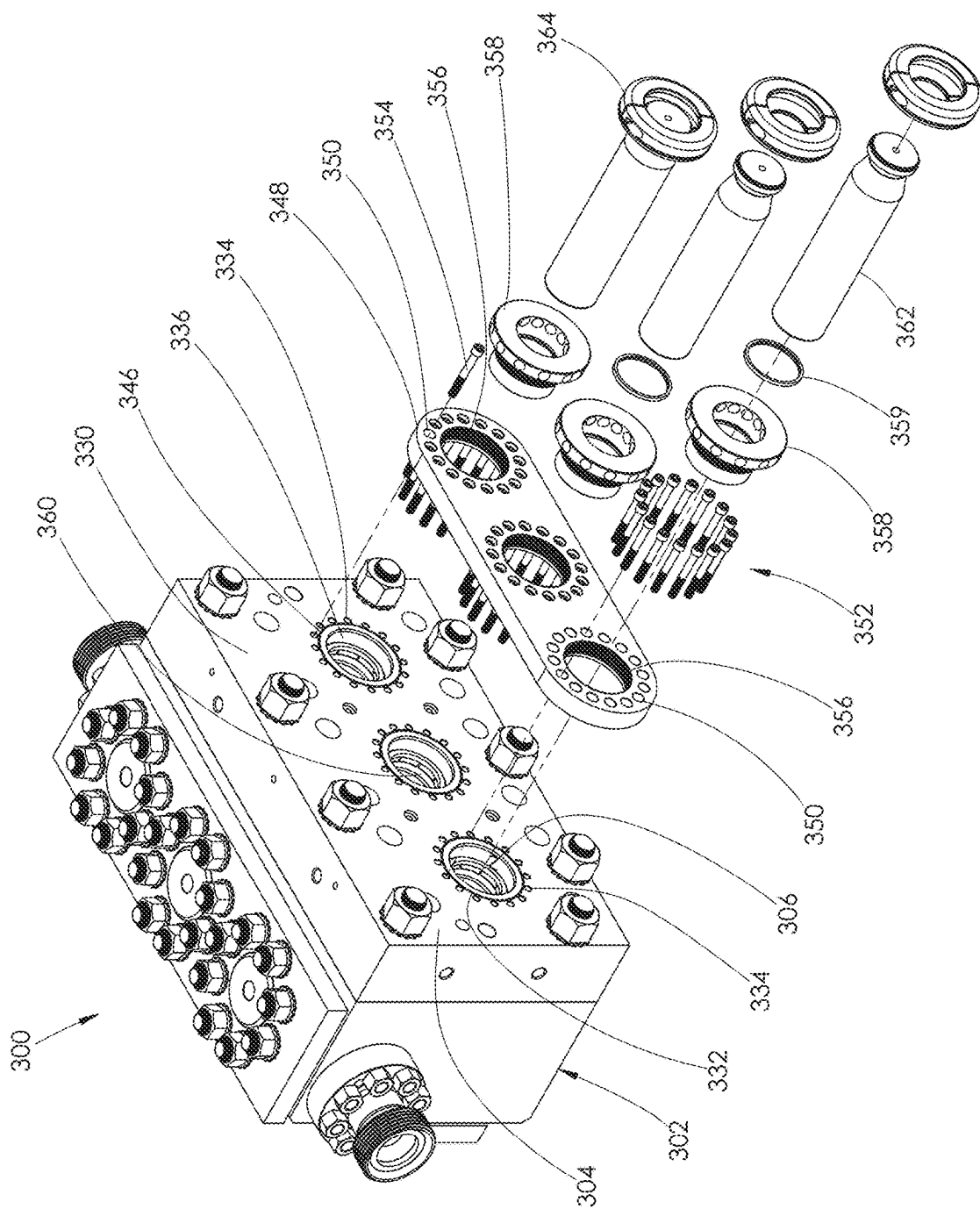


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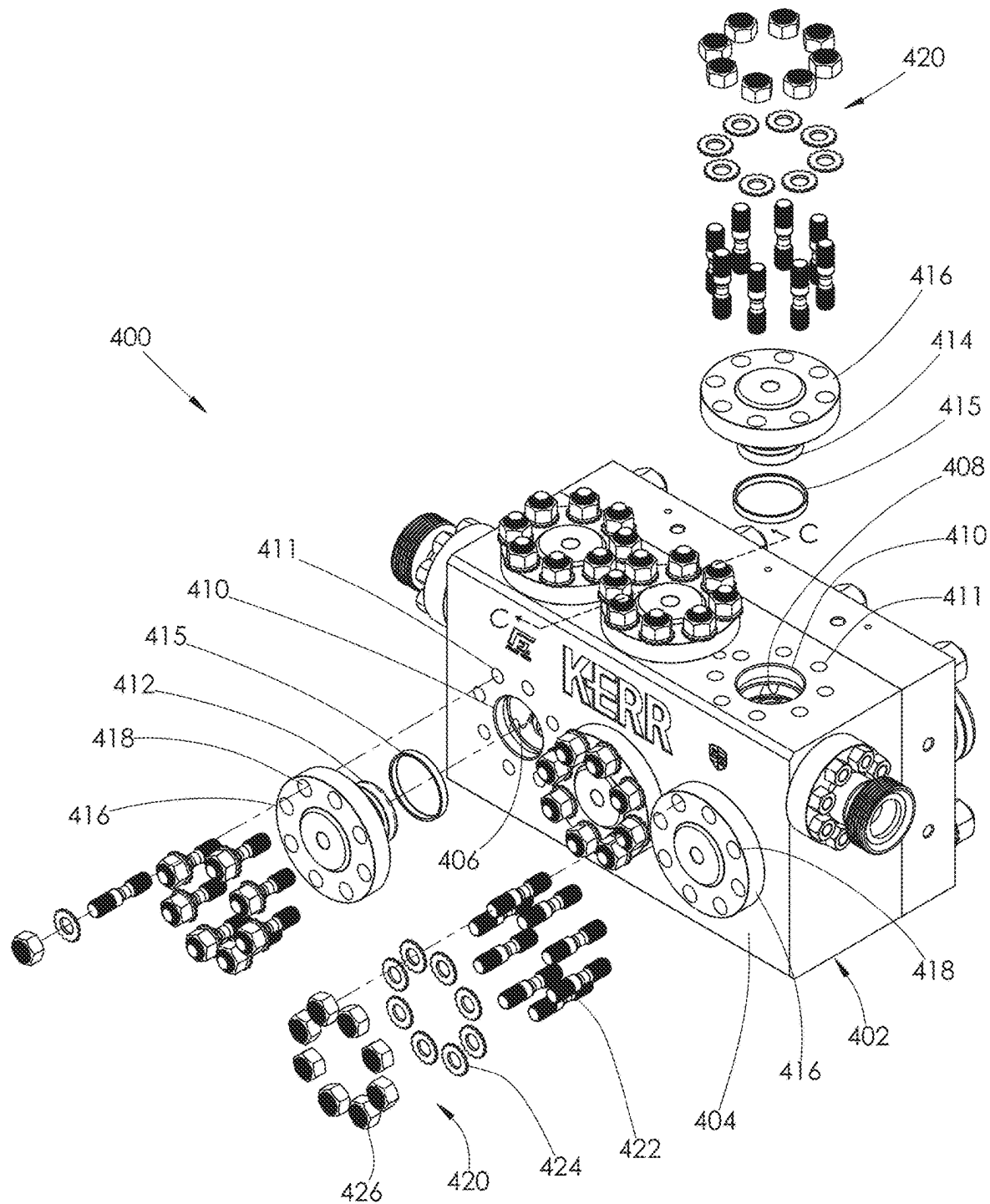


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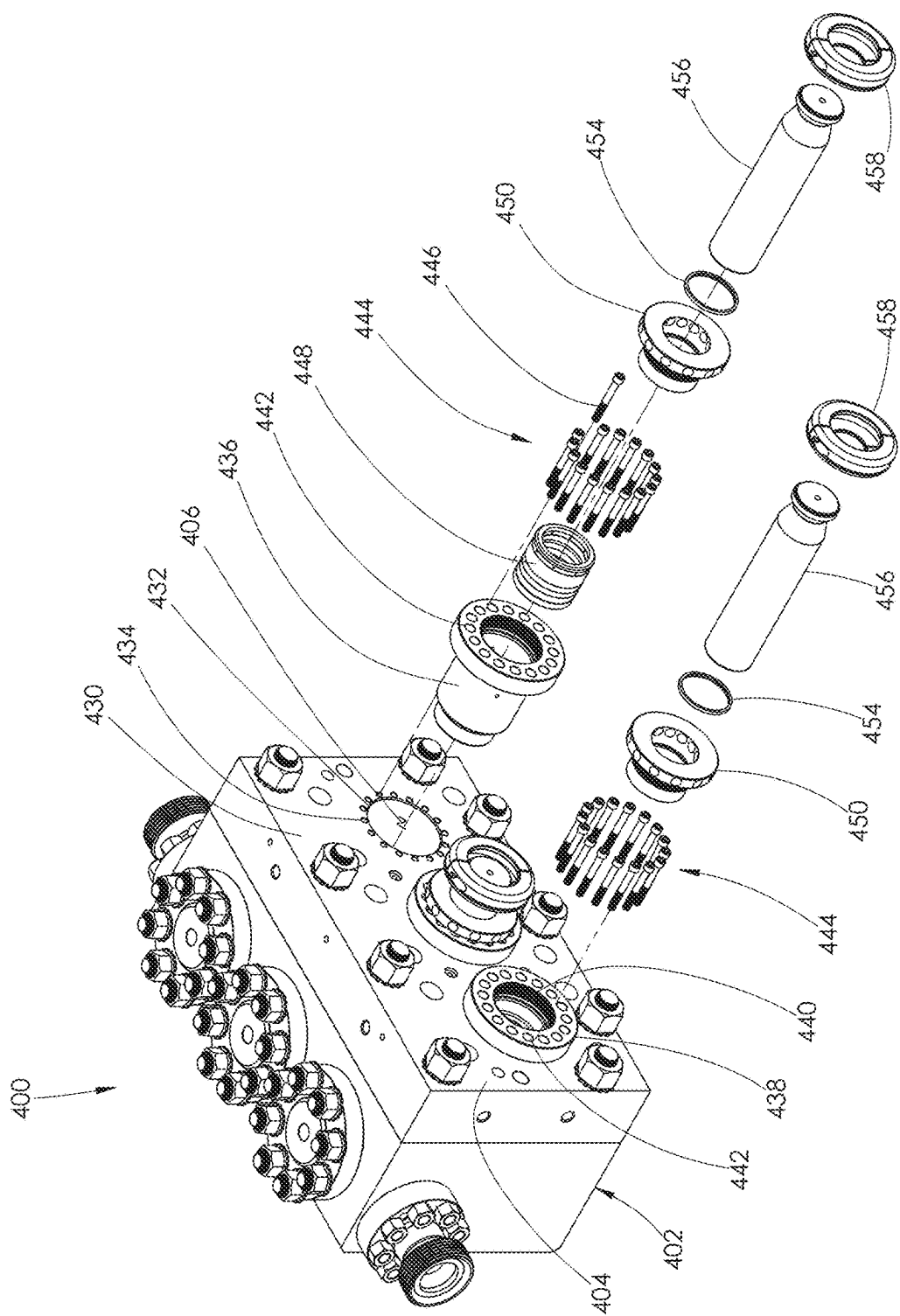


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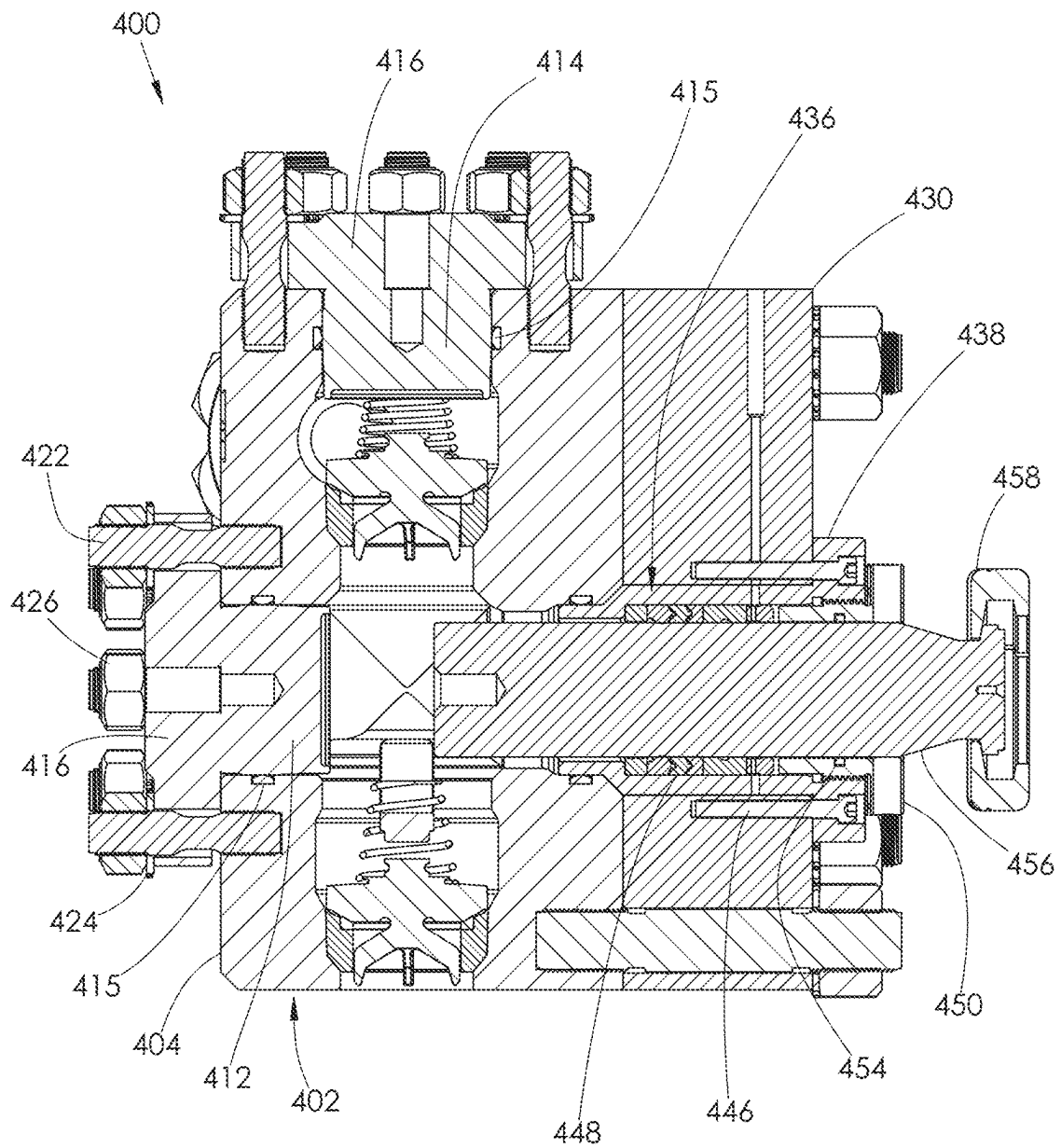


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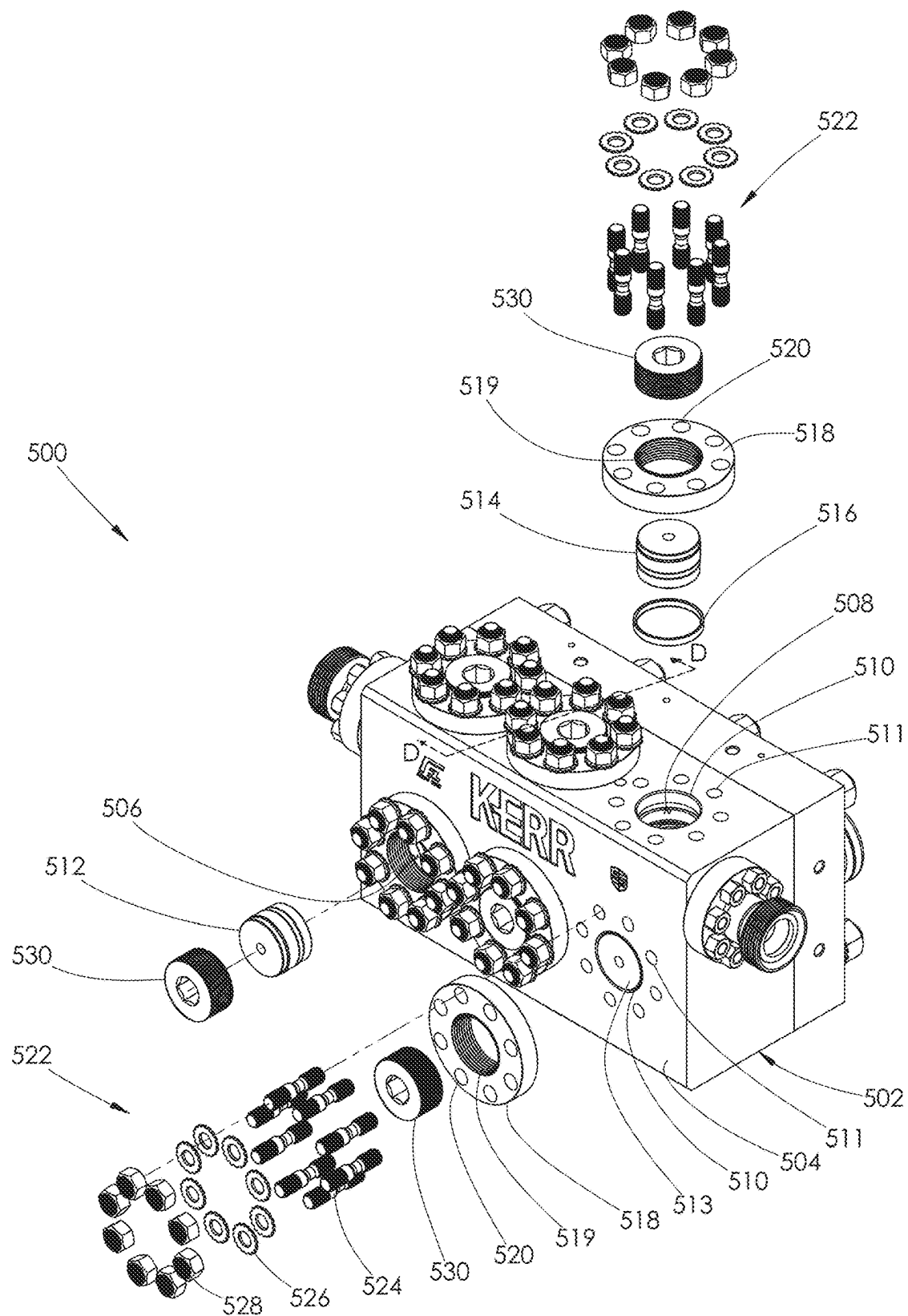


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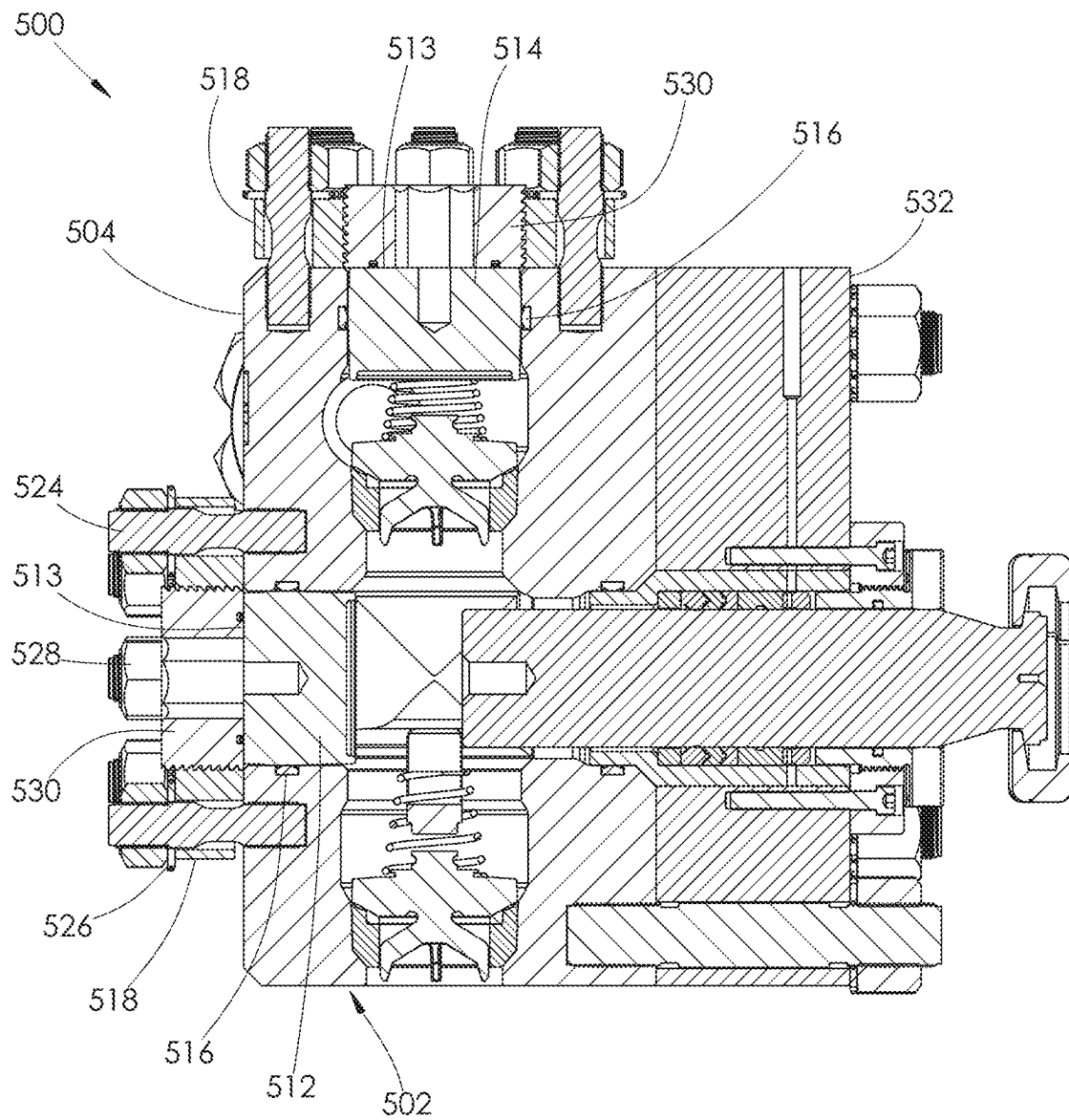


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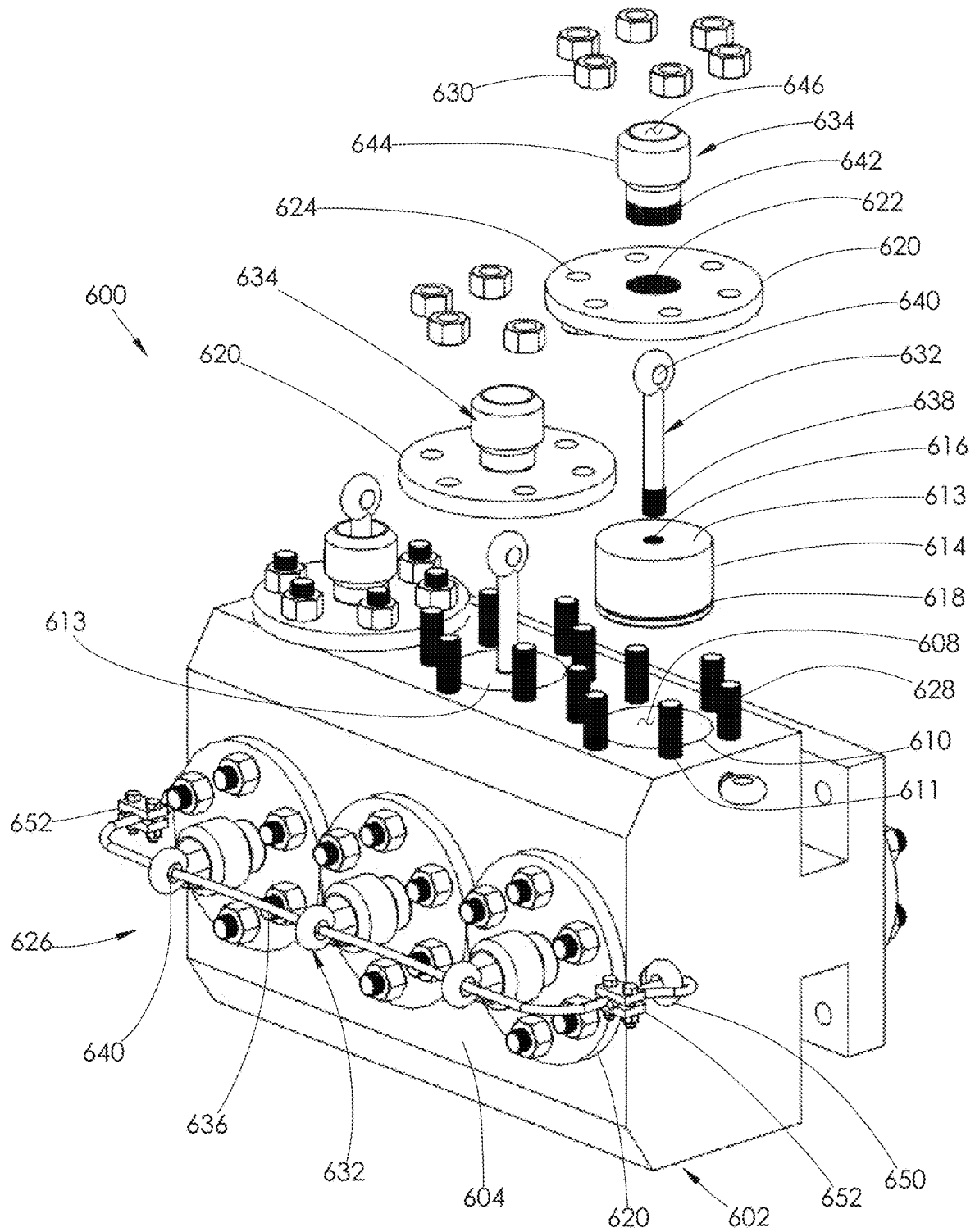


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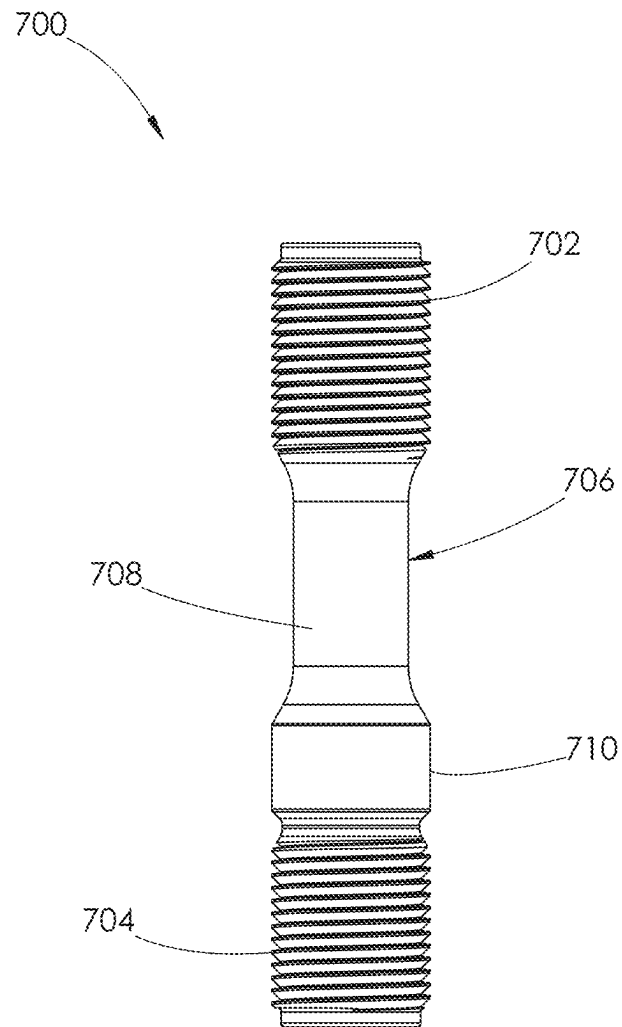


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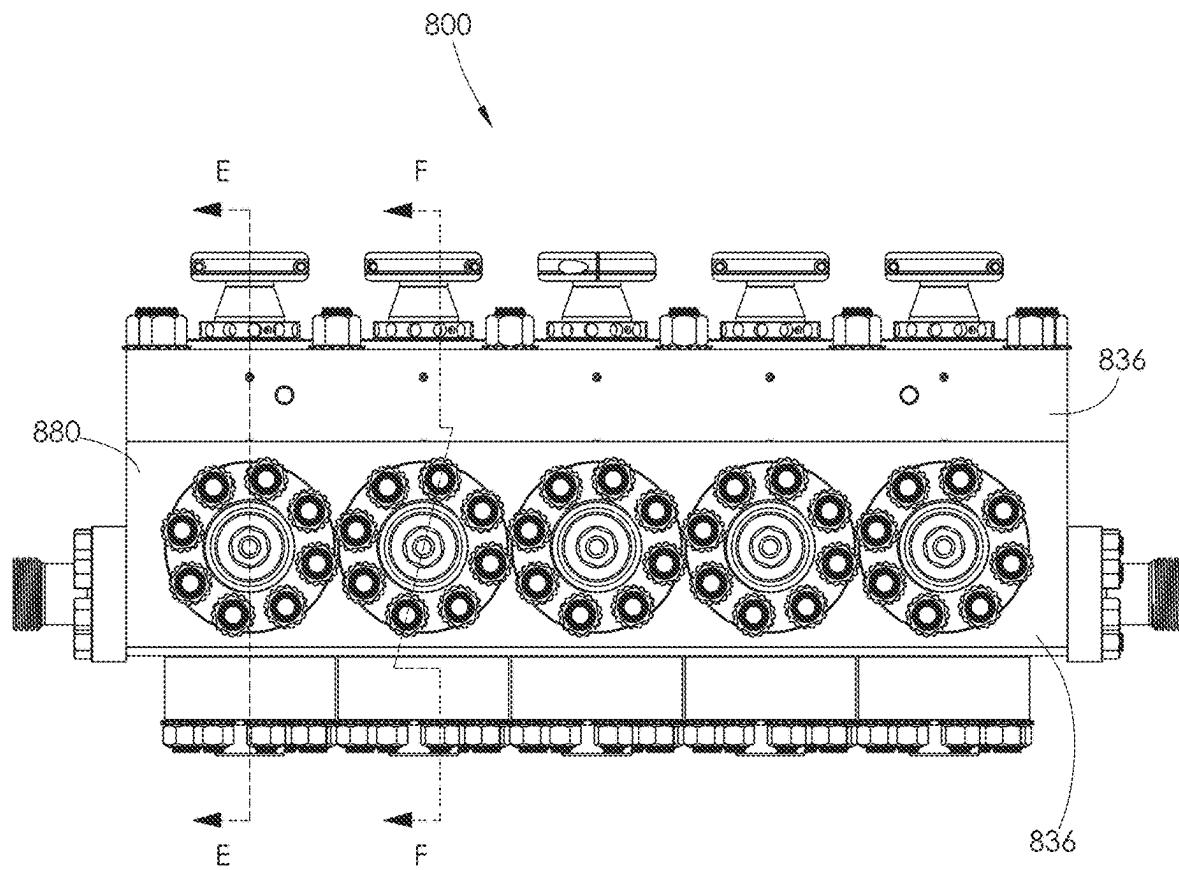


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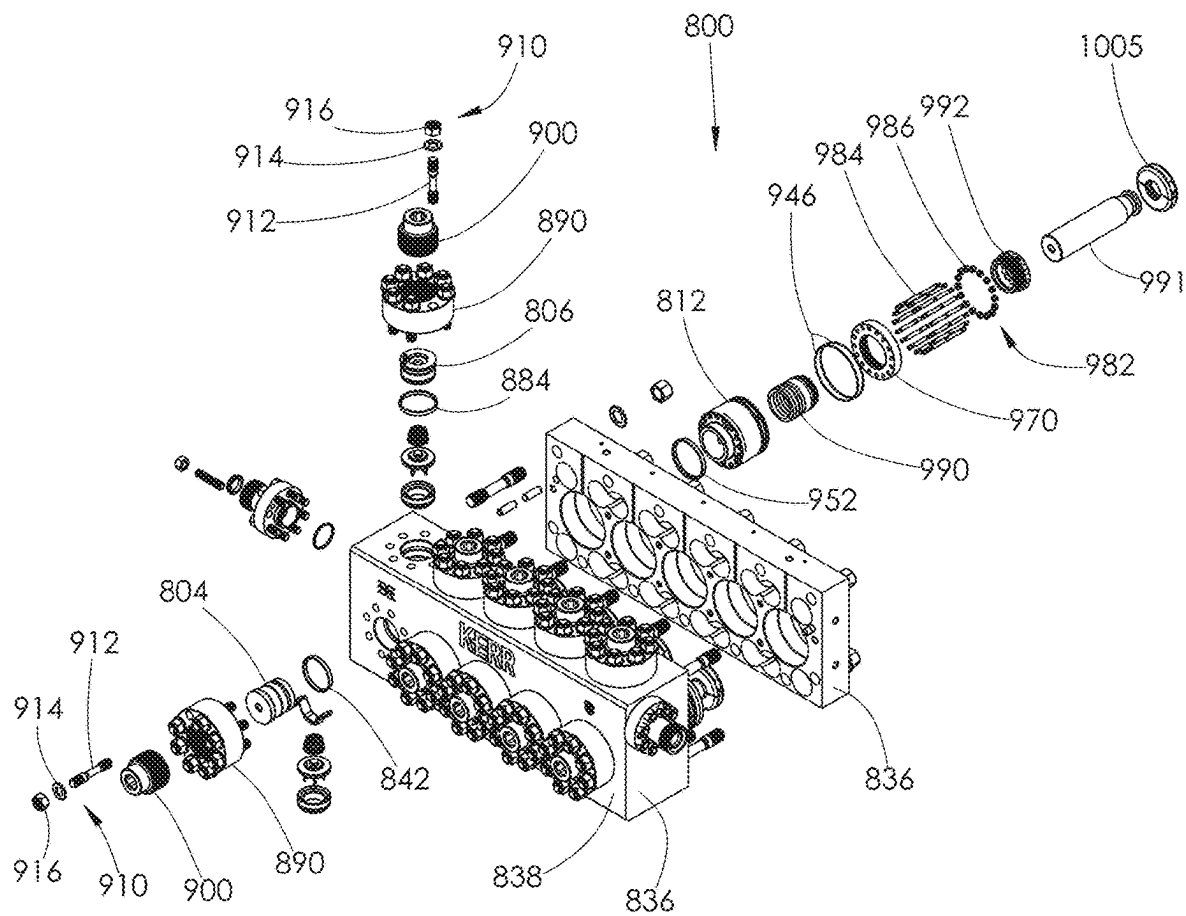


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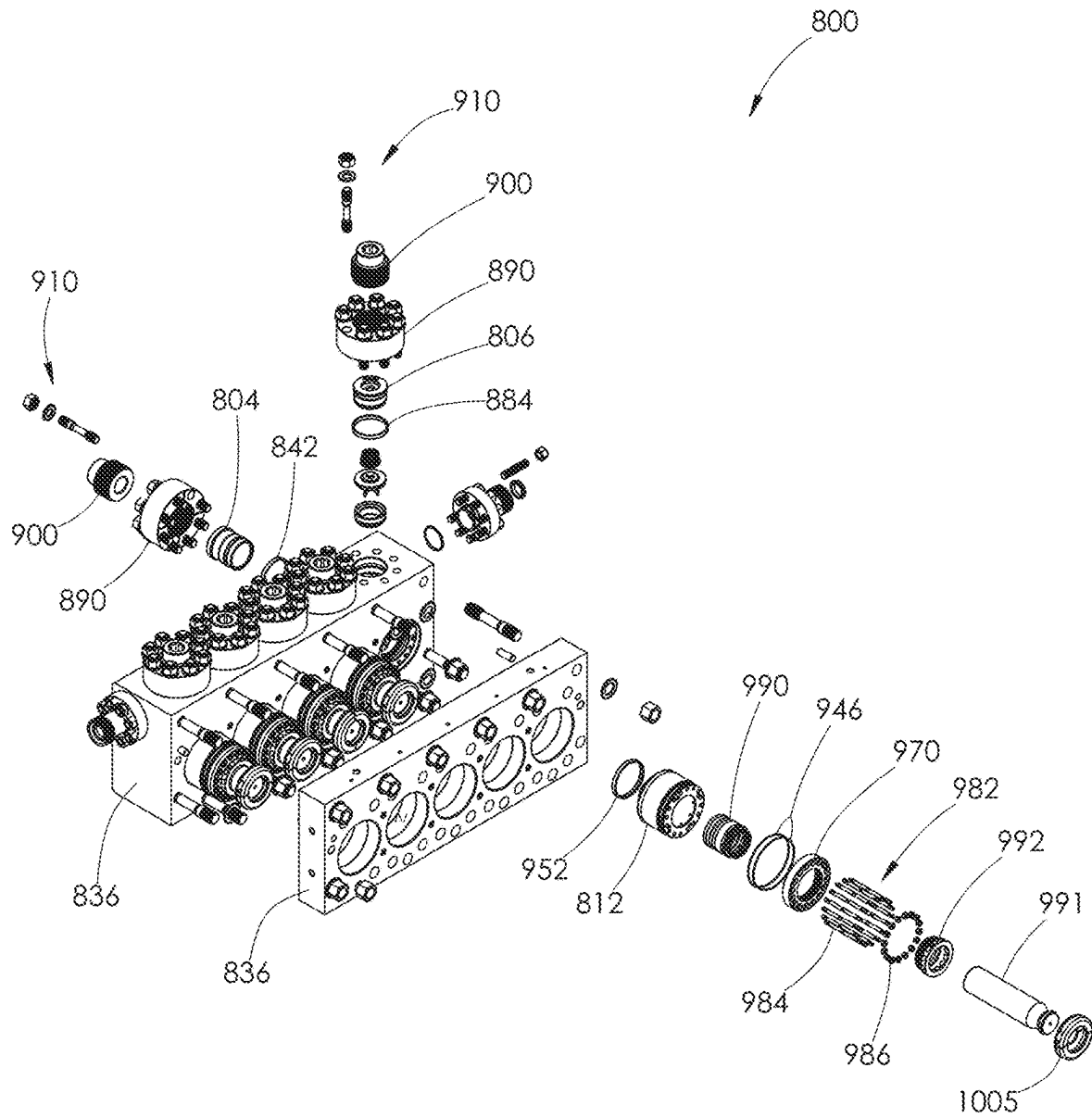


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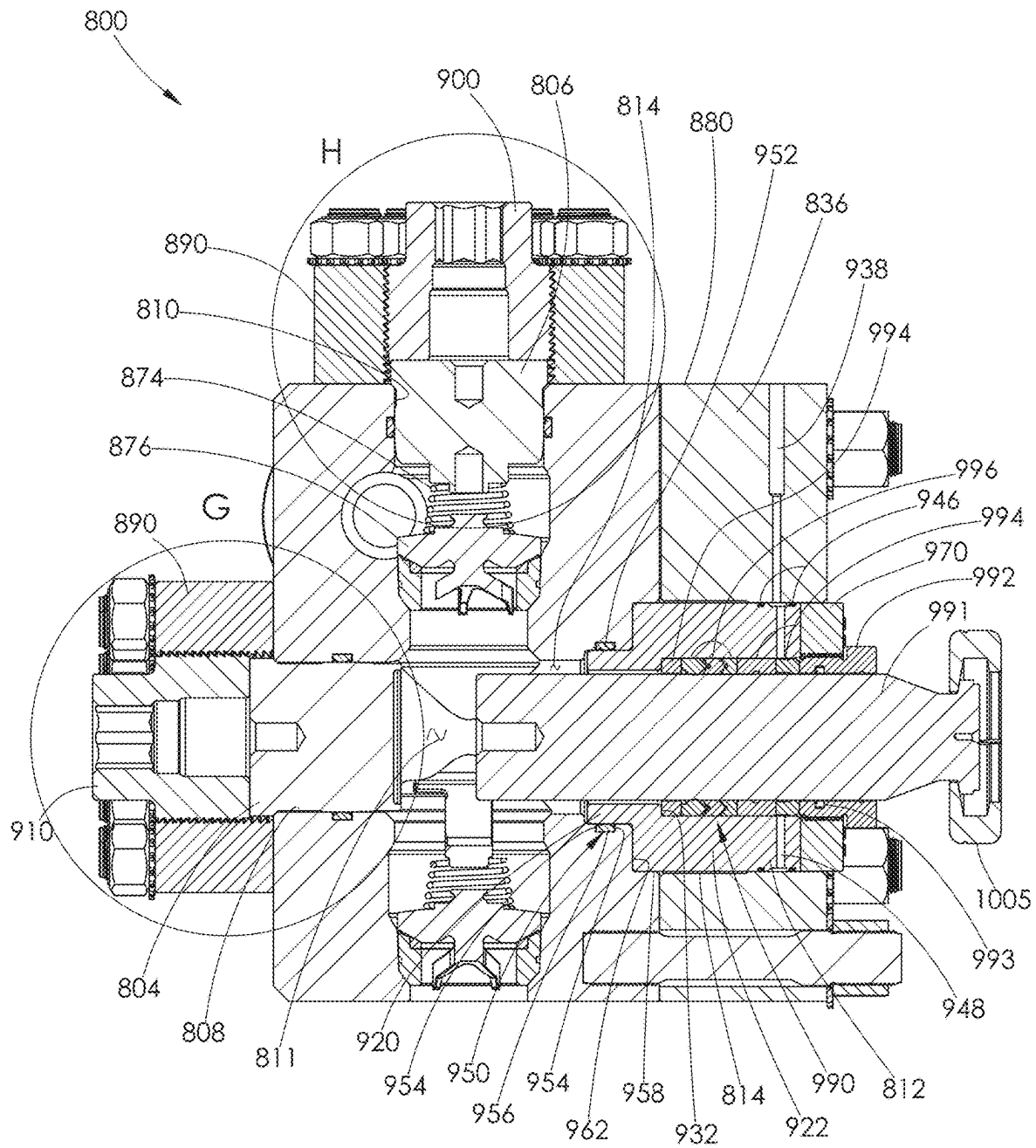


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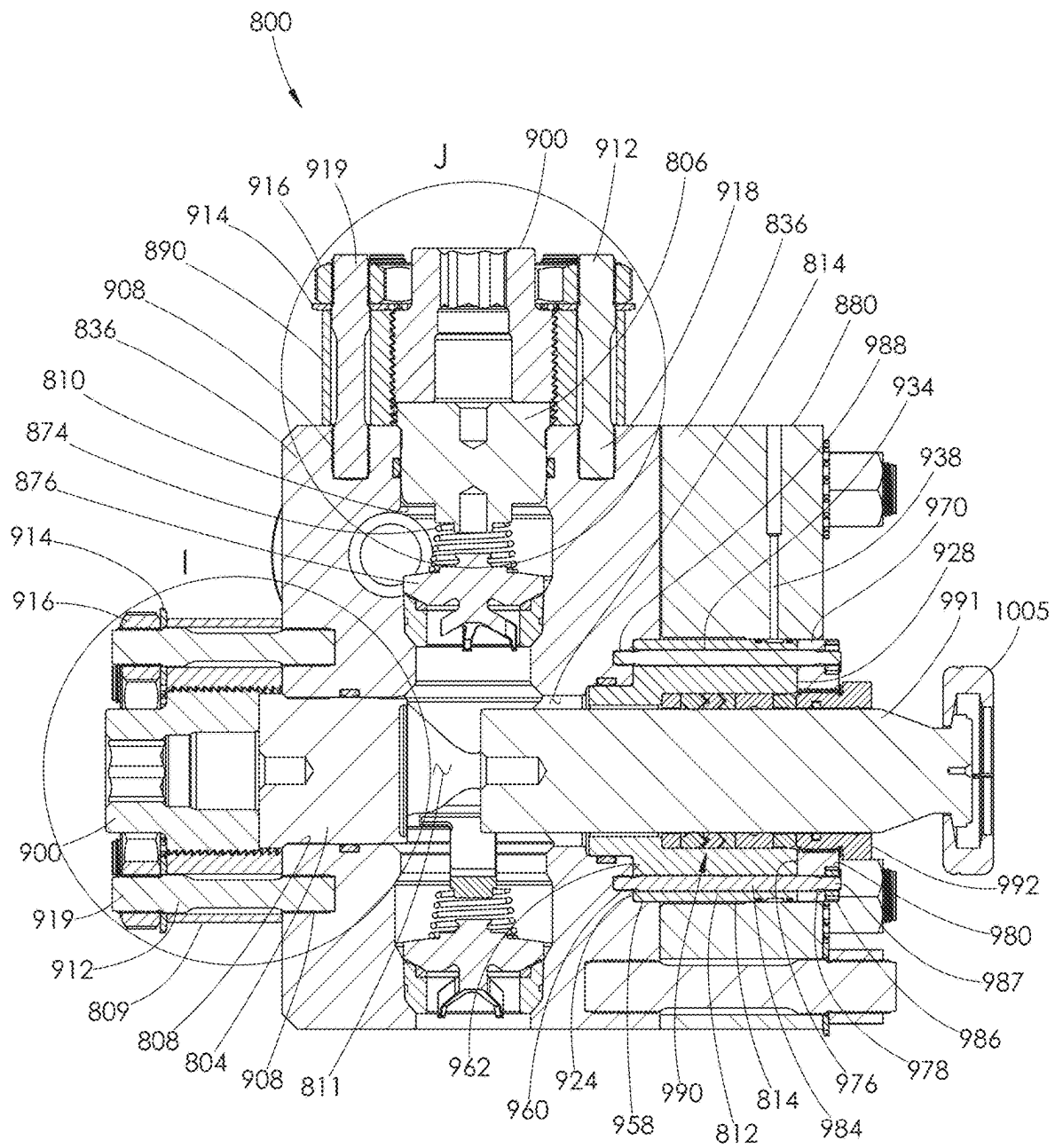


FIG. 20

FIG. 21

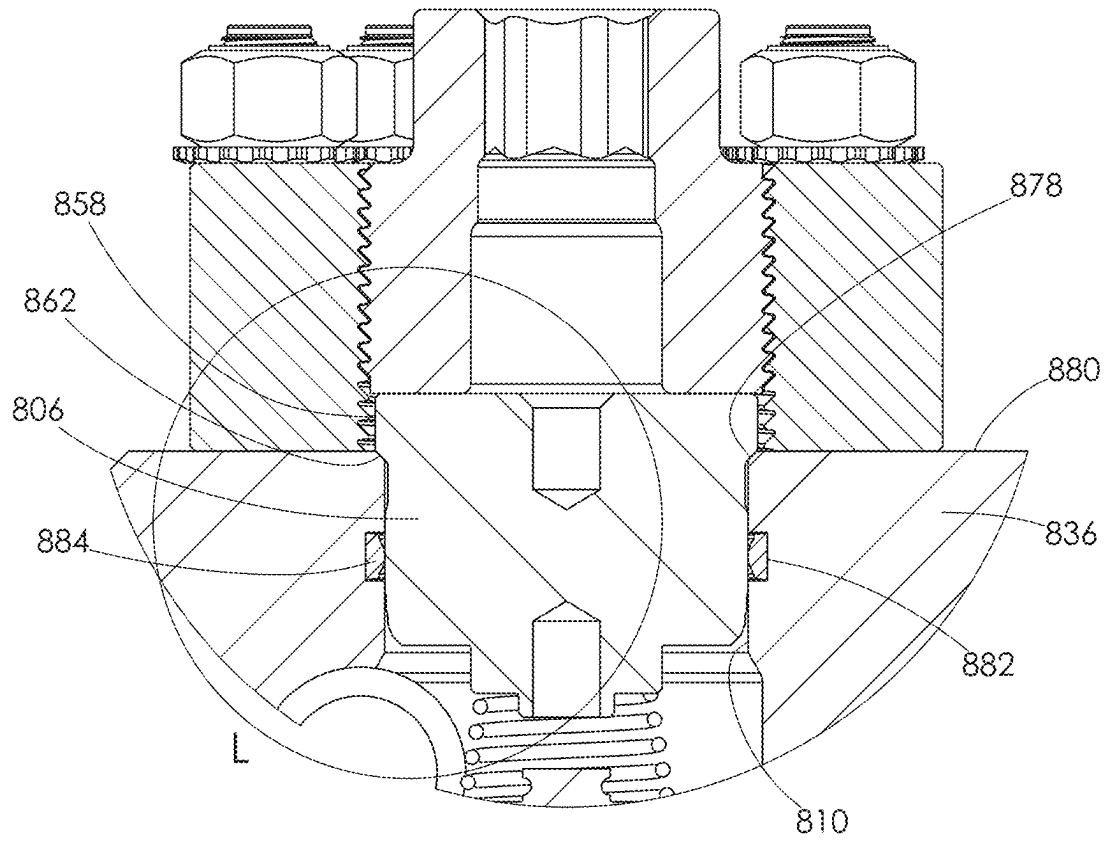


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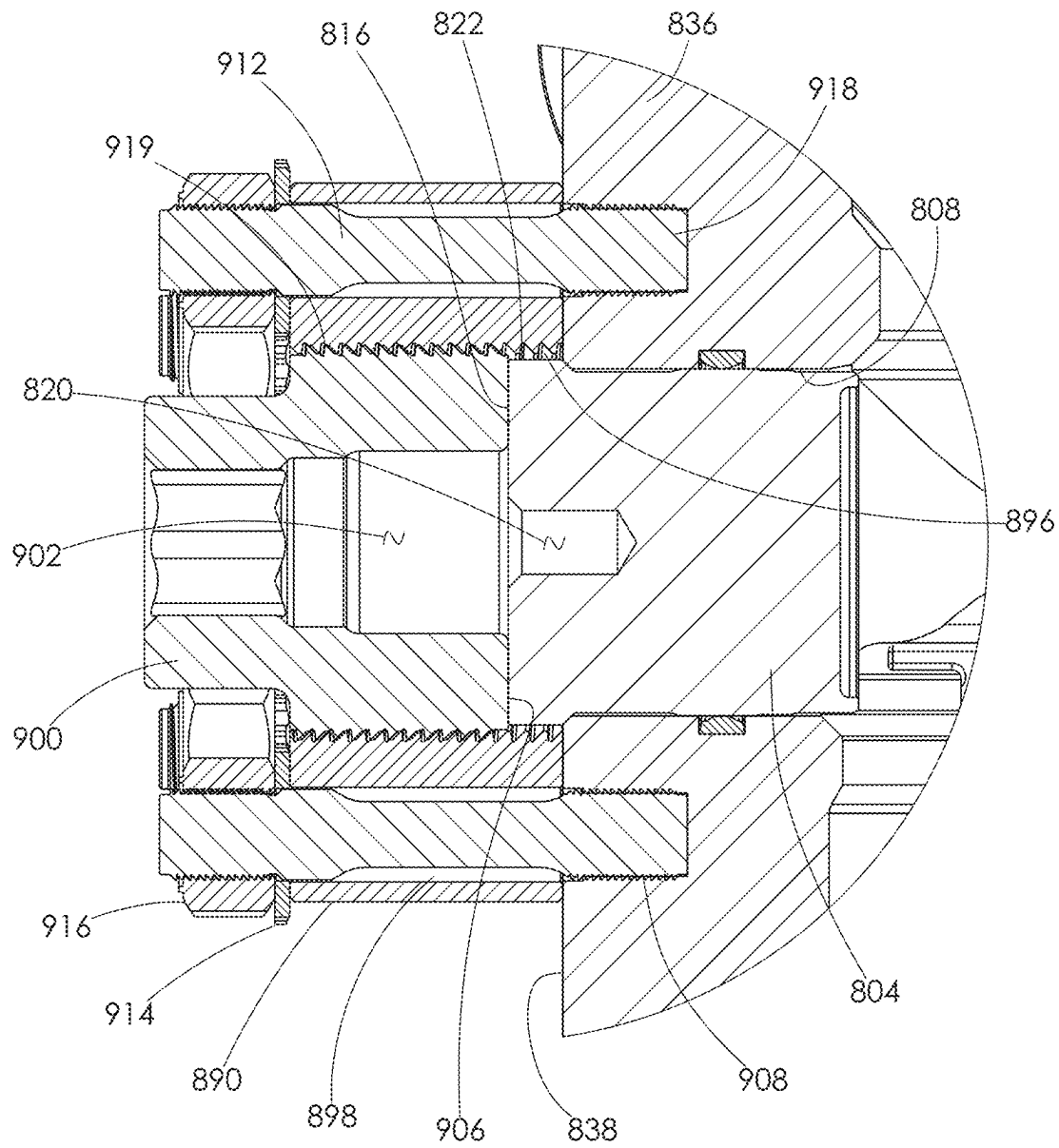


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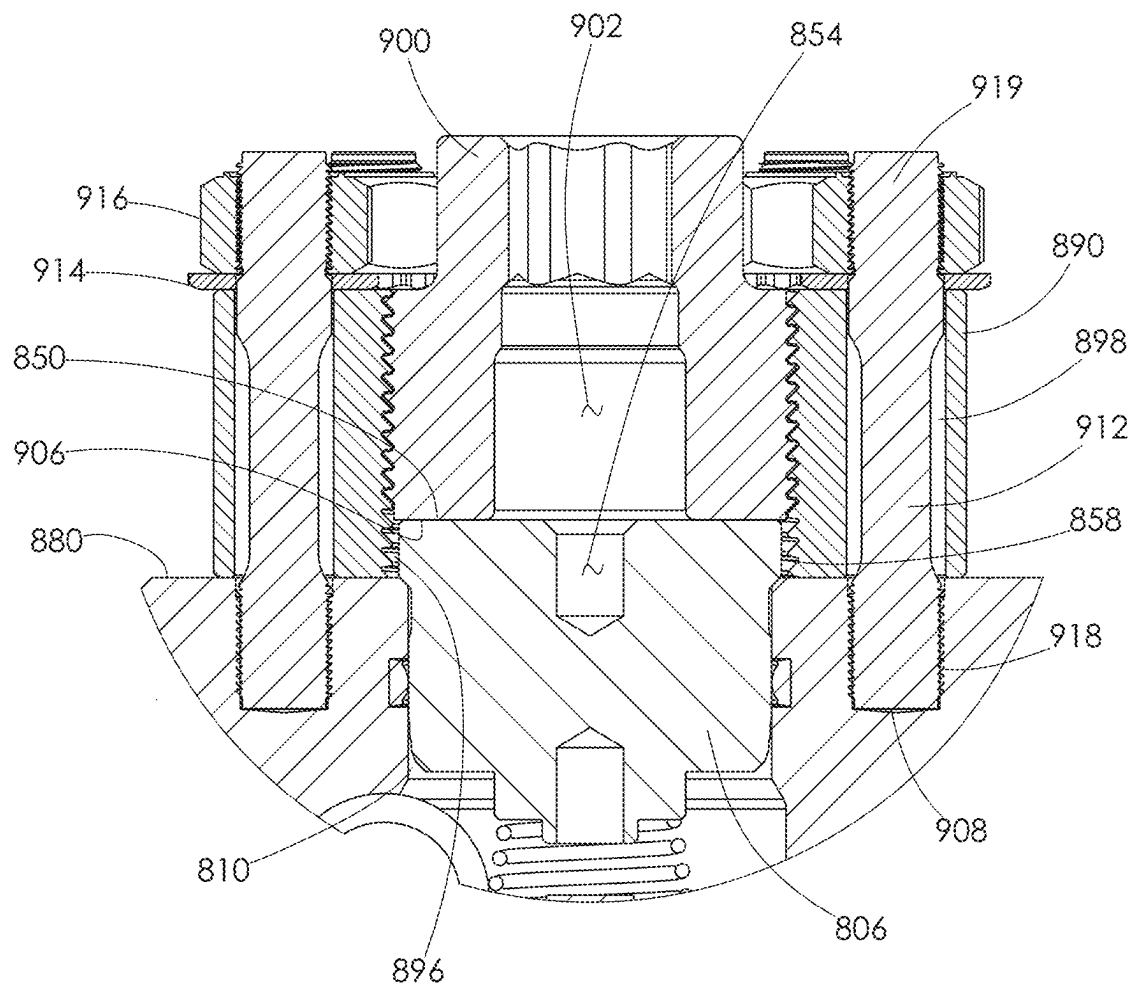


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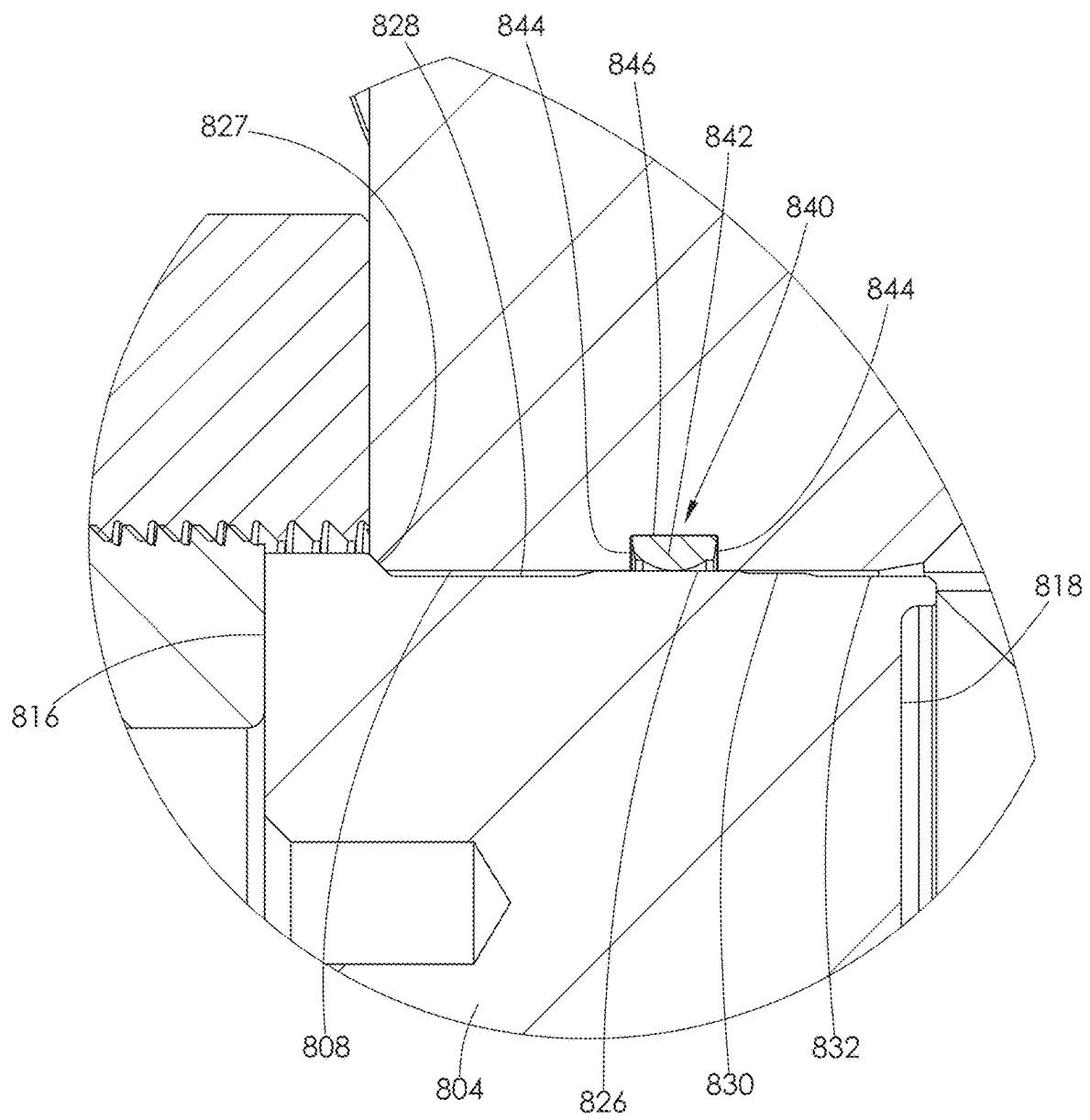


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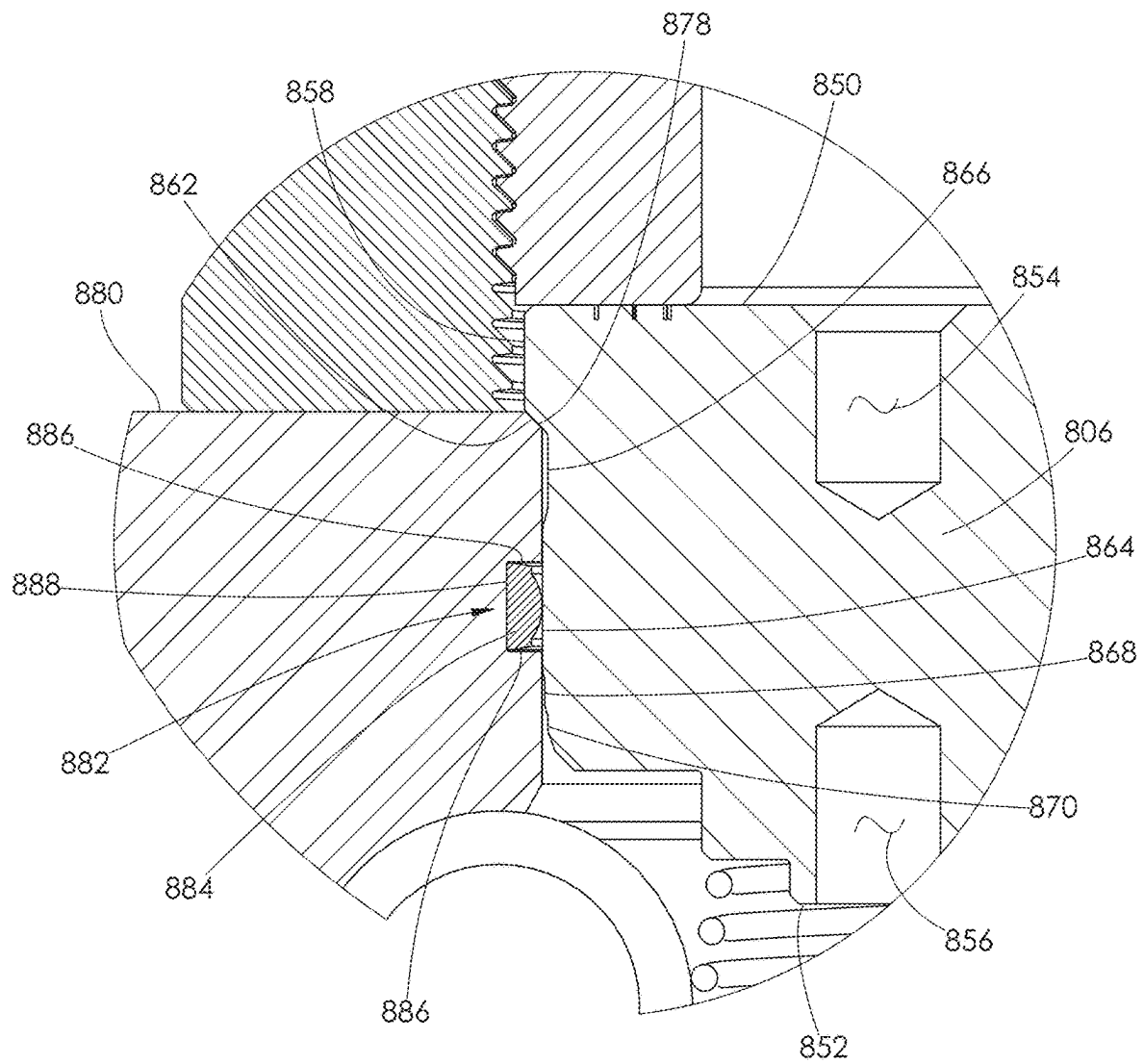


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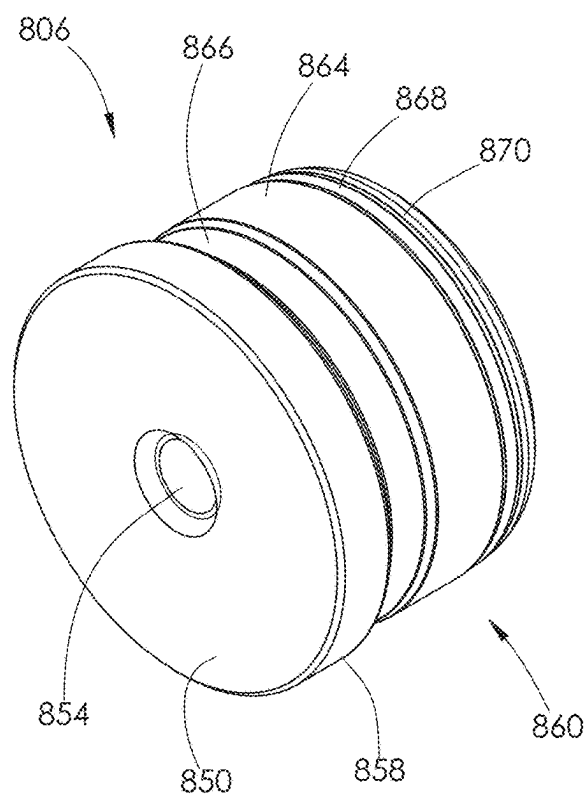


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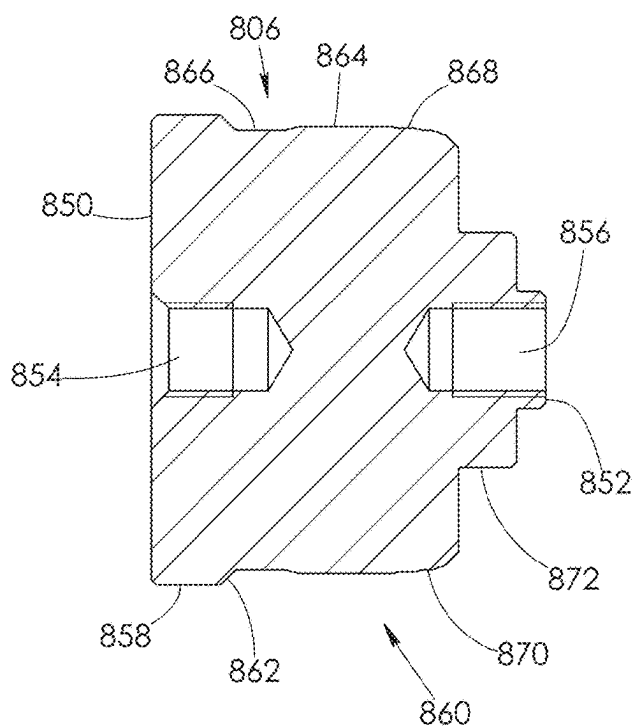


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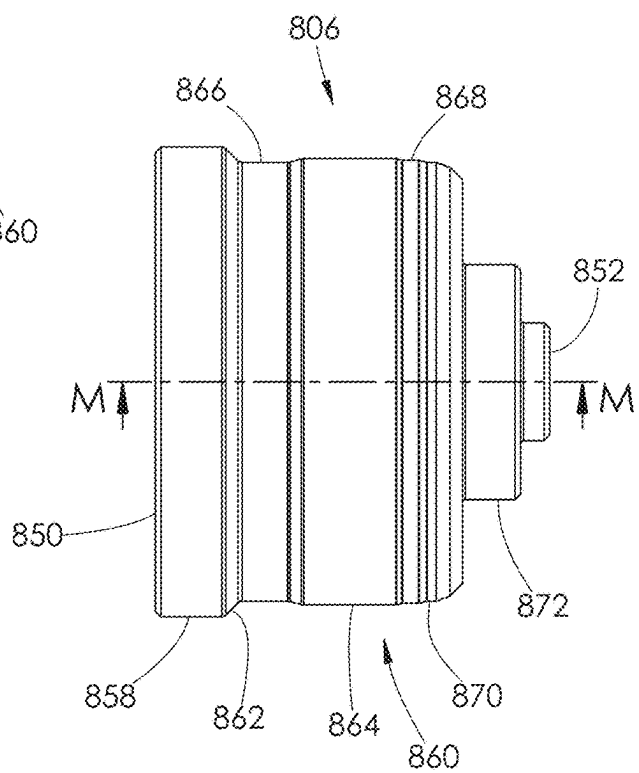


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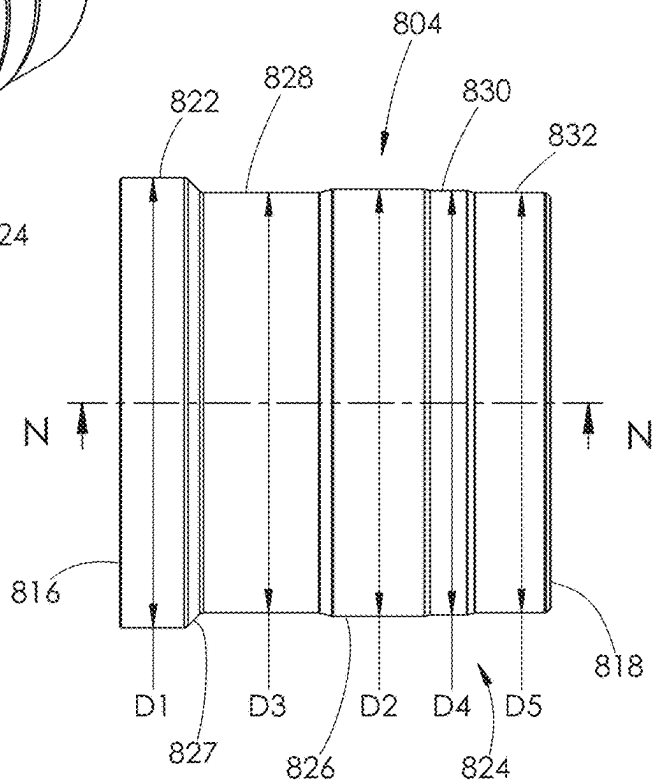
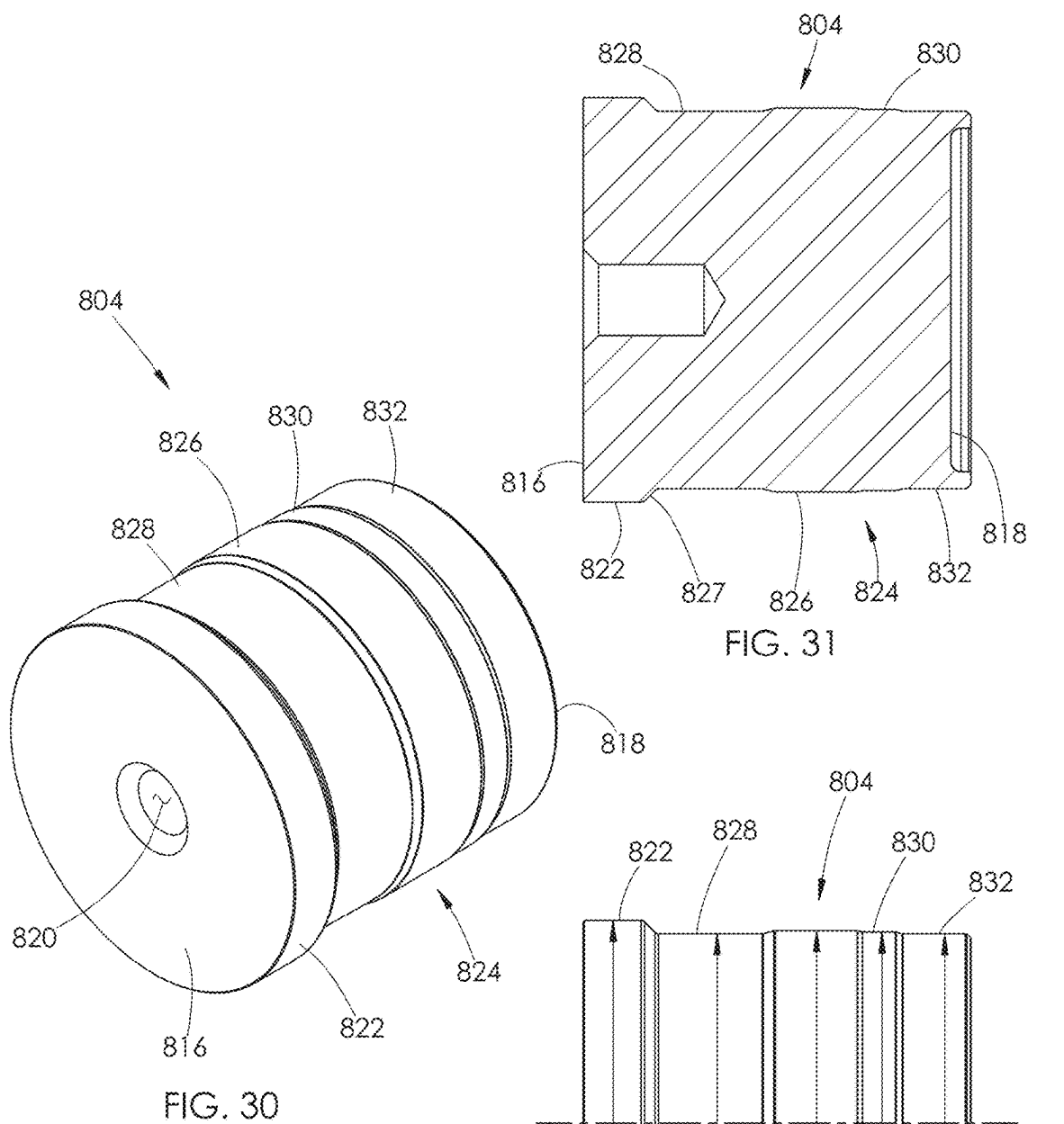


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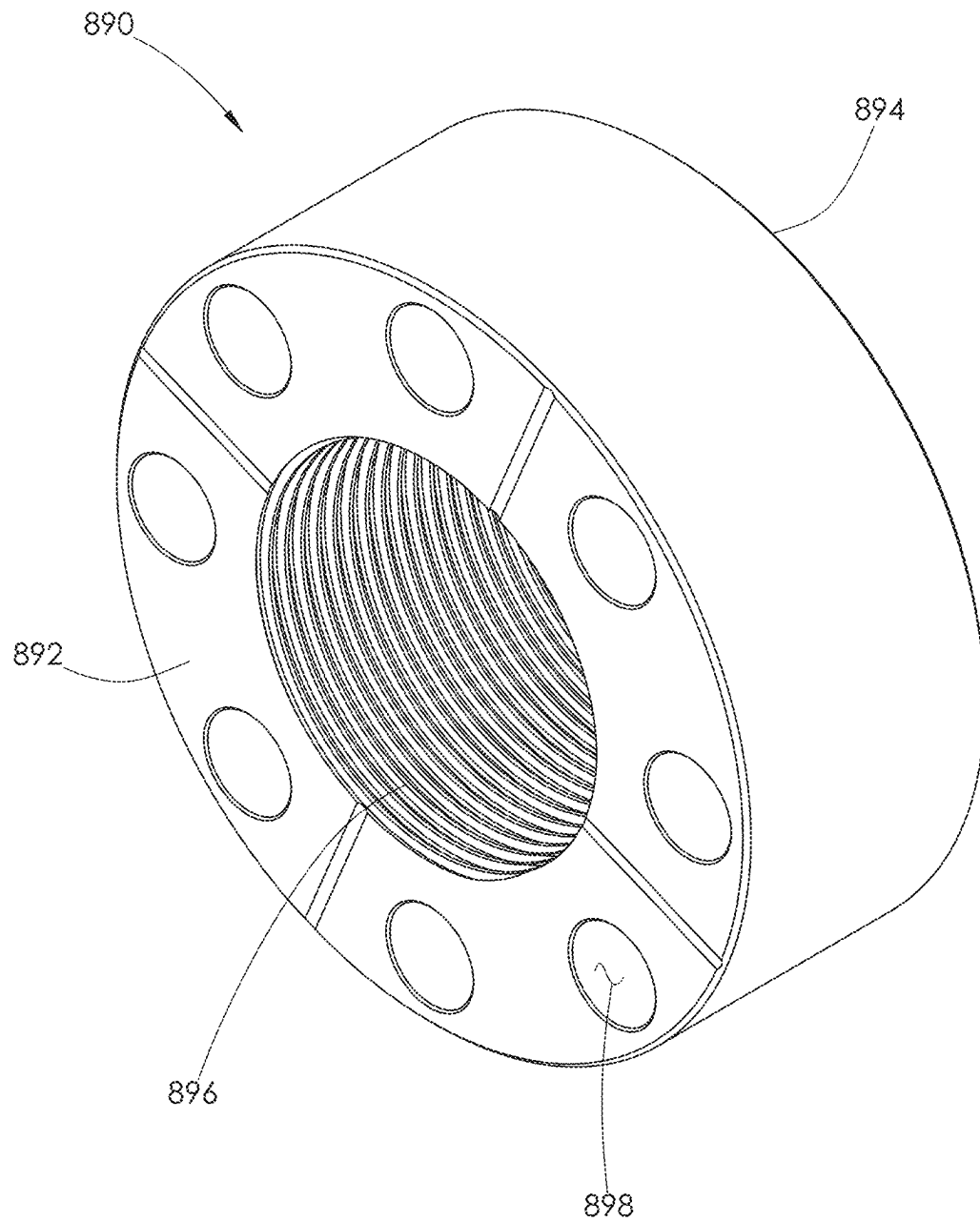
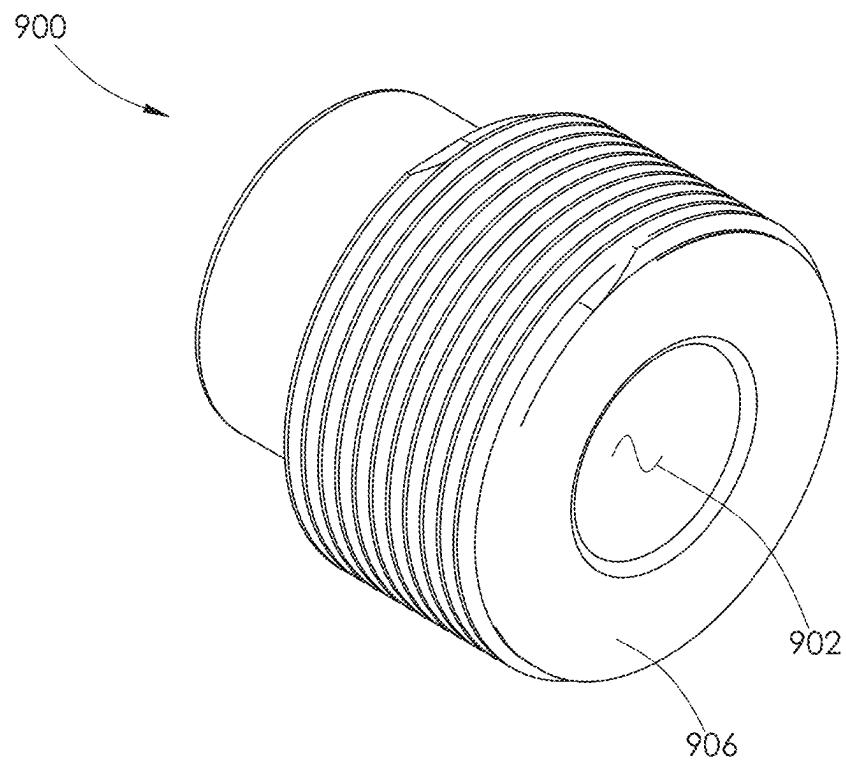
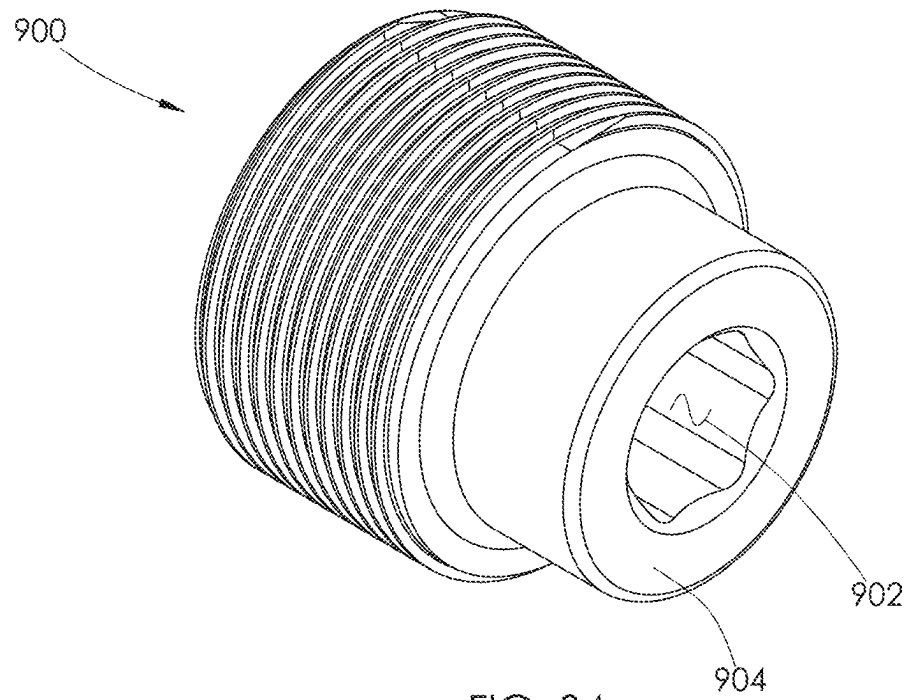


FIG. 33



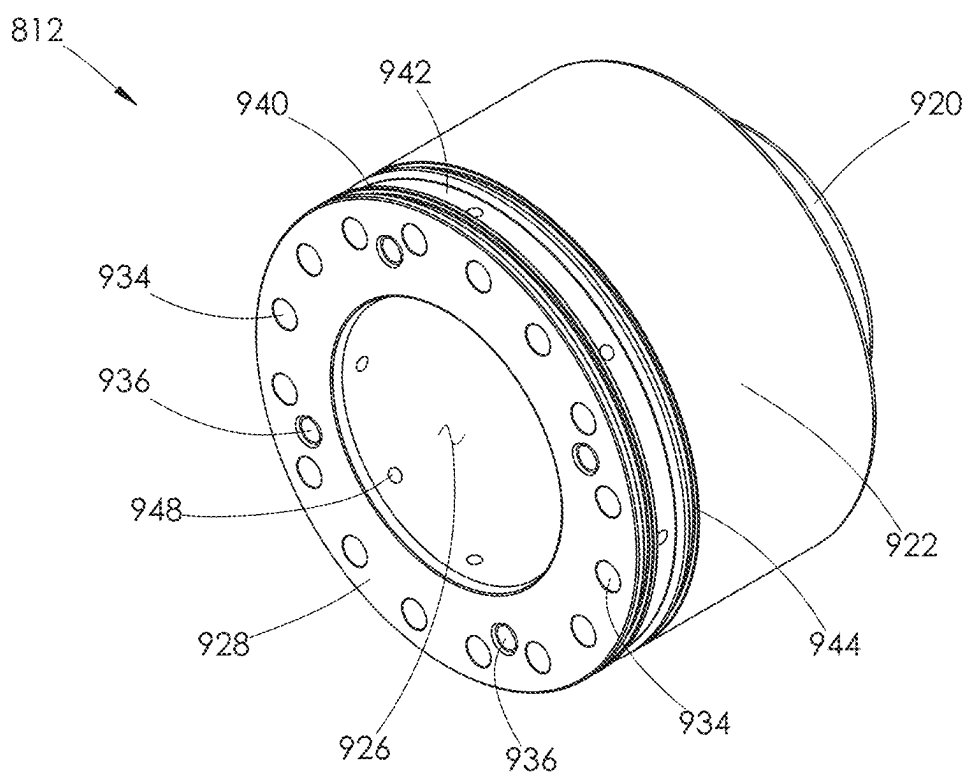


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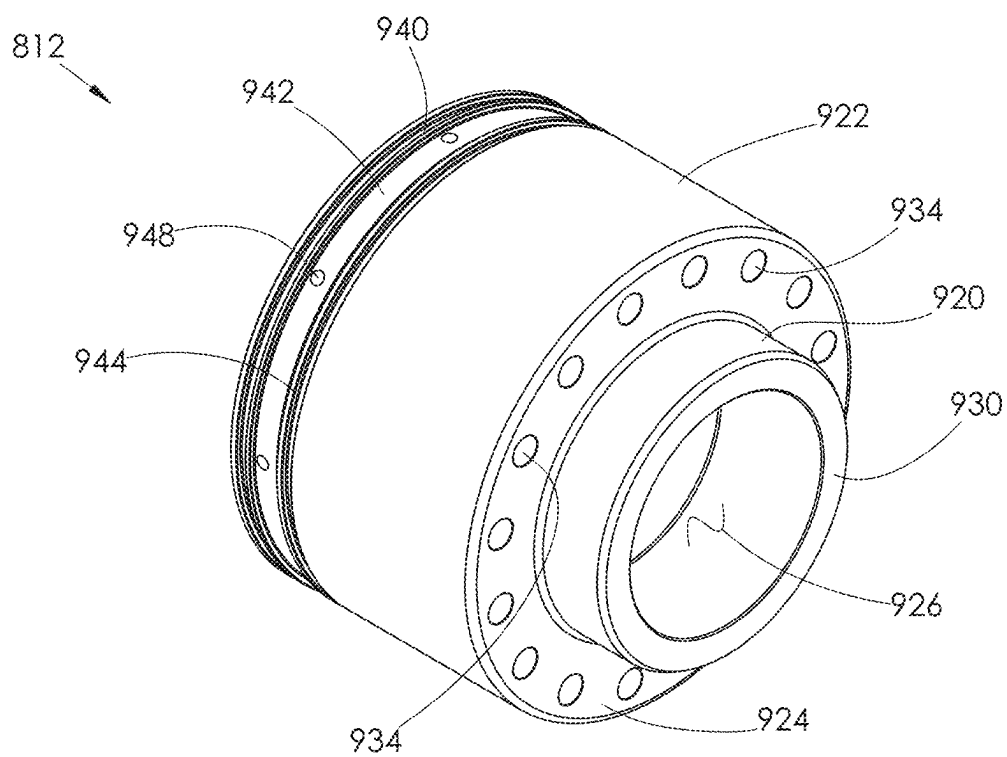


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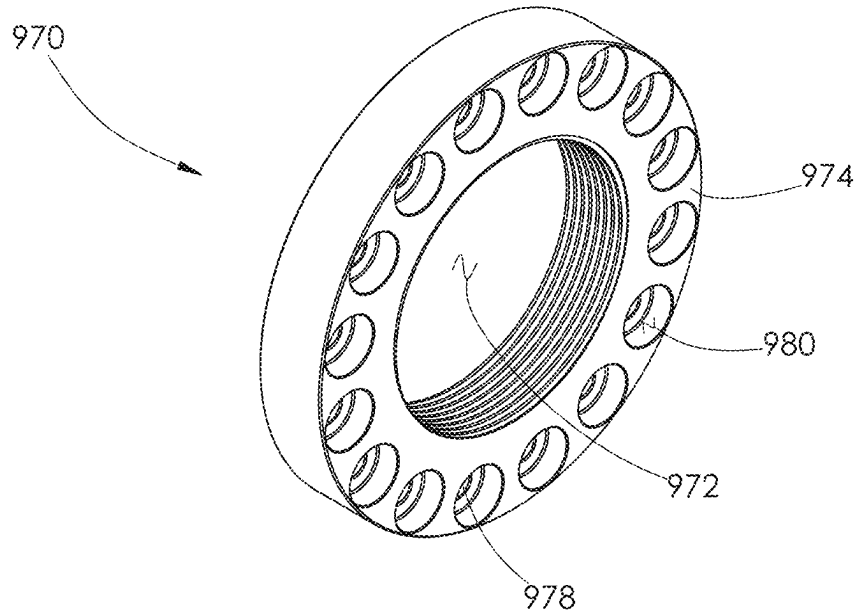


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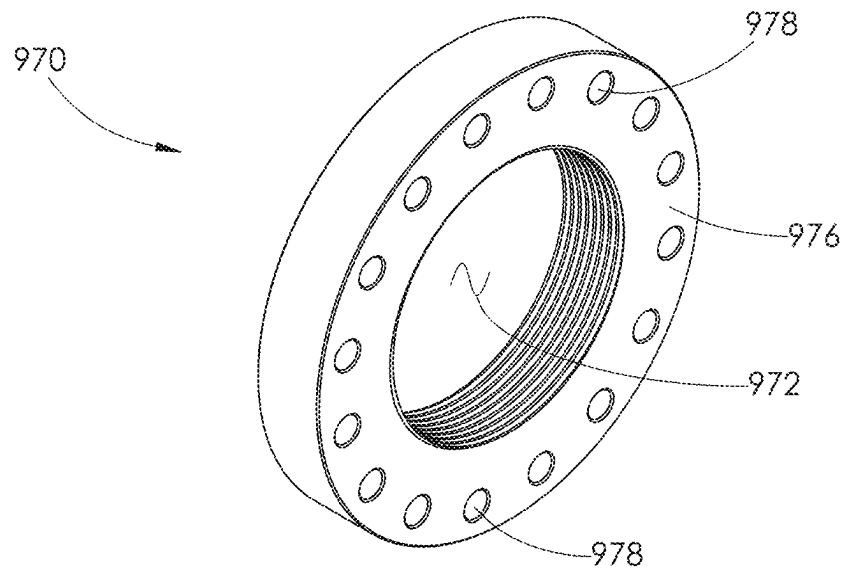


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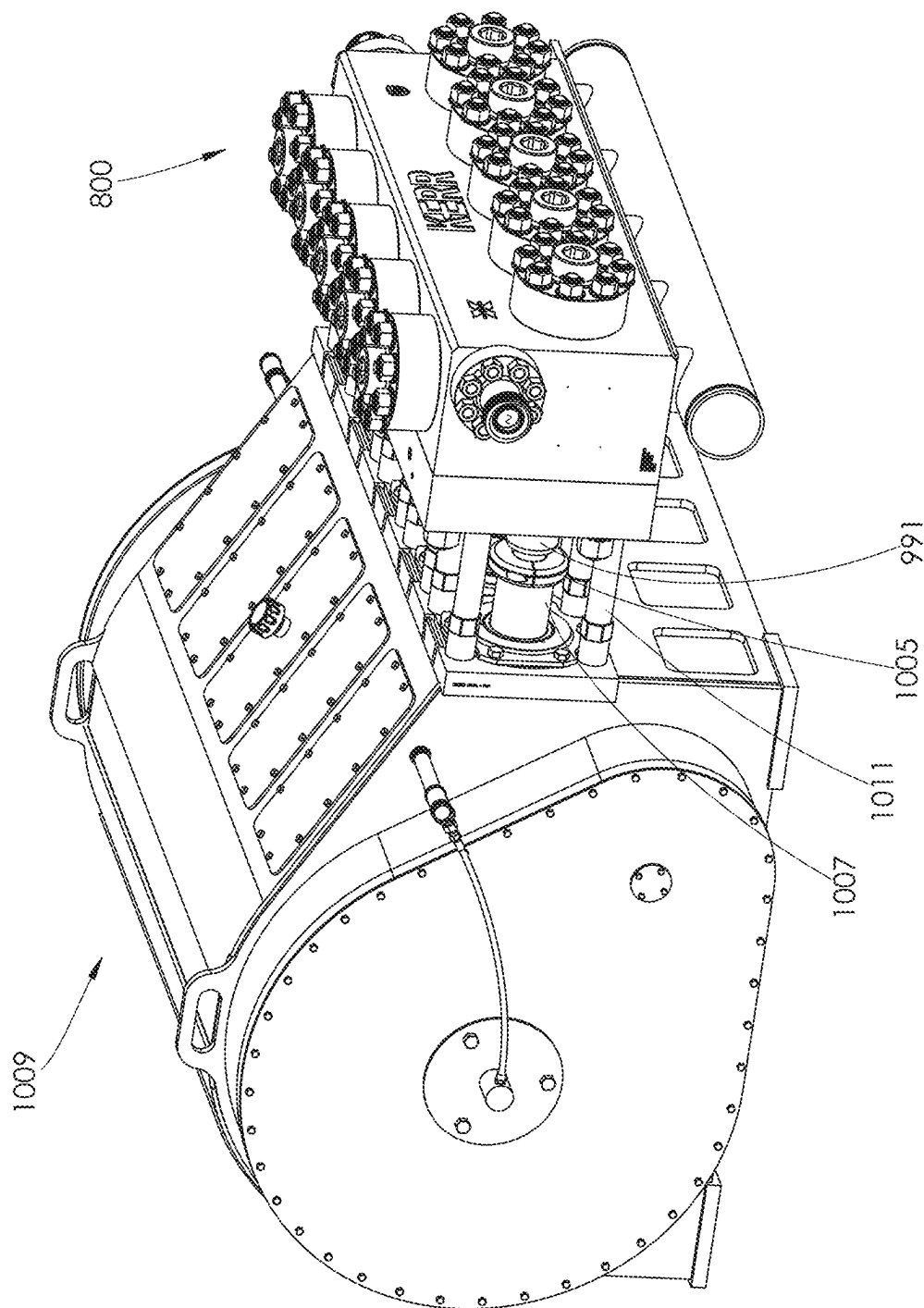


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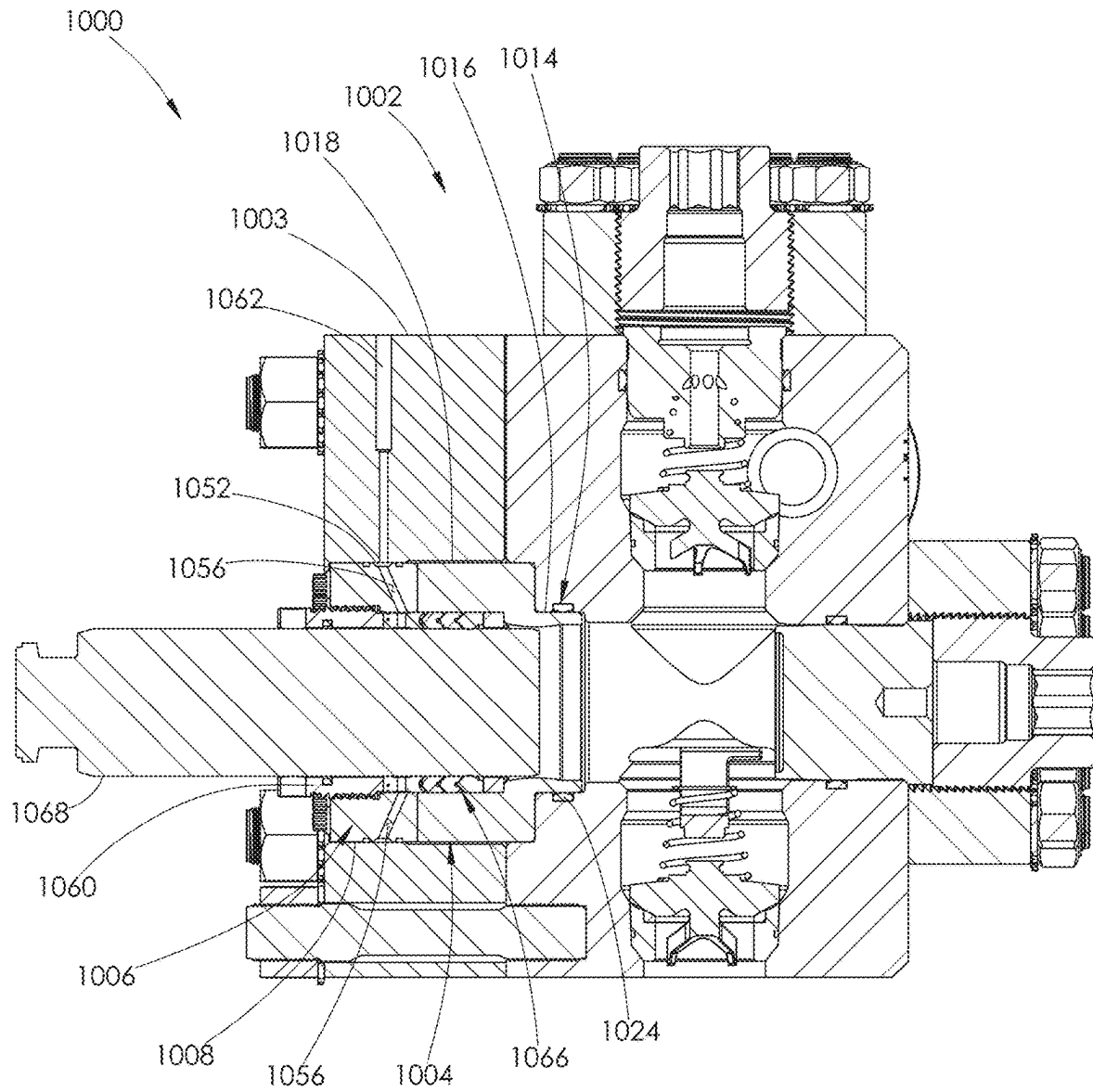


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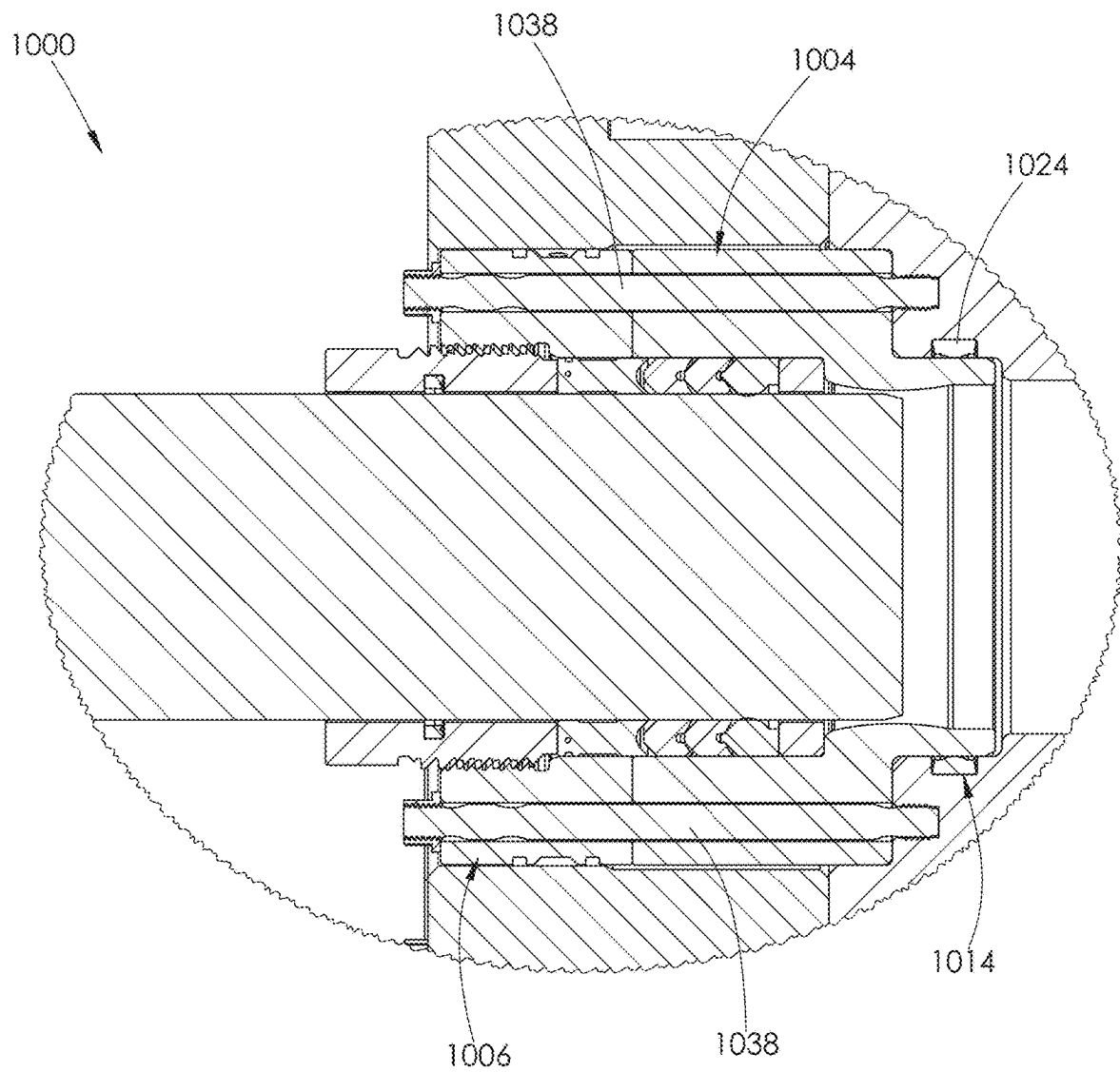


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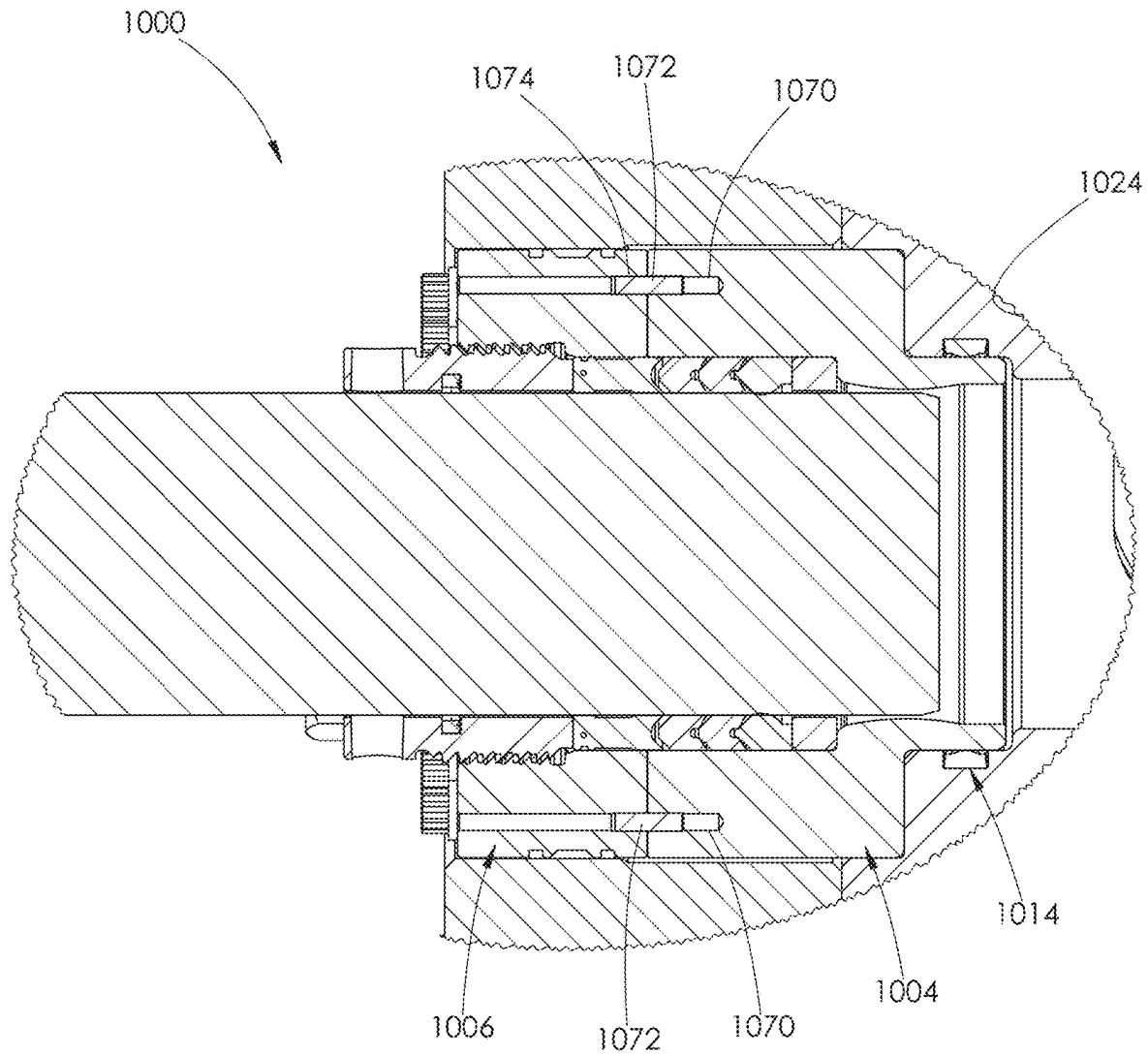


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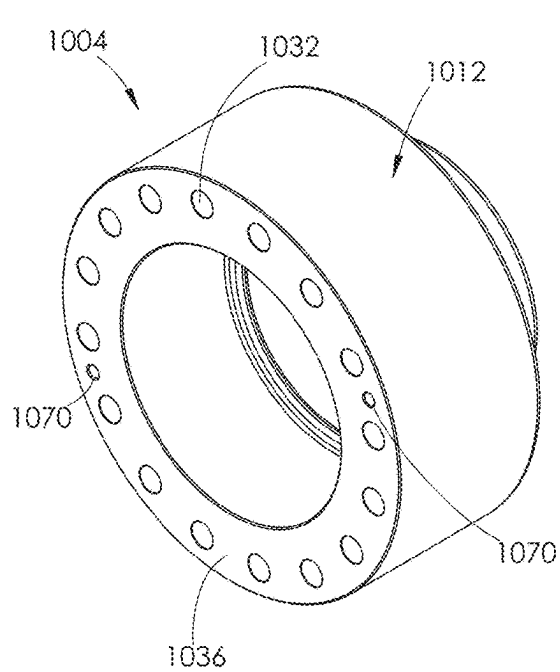


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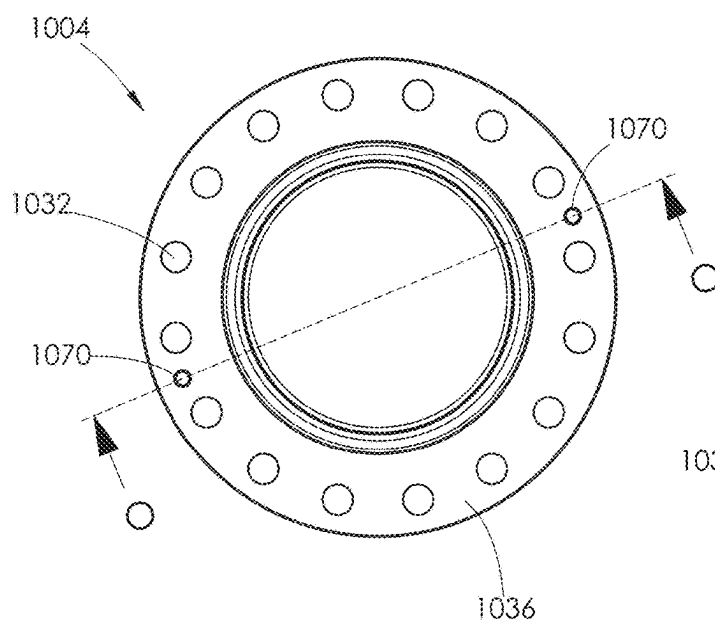
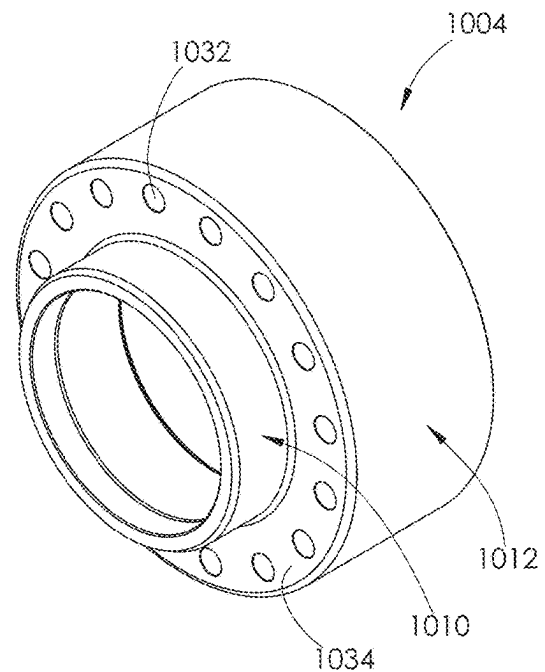


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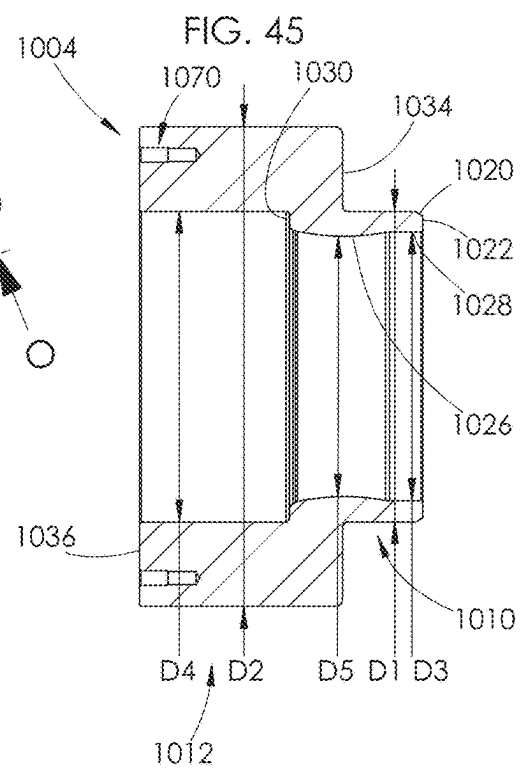


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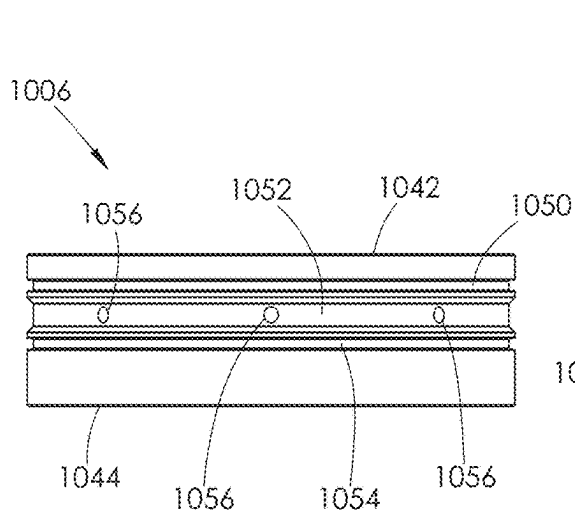


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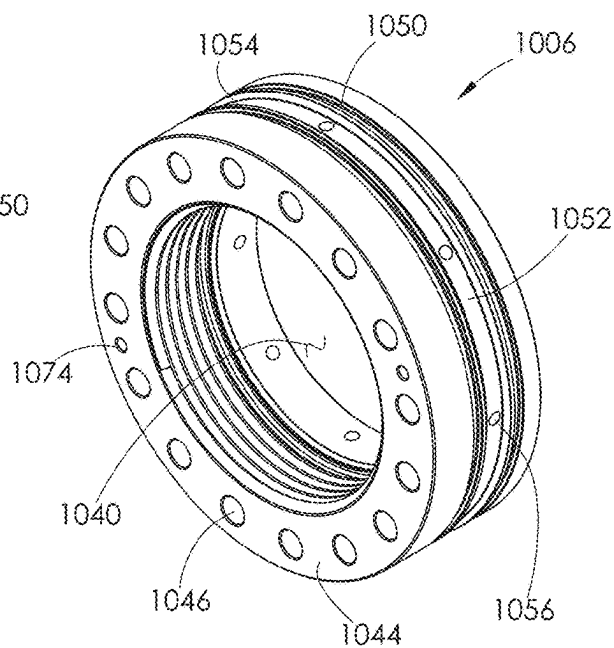


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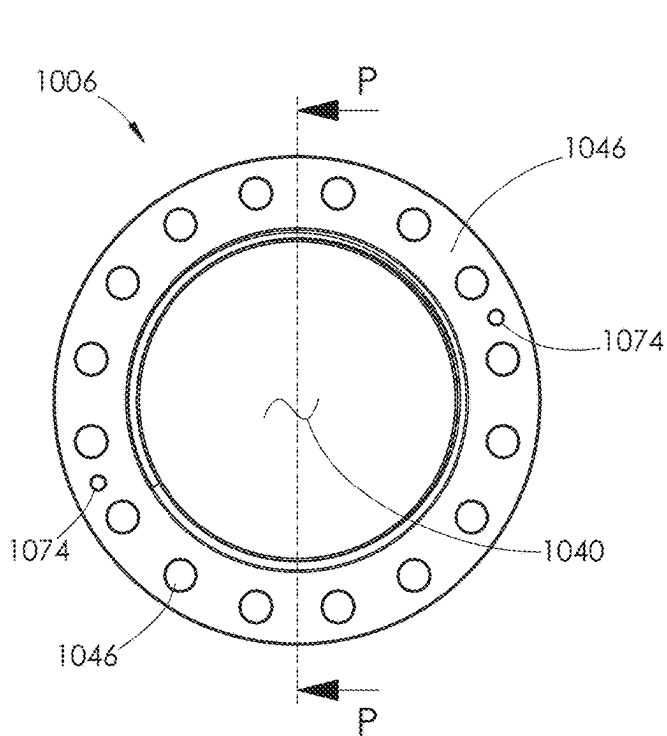


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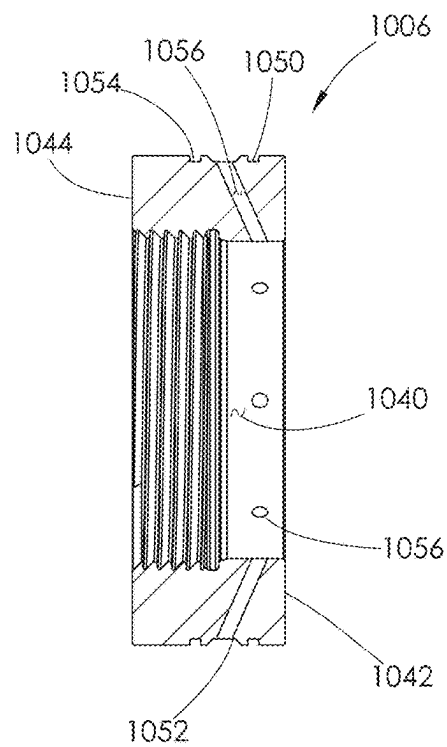


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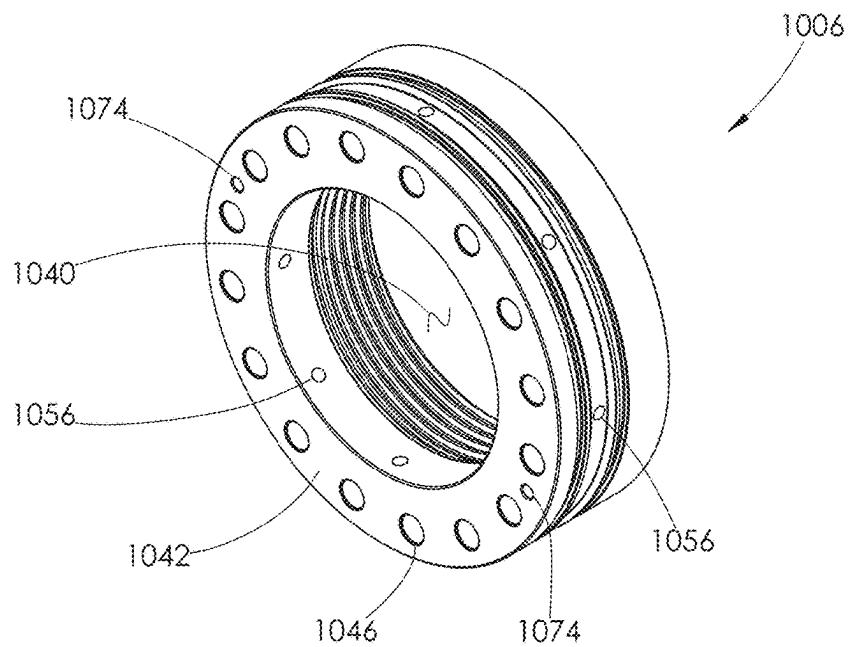


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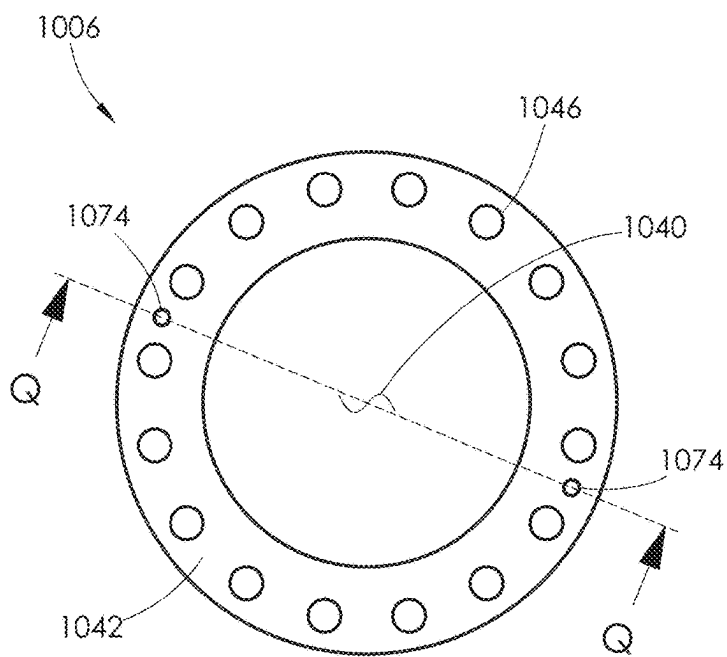


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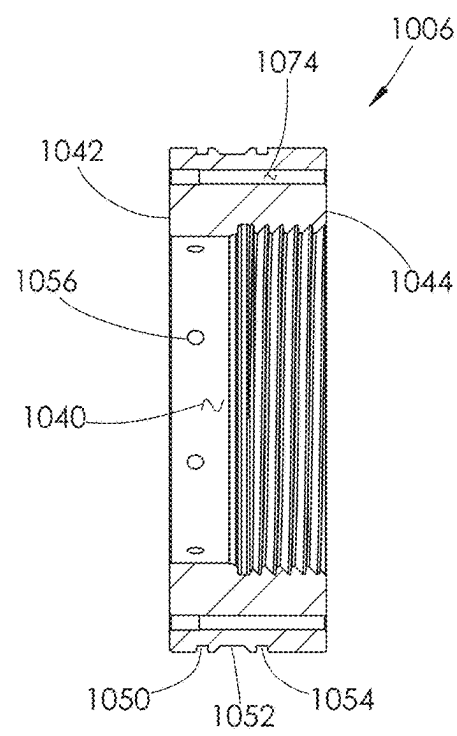


FIG. 54

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FLUID END ASSEMBLY

SUMMARY

The present disclosure is directed to a fluid end assembly. The fluid end assembly comprises a housing having an external surface and an internal chamber, a first conduit formed in the housing and having first and second sections, each section independently interconnecting the internal chamber and the external surface, and a second conduit formed in the housing, intersecting the first conduit and having third and fourth sections, each section independently interconnecting the internal chamber and the external surface. The fluid end assembly also comprises a non-threaded component installed within the third section, and a plurality of packing seals installed within the component. The fluid end assembly further comprises a retainer engaged with the component, and a fastening system installed within the retainer and the housing and configured to releasably hold the retainer against the component.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a partially exploded view of a first embodiment of a fluid end of the present invention. FIG. 1 shows a suction and discharge end of the fluid end.

FIG. 2 is a partially exploded view of a plunger end of the fluid end body shown in FIG. 1.

FIG. 3 is a cross-sectional view of the fluid end shown in FIG. 1, taken along line A-A.

FIG. 4 is a partially exploded view of a second embodiment of a fluid end.

FIG. 4 shows a suction and discharge end of the fluid end.

FIG. 5 is a partially exploded view of a plunger end of the fluid end body shown in FIG. 4.

FIG. 6 is a cross-sectional view of the fluid end shown in FIG. 4, taken along line B-B.

FIG. 7 is a partially exploded view of a third embodiment of a fluid end. FIG. 7 shows a suction and discharge end of the fluid end.

FIG. 8 is a partially exploded view of a plunger end of the fluid end body shown in FIG. 7.

FIG. 9 is a partially exploded view of a fourth embodiment of a fluid end. FIG. 9 shows a suction and discharge end of the fluid end.

FIG. 10 is a partially exploded view of a plunger end of the fluid end body shown in FIG. 9.

FIG. 11 is a cross-sectional view of the fluid end shown in FIG. 9, taken along line C-C.

FIG. 12 is a partially exploded view of a fifth embodiment of a fluid end. FIG. 12 shows a suction and discharge end of the fluid end.

FIG. 13 is a cross-sectional view of the fluid end shown in FIG. 12, taken along line D-D.

FIG. 14 is a partially exploded view of a sixth embodiment of a fluid end. FIG. 14 shows a suction and discharge end of the fluid end.

FIG. 15 is a side elevational view of one of the plurality of studs for use with the fluid ends.

FIG. 16 is a top plan view of a seventh embodiment of a fluid end.

FIG. 17 is a partially exploded front perspective view of the fluid end shown in FIG. 16.

FIG. 18 is a partially exploded rear perspective view of the fluid end shown in FIG. 16.

FIG. 19 is a cross-sectional view of the fluid end shown in FIG. 16, taken along line E-E.

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FIG. 20 is a cross-sectional view of the fluid end shown in FIG. 16, taken along line F-F.

FIG. 21 is an enlarged view of area G shown in FIG. 19.

FIG. 22 is an enlarged view of area H shown in FIG. 19.

FIG. 23 is an enlarged view of area I shown in FIG. 20.

FIG. 24 is an enlarged view of area J shown in FIG. 20.

FIG. 25 is an enlarged view of area K shown in FIG. 21.

FIG. 26 is an enlarged view of area L shown in FIG. 22.

FIG. 27 is a top perspective view of a discharge plug shown installed within the fluid end in FIGS. 19 and 20.

FIG. 28 is a cross-sectional view of the discharge plug shown in FIG. 29, taken along line M-M.

FIG. 29 is a side elevational view of the discharge plug shown in FIG. 27.

FIG. 30 is a top perspective view of a suction plug shown installed within the fluid end in FIGS. 19 and 20.

FIG. 31 is a cross-sectional view of the suction plug shown in FIG. 32, taken along line N-N.

FIG. 32 is a side elevational view of the suction plug shown in FIG. 30.

FIG. 33 is a top perspective view of a retainer shown attached to the fluid end in FIGS. 19 and 20.

FIG. 34 is a top perspective view of a retainer nut shown attached to the fluid end in FIGS. 19 and 20.

FIG. 35 is a bottom perspective view of the retainer nut shown in FIG. 34.

FIG. 36 is a rear perspective view of a stuffing box shown installed within the fluid end in FIGS. 19 and 20.

FIG. 37 is a front perspective view of the stuffing box shown in FIG. 36.

FIG. 38 is a rear perspective view of a retainer shown attached to the fluid end in FIGS. 19 and 20.

FIG. 39 is a front perspective view of the retainer shown in FIG. 38.

FIG. 40 is a side perspective view of the fluid end shown in FIG. 16 attached to a power end.

FIG. 41 is a cross-sectional view of an eighth embodiment of a fluid end.

FIG. 42 is an enlarged cross-sectional view of the plunger bore of the fluid end shown in FIG. 41. The cross-section is taken along a line extending through a different plane than that shown in FIG. 41.

FIG. 43 is an enlarged cross-sectional view of the plunger bore of the fluid end shown in FIG. 41. The cross-section is taken along a line extending through a different plane than that shown in FIGS. 41 and 42.

FIG. 44 is a perspective view of a second surface of another embodiment of a stuffing box.

FIG. 45 is a perspective view of a first surface of the stuffing box shown in FIG. 44.

FIG. 46 is a side elevational view of the second surface of the stuffing box shown in FIG. 44.

FIG. 47 is a cross-sectional view of the stuffing box shown in FIG. 46, taken along line O-O.

FIG. 48 is a top plan view of another embodiment of a retainer.

FIG. 49 is a perspective view of a second surface of the retainer shown in FIG. 48.

FIG. 50 is a side elevational view of the second surface of the retainer shown in FIG. 48.

FIG. 51 is a cross-sectional view of the retainer shown in FIG. 50, taken along line P-P.

FIG. 52 is a perspective view of a first surface of the retainer shown in FIG. 48.

FIG. 53 is a side elevational view of the first surface of the retainer shown in FIG. 48.

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FIG. 54 is a cross-sectional view of the retainer shown in FIG. 53, taken along line Q-Q.

DETAILED DESCRIPTION

Fluid end assemblies are typically used in oil and gas operations to deliver highly pressurized corrosive and/or abrasive fluids to piping leading to the wellbore. The assemblies are typically attached to power ends run by engines. The power ends reciprocate plungers within the assemblies to pump fluid throughout the fluid end. The power end used with the fluid end typically has a power output of at least 2,250 horsepower during hydraulic fracturing operations.

Fluid may be pumped through the fluid end at pressures that range from 5,000-15,000 pounds per square inch (psi). Fluid used in high pressure hydraulic fracturing operations is typically pumped through the fluid end at a minimum of 8,000 psi; however, fluid will normally be pumped through the fluid end at pressures around 10,000-15,000 psi during such operations.

In fluid end assemblies known in the art, the fluid flow passages or bores formed within the fluid end body are typically sealed by inserting a plug into each bore. A large retaining nut is then installed into each bore above the plug. The retaining nuts typically thread into internal threads formed in the walls of each bore.

In operation, the high level of fluid pressure pumping throughout the fluid end may cause the retaining nuts to back off or unthread from their installed position. When a retaining nut unthreads from its installed position, the plug it was retaining may be displaced by fluid pressure. Displacement of the plug allows fluid to leak around the plug and erode the walls of the bore. The internal threads formed in the bores for engagement with the retaining nuts are also known to crack over time. Erosion of the bore walls or cracking of the internal threads typically requires repair or replacement of the fluid end.

The present invention is directed to a plurality of different fluid ends having bores sealed without threading retaining nuts into the walls of each bore. As a result, the fluid ends of the present invention do not have internal threads formed in their bores proximate the bore openings. Removal of the internal threads eliminates the problems associated with the internal thread failures and the retaining nuts becoming unthreaded from the bores.

With reference to FIGS. 1 and 3, a first embodiment of a fluid end 100 is shown. The fluid end 100 comprises a housing or fluid end body 102 having a flat external surface 104 and a plurality of first and second bores 106, 108 formed adjacent one another therein, as shown in FIG. 1. The first bores 106 may also be referred to as vertical conduits, and the second bores 108 may also be referred to as horizontal conduits. Preferably, the number of first bores 106 equals the number of second bores 108. More preferably, each first bore 106 intersects its paired second bore 108 within the fluid end body 102 to form an internal chamber 112, as shown in FIG. 3. The first bore 106 comprises first and second sections positioned on opposite sides of the internal chamber 112. Likewise, the second bore 108 comprises third and fourth sections positioned on opposite sides of the internal chamber 112.

FIG. 1 shows five first and second bores 106, 108. In alternative embodiments, the number of sets of paired first and second bores in the fluid end body may be greater than five, or less than five. Thus, FIG. 4 shows a fluid end body that includes three sets of paired first and second bores. Each bore of each set of paired bores 106 and 108 terminates in

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a corresponding opening 110 formed in the external surface 104. The bores 106 and 108 and openings 110 exist in one-to-one relationship. A plurality of internally threaded openings 144 are formed in the body 102 and uniformly spaced around each bore opening 110, as shown in FIG. 1.

With reference to FIG. 3, each second bore 108 may have an intake opening 118 formed proximate the bottom end of the fluid end body 102. Each intake opening 118 is connected in one-to-one relationship to a corresponding coupler or pipe. These couplers or pipes are fed from a single common piping system (not shown).

A pair of valves 120 and 122 are positioned within each second bore 108. The valves 120, 122 route fluid flow within the body 102. The intake valve 120 blocks fluid backflow through the intake opening 118. The discharge valve 122 regulates fluid through one or more discharge openings 126. A plurality of couplers 127 may be attached to each discharge opening 126 for connection to a piping system (not shown), as shown in FIG. 1.

Continuing with FIGS. 1 and 3, the fluid end 100 further comprises a plurality of sets of components 128 and 130. The number of sets preferably equals the number of sets of paired first and second bores 106 and 108 formed in the body 102. The component 128 is positioned within a first bore 106, and the component 130 is positioned within its paired second bore 108. In one embodiment, the component 128 is a suction plug and the component 130 is a discharge plug. Each of the components 128 and 130 are substantially identical in shape and construction, and each is sized to fully block fluid flow within the respective bore 106, 108. A seal 136 is positioned around the outer surface of each component 128, 130 to block fluid from leaking from the bores 106, 108.

Each of the components 128 and 130 comprises a first section 138 joined to a second section 140. The first section 138 has a footprint sized to cover the bore opening 110 and the second section 140 is configured for removable receipt within one of the bores 106, 108. In one embodiment, the first section 138 is an enlarged plate and the second section 140 is a plug sized to be closely received within one of the bores 106, 108. When the component 128 or 130 is installed within one of the bores 106, 108, the first section 138 engages with the external surface 104 of the body 102. This engagement prevents longitudinal movement of the second section 140 within the bore 106 or 108 as shown in FIG. 3.

With reference to FIG. 1, the first section 138 may be formed as a circular structure having a plurality of notches 142 cut from its outer periphery. When each of the first sections 138 is engaged with the external surface 104 of the body 102, each of the notches 142 partially surrounds one of the openings 144 spaced around each bore opening 110.

Continuing with FIGS. 1 and 3, once each component 128, 130 is installed in the fluid end body 102, each of the components 128, 130 is secured in place by a retainer element 132 in a one-to-one relationship. Each retainer element 132 has a footprint sized to fully cover the first section 138 of the components 128 and 130. The retainer elements 132 shown in FIG. 1 are flat and cylindrical. A plurality of openings 146 are formed about the periphery of each retainer element 132. Each opening 146 is alignable with a corresponding one of the openings 144 in a one-to-one relationship.

Each of the retainer elements 132 is secured to the fluid end body 102 using a fastening system 134. The fastening system comprises a plurality of studs 148, a plurality of washers 150, and a plurality of nuts 152. Each stud 148 is externally threaded adjacent its first end 149, while each

opening 144 has internal threads that mate with those of the stud 148. Each stud 148 may be threaded into place within a corresponding one of the openings 144, in a one-to-one relationship.

Once a first stud 148 has been installed in the body 102 at its first end 149, its opposed second end 151 projects from the body's external surface 104. When each component 128 is positioned within its bore 106, each of its notches 142 at least partially surrounds a corresponding one of the studs 148. Likewise, when each component 130 is positioned within its bore 108, each of its notches 142 at least partially surrounds a corresponding one of the studs 148.

Each peripheral opening 146 formed in each of the retainer elements 132 is registerable with a corresponding one of the studs 148. The plurality of washers 150 and nuts 152 may be installed and torqued on each one of the studs 148. The plurality of washers 150 and nuts 152 hold the retainer element 132 against the first section 138 of the components 128, 130 and hold the first section 138 against the external surface 104 of the fluid end body 102. Because each of the retainer elements 132 is attached to the fluid end body 102 using the fastening system 134, no external threads are formed on the outer surface of each retainer element 132. Likewise, no internal threads are formed within the walls of each bore 106, 108.

With reference to FIGS. 2 and 3, a plunger end 154 of the fluid end 100 is shown. The plurality of first bores 106 terminate at openings 156 formed on the external surface 104 of the plunger end 154. An internal seat 159 is formed in the walls of each of the bores 106 proximate each of the bore openings 156. A plurality of threaded openings 161 are formed in each of the internal seats 159, as shown in FIG. 2.

A component 158 is positioned within each first bore 106 through each of the openings 156. Each of the components 158 is tubular and sized to be closely received within each bore 106. In one embodiment, the components 158 are stuffing box sleeves.

With reference to FIG. 3, each of the components 158 may have a first or rear section 160 that joins a second or front section 162 via a tapered section 164. The first section 160 may have a larger diameter than the second section 162. When each of the components 158 is installed within each of the bores 106, the tapered section 164 engages a tapered seat 166 formed in the walls of each bore 106. This engagement prevents longitudinal movement of each component 158 within each bore 106. A seal 167 is positioned around the outer surface of the second section 162 of each of the components 158 in order to block fluid from leaking from the bores 106.

Once installed within the body 102, each component 158 is secured in place by a retainer element 170 in a one-to-one relationship. Each of the retainer elements 170 is sized to be closely received within each bore 106 and engage a top surface 171 of each component 158, as shown in FIG. 3. Each of the retainer elements 170 shown in FIG. 2 has a cylindrical body and a threaded central opening 172. A plurality of openings 174 are formed about the periphery of each of the retainer elements 170. The openings 174 are uniformly spaced around each central opening 172.

A plurality of ports 175 may be formed in an outer surface of each retainer element 170 that are orthogonal to the plurality of openings 174. At least one seal 176 may also be disposed around the outer surface of each of the retainer elements 170. The seal 176 helps block fluid from leaking from the bores 106.

Each of the retainer elements 170 is secured to the fluid end body 102 using a fastening system 178. The fastening system 178 comprises a plurality of threaded screws 180. The screws 180 are preferably socket-headed cap screws.

The fastening system 178 secures each retainer element 170 to each internal seat 159. When each retainer element 170 is positioned within each bore 106, each of the peripheral openings 174 is alignable with a corresponding one of the openings 161 in a one-to-one relationship. Each of the screws 180 is registerable within one of the openings 161 in the seat 159 and one of the peripheral openings 174 in the retainer element 170.

The screws 180 may be torqued as desired to tightly attach each of the retainer elements 170 to each internal seat 159 and securely hold each component 158 within each bore 106. Because each of the retainer elements 170 is attached to the fluid end body 102 using the fastening system 178, no external threads are formed on the outer surface of each of the retainer elements 170. Likewise, no internal threads are formed within the walls of each bore 106 on the plunger end 154 of the body 102.

Continuing with FIGS. 2-3, a plurality of packing seals 181 may be positioned within each of the components 158 and each of the retainer elements 170 to prevent fluid from leaking from the bores 106. At least one of the packing seals 181 may have a plurality of ports 179 formed in its outer periphery, as shown in FIG. 2. The ports 179 provide an exit for fluid trapped within the packing seals 181. Fluid exiting the ports 179 may exit the retainer element 170 through the ports 175.

A packing nut 182 may also be threaded into the central opening 172 of each of the retainer elements 170 in a one-to-one relationship. The packing nut 182 has a threaded section 183 joined to a body 184. The body 184 shown in FIG. 2 is cylindrical. However, the body 184 may also be square or rectangular shaped. A central passage 185 extends through the threaded section 183 and the body 184. The threaded section 183 of the packing nut 182 is threaded into the central opening 172 of the retainer element 170.

When installed within each of the retainer elements 170, each of the packing nuts 182 engages with and compresses the packing seals 181 installed within each component 158 and retainer element 170, as shown in FIG. 3. Compression of the packing seals 181 helps prevent fluid from leaking past the seals 181. A seal 186 may also be positioned within the central passage 185 of each of the packing nuts 182 to further seal fluid from leaking from the bores 106.

A plurality of holes 187 are formed around the outer surface of each of the packing nut bodies 184. The holes 187 serve as connection points for a spanner wrench that may be used to tightly thread the packing nut 182 into the central opening 172 of each of the retainer elements 170.

A plunger 188 may also be installed within each bore 106 in a one-to-one relationship. When a plunger 188 is installed within a bore 106, the plunger 188 is positioned within the component 158, the retainer element 170, and the packing nut 182, as shown in FIG. 3. Each of the plungers 188 projects from the plunger end 154 of the fluid end body 102 and is attached to a separate power end. As discussed above, the power end reciprocates each of the plungers 188 within the fluid end body 102 so as to pump fluid throughout the body. Each of the plungers 188 may be attached to the power end via a clamp 190 in a one-to-one relationship.

Several kits are useful for assembling the fluid end 100. A first kit comprises a plurality of the components 128 or 130, a plurality of the retainer elements 132, and the fastening system 134. A second kit may comprise the

plurality of components 158, a plurality of the retainer elements 170, and the fastening system 178. The second kit may further comprise a plurality of the packing seals 181, a plurality of the packing nuts 182, and a plurality of the plungers 188. Each of the kits may be assembled using the fluid end body 102.

With reference to FIGS. 4 and 6, a second embodiment of a fluid end 200 is shown. The fluid end 200 comprises a housing or fluid end body 202 having a flat external surface 204 and a plurality of first and second bores 206, 208 formed adjacent one another therein, as shown in FIG. 4. Each bore of each set of paired bores 206 and 208 terminates in a corresponding opening 210 formed in the external surface 204. A plurality of threaded openings 211 are formed in the body 202 and uniformly spaced around each opening 210. The internal functions of the fluid end 200 are identical to those described with reference to fluid end 100, shown in FIG. 3.

The fluid end 200 further comprises a plurality of sets of components 212 and 214. The number of sets preferably equals the number of set of paired first and second bores 206 and 208 formed in the body 202. The component 212 is positioned within a first bore 206, and the component 214 is positioned within its paired second bore 208. In one embodiment, the component 212 is a suction plug and the component 214 is a discharge plug.

Each of the components 212 and 214 is substantially identical in shape and construction, and is sized to fully block fluid flow within the respective bore 206, 208. A seal 216 is positioned around the outer surface of each component 212, 214 to block fluid from leaking from the bores 206, 208.

As shown in FIG. 4, a top surface 213 of each component 212, 214 may sit flush with the external surface 204 of the body 202 when installed within a respective bore 206, 208. Each of the components 212 and 214 may engage with internal seats (not shown) formed in the walls of each of the bores 206, 208. Such engagement helps prevent longitudinal movement of the components 212, 214 within the respective bore 206, 208.

Once installed within the fluid end body 202, each component 212 and 214 is secured in place by a retainer element 218 in a one-to-one relationship. Each of the retainer elements 218 has a footprint sized to cover a single bore opening 210. The retainer elements 218 shown in FIG. 4 are flat and cylindrical. A plurality of openings 220 are formed about the periphery of each retainer element 218. Each peripheral opening 220 is alignable with a corresponding one of the openings 211 in a one-to-one relationship, as shown in FIG. 4.

The retainer elements 218 are secured to the external surface 204 of the fluid end body 202 by a fastening system 222. The fastening system 222 comprises a plurality of externally threaded studs 224, a plurality of washers 226, and a plurality of internally threaded nuts 228. Each stud 224 is externally threaded adjacent its first end 230, while each opening 211 has internal threads that mate with those of the stud 224. Each stud 224 may be threaded into place within a corresponding one of the openings 211, in a one-to-one relationship.

Once a first stud 224 has been installed in the body 202 at its first end 230, its opposed second end 232 projects from the body's external surface 204. Each peripheral opening 220 formed in the retainer elements 218 is registerable with a corresponding one of the studs 224. The plurality of washers 226 and nuts 228 may be installed and torqued on each of the studs 224. The plurality of washers 226 and nuts

228 hold the retainer elements 218 against the external surface 204 of the fluid end body 202. Because each of the retainer elements 218 is attached to the fluid end body 202 using the fastening system 222, no external threads are formed on the outer surface of each retainer element 218. Likewise, no internal threads are formed within the walls of each bore 206 and 208.

With reference to FIGS. 5-6, a plunger end 234 of the fluid end 200 is shown. The plurality of first bores 206 terminate at openings 236 formed on the external surface 204 of the plunger end 234. The plunger end 234 of the fluid end body 202 is similar to the plunger end 154 of fluid end body 102, shown in FIGS. 2-3, except that an internal seat 159 is not formed within each bore 206. Instead, a plurality of internally threaded openings 238 are formed in the external surface 204 of the fluid end body 202 that are uniformly spaced around each bore opening 236.

A component 240 is positioned within each first bore 206 through each of the openings 236 in a one-to-one relationship. Each of the components 240 is tubular and sized to be closely received within each bore 206. In one embodiment, the components 240 are stuffing box sleeves.

With reference to FIG. 6, each of the components 240 may have a first or rear section 242 that joins a second or front section 244 via a tapered section 246. The first section 242 may have a larger diameter than the second section 244. When each of the components 240 is installed within each of the bores 206, the tapered section 246 engages a tapered seat 248 formed in the walls of each bore 206. This engagement prevents longitudinal movement of each component 240 within each bore 206. A seal 250 is positioned around the outer surface of the second section 244 of each of the components 240 to block fluid from leaking from the bores 206.

Once installed within the body 202, a top surface 252 of each of the components 240 may sit flush with the external surface 204 of the body 202. Each of the components 240 is secured in place within each bore 206 by a retainer element 254 in a one-to-one relationship. The retainer elements 254 shown in FIG. 5 have a cylindrical body and a threaded central opening 256. A plurality of openings 258 are formed about the periphery of each of the retainer elements 254. The openings 258 are uniformly spaced around each central opening 256.

The retainer elements 254 are secured to the external surface 204 of the fluid end body 202 using a fastening system 260. The fastening system 260 comprises a plurality of threaded screws 262. The screws 262 are preferably socket-headed cap screws. When each retainer element 254 is positioned over each bore opening 236, each of the peripheral openings 258 is alignable with a corresponding one of the openings 238 in a one-to-one relationship. Each of the screws 262 is registerable within one of the openings 238 in the body 202 and one of the peripheral openings 258 in each of the retainer elements 254.

The screws 262 may be torqued as desired to tightly attach each of the retainer elements 254 to the body 202 and securely hold each of the components 240 within each bore 206. Because each of the retainer elements 254 is attached to the fluid end body 202 using the fastening system 260, no external threads are formed on the outer surface of each retainer element 254. Likewise, no internal threads are formed within the walls of each bore 206 on the plunger end 234 of the body 202.

Similar to the plunger end 154 shown in FIG. 2, a plurality of packing seals 264 may be positioned within each of the components 240. A packing nut 266 may thread into the

central opening 256 of each retainer element 254 and compress the packing seals 264. A seal 267 may also be positioned within each packing nut 266. Additionally, a plurality of plungers 268 may be disposed within each component 240, retainer element 254, and packing nut 266. Each of the plungers 268 may be attached to a power end via a clamp 270.

In alternative embodiments, the components 212, 214, and 240 may not be flush with the external surface 204 of the body 202 when installed in the respective bores 206, 208. In such case, a flange or ledge may be formed on each of the retainer elements 218 or 254 on its side facing the component 212, 214, or 240. The flange or ledge may be installed within the bores 206, 208 so that it tightly engages the top surface 213 or 252 of the components 212, 214, or 240.

Likewise, if the components 212, 214, or 240 project from the external surface 204 of the body 202 when installed within the respective bores 206, 208, the retainer elements 218 or 254 can be modified to accommodate the component 212, 214, or 240. For example, a cut-out may be formed in the retainer element 218 or 254 for closely receiving the portion of the component 212, 214, or 240 projecting from the body 202. The area of the retainer element 218 or 254 surrounding the cut-out will engage the external surface 204 of the body 202.

Several kits are useful for assembling the fluid end 200. A first kit comprises a plurality of the components 212 or 214, a plurality of retainer elements 218, and the fastening system 222. A second kit may comprise the plurality of components 240, a plurality of the retainer elements 254, and the fastening system 260. The second kit may further comprise a plurality of packing seals 264, a plurality of packing nuts 266, and a plurality of plungers 268. Each of the kits may be assembled using the fluid end body 202.

Turning now to FIG. 7, a third embodiment of a fluid end 300 is shown. The fluid end 300 comprises a housing or fluid end body 302 having a flat external surface 304 and a plurality of first and second bores 306, 308 formed adjacent one another therein. Each bore of each set of paired bores 306 and 308 terminates in a corresponding opening 310 formed in the external surface 304. A plurality of threaded openings 311 are formed in the body 302 and uniformly spaced around each bore opening 310. The internal functions of the fluid end 300 are identical to those described with reference to fluid end 100, shown in FIG. 3.

The fluid end 300 further comprises a plurality of sets of components 312 and 314. The number of sets preferably equals the number of set of paired first and second bores 306 and 308 formed in the body 302. The component 312 is positioned within a first bore 306, and the component 314 is positioned within its paired second bore 308. In one embodiment, the component 312 is a suction plug and the component 314 is a discharge plug. A seal 315 is positioned around each of the components 312, 314 to block fluid from leaking from the respective bores 306, 308.

The components 312 and 314 have the same shape and construction as the components 212 and 214 shown in FIGS. 4 and 6. Each of the components 312 and 314 may engage with internal seats (not shown) formed in the walls of each of the bores 306, 308. Such engagement helps prevent longitudinal movement of the components 312, 314 within the respective bores 306, 308.

Once installed within the body 302, a top surface 313 of each of the components 312, 314 may sit flush with the external surface 304 of the body 302. Each of the components 312, 314 is secured within each respective bore 306, 308 by a retainer element 316. Each of the retainer elements

316 shown in FIG. 7 is a large rectangular plate having a footprint sized to cover a plurality of adjacent bore openings 310 at one time. A plurality of openings 318 are formed in each retainer element 316 that are alignable with a corresponding one of the openings 311 in a one-to-one relationship.

Each of the retainer elements 316 is secured to the external surface 304 of the fluid end body 302 by a fastening system 320. The fastening system 320 comprises a plurality of externally threaded studs 322, a plurality of washers 324, and a plurality of internally threaded nuts 326. The fastening system 320 secures each of the retainer elements 316 on the fluid end body 302 in the same way as described with reference to the fastening system 222 used with the fluid end 200.

Because each of the retainer elements 316 is attached to the fluid end body 302 using the fastening system 320, no external threads are formed in the retainer element 316. Likewise, no internal threads are formed within the walls of each bore 306 and 308. When the retainer elements 316 are installed on the fluid end body 302, the edges of the retainer element 316 may extend far enough so as to sit flush with the edges of the fluid end body 302. In alternative embodiments, the retainer element 316 may have different shapes or sizes. For example, the retainer element 316 may be large enough so as to cover an entire side surface of the fluid end body 302. Alternatively, the retainer elements 316 may have rounded edges, as shown in FIG. 8.

Turning to FIG. 8, a plunger end 330 of the fluid end 300 is shown. The plurality of first bores 306 terminate at openings 332 formed on the external surface 304 of the plunger end 330. A plurality of internally threaded openings 334 are formed in the external surface 304 that are uniformly spaced around each bore opening 332.

A component 336 is positioned within each first bore 306 through each of the openings 332. Each of the components 336 is tubular and sized to be closely received within each bore 306. In one embodiment, the components 336 are stuffing box sleeves. The components 336 have the same shape and construction as the components 240, shown in FIGS. 5-6.

Once installed within the body 302, a top surface 346 of each of the components 336 may sit flush with the external surface 304 of the body 302. Each of the components 336 is secured within each bore 306 by a single retainer element 348. The retainer element 348 shown in FIG. 8 is a large oval plate having a footprint sized to cover a plurality of adjacent bore openings 332 formed on the plunger end 330 of the fluid end body 302. A plurality of openings 350 are formed in the retainer element 348 that are alignable with a corresponding one of the openings 334 in a one-to-one relationship.

In alternative embodiments, the retainer element 348 may have different shapes or sizes. For example, the retainer element 348 may be large enough so as to cover an entire side surface of the fluid end body 302. Alternatively, the retainer element 348 may have squared edges, as shown in FIG. 7.

The retainer element 348 is secured to the external surface 304 of the fluid end body 302 by a fastening system 352. The fastening system 352 comprises a plurality of screws 354. The fastening system 352 secures the retainer element 348 on the fluid end body 302 in the same way as described with reference to the fastening system 260 used with the fluid end 200 and shown in FIGS. 5-6.

Because the retainer element 348 is attached to the fluid end body 302 using the fastening system 352, no external

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threads are formed in the retainer element **348**. Likewise, no internal threads are formed within the walls of each bore **306**.

A central threaded opening **356** is formed in the center of each grouping of openings **350** in the retainer element **348**. The openings **356** are alignable with each bore opening **332** in a one-to-one relationship. A single packing nut **358** may thread into each central opening **356**. A seal **359** may be positioned within each packing nut **358**.

Similar to the plunger end **234** shown in FIGS. 5-6, a plurality of packing seals **360** may be positioned within each component **336**. Each of the packing nuts **358** may compress the packing seals **360** when installed within the retainer element **348**. A plurality of plungers **362** may be disposed within each component **336**, the retainer element **348**, and each packing nut **358**. Each of the plungers **362** may be connected to a power end via a clamp **364**. A cross-sectional view of the fluid end **300** looks identical to the cross-sectional view of the fluid end **200**, shown in FIG. 6.

Several kits are useful for assembling the fluid end **300**. A first kit comprises a plurality of the components **312** or **314**, a retainer element **316**, and the fastening system **320**. A second kit may comprise a plurality of the components **336**, a retainer element **348**, and the fastening system **352**. The second kit may further comprise a plurality of the packing seals **360**, a plurality of the packing nuts **358**, and a plurality of the plungers **362**. Each of the kits may be assembled using the fluid end body **302**.

With reference to FIGS. 9 and 11, a fourth embodiment of a fluid end **400** is shown. The fluid end **400** comprises a housing or fluid end body **402** having a flat external surface **404** and a plurality of first and second bores **406**, **408** formed adjacent one another therein, as shown in FIG. 9. Each bore of each set of paired bores **406** and **408** terminates in a corresponding opening **410** formed in the external surface **404**. A plurality of threaded openings **411** are formed in the body **402** and uniformly spaced around each opening **410**. The internal functions of the fluid end **400** are identical to those described with reference to fluid end **100**, shown in FIG. 3.

The fluid end **400** further comprises a plurality of sets of components **412** and **414**. The number of sets preferably equals the number of set of paired first and second bores **406** and **408** formed in the body **402**. The component **412** is positioned within a first bore **406**, and the component **414** is positioned within its paired second bore **408**. In one embodiment, the component **412** is a suction plug and the component **414** is a discharge plug. A seal **415** is positioned around the outer surface of each of the components **412**, **414** to block fluid from leaking from the respective bores **406**, **408**.

The components **412** and **414** have substantially the same shape and construction as the components **212** and **214** shown in FIGS. 4 and 6. However, in contrast to the components **212**, **214**, each of the components **412** and **414** is joined to a single retainer element **416**.

The components **412**, **414** may be welded or fastened to the center of the back surface of each retainer element **416**. Alternatively, each of the components **412** or **414** and a corresponding retainer element **416** may be machined as a single piece, as shown in FIG. 11. Each of the retainer elements **416** secures each of the components **412**, **414** within the respective bores **406**, **408**. The retainer elements **416** also prevent the components **412**, **414** from moving longitudinally within the respective bores **406**, **408**.

A plurality of openings **418** are formed about the periphery of each retainer element **416**. Each peripheral opening

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418 is alignable with a corresponding one of the openings **411** in a one-to-one relationship, as shown in FIG. 9.

The retainer elements **416** are secured to the external surface **404** of the body **402** using a fastening system **420**. The fastening system **420** comprises a plurality of externally threaded studs **422**, a plurality of washers **424**, and a plurality of internally threaded nuts **426**. The fastening system **420** secures the retainer elements **416** to the fluid end body **402** in the same way as described with reference to the fastening system **222** used with the fluid end **200**.

Because the retainer elements **416** are attached to the fluid end body **402** using the fastening system **420**, no external threads are formed in the retainer elements **416**. Likewise, no internal threads are formed within the walls of each bore **406** and **408**.

Turning now to FIGS. 10-11, a plunger end **430** of the fluid end **400** is shown. The plurality of first bores **406** terminate at openings **432** formed on the external surface **404** of the plunger end **430**. A plurality of internally threaded openings **434** are formed in the external surface **404** that are uniformly spaced around each bore opening **432**. A component **436** is positioned within each first bore **406** through each of the openings **432**. Each of the components **436** is tubular and sized to be closely received within each bore **406**. In one embodiment, the components **436** are stuffing box sleeves. The components **436** have substantially the same shape and construction as the components **240**, shown in FIGS. 5-6. However, in contrast to the components **240**, each of the components **436** is joined to a single retainer element **438**.

The components **436** may be welded or fastened to the center of the back surface of each retainer element **438**. Alternatively, each of the components **436** and a corresponding retainer element **438** may be machined as a single piece, as shown in FIG. 11. Each of the retainer elements **438** secures each of the components **436** within the bores **406**. The retainer elements **438** also prevent the components **436** from moving longitudinally within the bores **406**.

A threaded central opening **440** is formed within each retainer element **438**. A plurality of threaded openings **442** are formed about the periphery of each of the retainer elements **438** and are uniformly spaced around each central opening **440**. Each peripheral opening **442** is alignable with a corresponding one of the openings **434** in a one-to-one relationship, as shown in FIG. 10.

The retainer elements **438** are secured to the external surface **404** of the body **402** using a fastening system **444**. The fastening system **444** comprises a plurality of screws **446**. The fastening system **444** secures the retainer elements **438** to the fluid end body **402** in the same way as described with reference to the fastening system **260** used with the fluid end **200** and shown in FIGS. 5-6.

Because the retainer elements **438** are attached to the fluid end body **402** using the fastening system **444**, no external threads are formed in the retainer elements **416**. Likewise, no internal threads are formed within the walls of each bore **406** on the plunger end **430** of the body **402**.

Like the plunger end **330** of fluid end **300**, the fluid end **400** may also comprise a plurality of packing seals **448**, a plurality of packing nuts **450**, each housing a seal **454**, and a plurality of plungers **456**. Each plunger **456** may be connected to a power end via a clamp **458**.

Several kits are useful for assembling the fluid end **400**. A first kit comprises a plurality of the components **412** or **414**, a plurality of the retainer elements **416**, and the fastening system **420**. A second kit may comprise a plurality of the components **436**, a plurality of the retainer elements

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438, and the fastening system 444. The second kit may further comprise a plurality of the packing seals 448, a plurality of the packing nuts 450 and a plurality of the plungers 456. Each of the kits may be assembled using the fluid end body 402.

With reference to FIGS. 12-13, a fifth embodiment of a fluid end 500 is shown. The fluid end 500 comprises a housing or fluid end body 502 having a flat external surface 504 and a plurality of first and second bores 506, 508 formed adjacent one another therein, as shown in FIG. 12. Each bore of each set of paired bores 506 and 508 terminates in a corresponding opening 510 formed in the external surface 504. A plurality of threaded openings 511 are formed in the body 502 and uniformly spaced around each opening 510. The internal functions of the fluid end 500 are identical to those described with reference to fluid end 100, shown in FIG. 3.

The fluid end 500 further comprises a plurality of sets of components 512 and 514. The number of sets preferably equals the number of set of paired first and second bores 506 and 508 formed in the body 502. The component 512 is positioned within a first bore 506, and the component 514 is positioned within its paired second bore 508. In one embodiment, the component 512 is a suction plug and the component 514 is a discharge plug. The components 512 and 514 have the same shape and construction as the components 212 and 214 shown in FIGS. 4 and 6. A seal 516 is positioned around the outer surface of each component 512, 514 to block fluid from leaking from the bores 506, 508.

As shown in FIG. 12, a top surface 513 of each of the components 512, 514 may sit flush with the external surface 504 of the body 502 when installed within a respective bore 506, 508. Each of the components 512 and 514 may engage with internal seats (not shown) formed in the walls of each of the bores 506, 508. Such engagement helps prevent longitudinal movement of the components 512, 514 within the respective bore 506, 508.

Once installed within the fluid end body 502, each component 512 and 514 is secured in place by a retainer element 518 in a one-to-one relationship. Each of the retainer elements 518 has a footprint sized to cover a single bore opening 510. The retainer elements 518 shown in FIG. 12 are flat and cylindrical and each have a central threaded opening 519. A plurality of openings 520 are formed about the periphery of each retainer element 518 and are uniformly spaced around each central opening 519. Each peripheral opening 520 is alignable with a corresponding one of the openings 511 in a one-to-one relationship, as shown in FIG. 12.

The retainer elements 518 are secured to the external surface 504 of the fluid end body 502 by a fastening system 522. The fastening system 522 comprises a plurality of externally threaded studs 524, a plurality of washers 526, and a plurality of internally threaded nuts 528. The fastening system 522 secures the retainer elements 518 to the fluid end body 502 in the same way as described with reference to the fastening system 222 used with the fluid end 200 shown in FIGS. 4 and 6.

Each central opening 519 formed in each retainer element 518 is alignable with each corresponding bore opening 510 in a one-to-one relationship. A retaining nut 530 may thread into each central opening 519 to cover each bore opening 510. Using a threaded retaining nut 530 with the retainer element 518 allows access to each bore opening 510 without having to remove the retainer elements 518 from the fluid end body 502.

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While the fluid end 500 uses a threaded retaining nut 530, the retaining nut 530 is not threaded into the walls of the bores 506, 508. Thus, any failures associated with the retaining nut 530 may be experienced in the retainer element 518, which is easily replaceable. This similar configuration is used on the plunger end 234 of the fluid end 200 shown in FIGS. 5-6. Such configuration is shown again on a plunger end 532 of the fluid end body 502 in FIG. 13.

A kit is useful for assembling the fluid end 500. The kit may comprise a plurality of the components 512 or 514, a plurality of the retainer elements 518, and the fastening system 522. The kit may further comprise a plurality of retaining nuts 530. The kit may be assembled using the fluid end body 502.

Turning now to FIG. 14, a sixth embodiment of a fluid end 600 is shown. The fluid end 600 comprises a housing or fluid end body 602 having a flat external surface 604 and a plurality of first bores (not shown) and second bores 608 formed adjacent one another therein. Each bore of each set of paired bores terminates in a corresponding opening 610 formed in the external surface 604. A plurality of threaded openings 611 are formed in the body 602 and uniformly spaced around each opening 610. The internal functions of the fluid end 600 are identical to those described with reference to fluid end 100, shown in FIG. 3.

The fluid end 600 further comprises a plurality of sets of components 614. The component 614 is positioned within a second bore 608. The components positioned within each first bore are not shown in FIG. 14. However, such components are identical in shape and construction to the components 614.

The number of sets of components preferably equals the number of set of paired first bores (not shown) and second bores 608 formed in the body 602. In one embodiment, the component positioned within a first bore is a suction plug, and the component 614 positioned within its paired second bore 608 is a discharge plug. The components 614 have a substantially similar shape and construction as the components 212 and 214 shown in FIGS. 4 and 6, except that a threaded hole 616 is formed in a top surface 613 of each component 614. A seal 618 is positioned around the outer surface of each component 614 to block fluid from leaking from the bores 608.

The top surface 613 of each component 614 may sit flush with the external surface 604 of the body 602 when installed within a bore 608. Each of the components 614 may engage with internal seats (not shown) formed in the walls of each of the bores 608. This engagement helps prevent longitudinal movement of the components 614 within the bore 608. Likewise, the components positioned within the first bores (not shown) may engage internal seats formed within the walls of the first bores.

Once installed within the fluid end body 602, each component 614 is secured by a retainer element 620 in a one-to-one relationship. Likewise, the components positioned within the first bores (not shown) are each secured by one of the retainer elements 620. Each of the retainer elements 620 has a footprint sized to cover a single bore opening 610. The retainer elements 620 shown in FIG. 14 are flat and cylindrical and each have a central threaded opening 622. A plurality of openings 624 are formed about the periphery of each retainer element 620 and are uniformly spaced around each central opening 622. Each peripheral opening 624 is alignable with a corresponding one of the openings 611 in a one-to-one relationship.

The retainer elements 620 are secured to the external surface 604 of the fluid end body 602 by a fastening system

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626. The fastening system 626 comprises a plurality of externally threaded studs 628, a plurality of washers (not shown), and a plurality of internally threaded nuts 630. The fastening system 626 secures the retainer elements 620 to the fluid end body 602 in the same way as described with reference to the fastening system 222 used with the fluid end 200 shown in FIGS. 4 and 6.

The fastening system 626 may further comprise a plurality of eye bolts 632, a plurality of handles 634, and a cable 636. Each eye bolt 632 has external threads 638 formed on its first end and an eye 640 formed on its opposite second end. The threaded end 638 of each eye bolt 632 threads into each hole 616 formed in each component 614 in a one-to-one relationship. Once installed within each hole 614, the eye 640 of each eyebolt 632 projects through the central opening 622 formed in each retainer element 620.

Each of the handles 634 has a threaded section 642 joined to a cylindrical body 644. A central passage 646 extends through the threaded section 642 and the body 644. Each of the threaded sections 642 may be installed within the central opening 622 of each of the retainer elements 620 such that each eye bolt 632 is disposed within the central passage 646. Once one of the handles 634 is installed in a retainer element 620, the eye bolt 632 projects from the handle 634. The handle 634 helps support the eye bolt 632 and provides a grip to assist in installation or removal of a retainer element 620 on the fluid end body 602.

The cable 636 may be disposed through each eye 640 of each eye bolt 632. Each of the eye bolts 632 may be oriented to facilitate the passage of the cable 636 through each eye 640. The ends of the cable 636 may be attached to the external surface 604 of the fluid end body 602 using eye bolts 650 and clamps 652. The cable 636 is preferably made of a stiff and tough material, such as high-strength nylon or steel.

In operation, the eyebolts 632 and cable 636 tether each of the retaining elements 620 and components 614, in case of failure of the retainer elements 620, a portion of the fastening system 626, or the fluid end body 602. When a failure occurs, the large pressure in the fluid end body 602 will tend to force the components 614 out of their respective bores 608 with a large amount of energy. The cable 636 helps to retain the components 614 within the bores 608 in the event of a failure. The cable 636 also helps to retain the retainer elements 620 in position in the event of a failure. The fastening systems 134, 222, 320, 420, and 522 used with fluid ends 100, 200, 300, 400, and 500 may also be configured for use with the eye bolts 632, handles 634 and cable 636.

In alternative embodiments, the handles 634 may not be used. A single eye bolt 632 may also be formed integral with a single component 614. A single cable 636 may also be used through each of the eyebolts 632. Each cable 636 would independently attach to the external surface 604 of the fluid end body 602.

Several kits are useful for assembling the fluid end 600. A first kit comprises a plurality of the components 614, a plurality of the retainer elements 620, and the fastening system 626. The kit may be assembled using the fluid end body 602.

Turning to FIGS. 16-20, a seventh embodiment of the fluid end 800 is shown. The fluid end 800 comprises a housing or fluid end body 836. The fluid end 800 is similar to the fluid ends 100 and 500, shown in FIGS. 1-3, 12 and 13, with the below described exceptions.

With reference to FIGS. 19, 20, and 30-32, a component 804 installed within a suction bore 808 may also be referred

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to as a suction plug 804. The suction bore 808 may also be characterized as a fourth section of the horizontal conduit. The suction plug 804 is sized to form a barrier to fluid flow within the suction bore 808. Each of the suction plugs 804 comprises a cylindrical body having opposed top and bottom surfaces 816 and 818. The suction plug 804 is of single-piece construction and is substantially solid with the exception of a threaded hole 820 formed in its top surface 816. The suction plug 804 includes an upper or first portion 822 joined to a lower or second portion 824 by a tapered portion 827.

Continuing with FIGS. 30-32, the upper portion 822 has an outer diameter, D1 and the lower portion 824 has an outer diameter, D2. The lower portion 824 has a reduced diameter relative to that of the upper portion 822, meaning the diameter D1 is greater than the diameter D2. As shown in FIGS. 30-32, the lower portion 824 has a length that is more than three times a length of the upper portion 822. The lower portion 824 includes a plurality of sections along its length; the sections have several different diameters. The section of greatest diameter is situated midway along the length of the lower portion 824, and presents an external sealing surface 826. The sealing surface 826 may be considered the diameter D2 of the lower portion 824.

First and second sections 828 and 830, having outer diameters D3 and D4, are formed on opposite sides of the sealing surface 826 and diameter D2. Each of the sections 828 and 830 has a reduced diameter relative to that of the sealing surface 826. Thus, the diameter D2 is greater than the diameters D3 and D4. A third section 832 extends between the second section 830 and the bottom surface 818 and has an outer diameter, D5. The third section 832 has a reduced diameter relative to that of the second section 830. Thus, the diameter D4 is greater than the diameter D5.

With reference to FIG. 21, a plurality of beveled corners 834 are formed in the fluid end body 836 at the intersection of a front surface 838 and the walls surrounding the opening of each suction bore 808. When a suction plug 804 is installed within one of the suction bores 808, the tapered portion 827 of the plug 804 engages the beveled corners 834. Such engagement prevents further axial movement of the plug 804 within the bore 808. The upper portion 822 of the plug 804, which includes diameter D1, projects from a front external surface 838 of the fluid end body 836 when installed within one of the bores 808. Thus, upon installation of the suction plug 804 within the suction bore 808, the difference between the diameters D1 and D2 prevents further axial movement of the suction plug 804 into the fluid end body 836 along the longitudinal axis of the suction bore 808. In alternative embodiments, the upper portion of the suction plug may engage the front surface of the fluid end body. In further alternative embodiments, axial movement of the suction plug within the bore may be prevented by engagement of the bottom surface of the plug with the walls surrounding the bore or a protrusion from the walls surrounding the bore.

Turning back to FIGS. 30-32, the outer surface of the plug 804 includes no annular recess for housing a seal. Instead, an annular recess 840 is formed in the walls surrounding each of the suction bores 808 adjacent the front surface 838 of the fluid end body 836, as shown in FIG. 21. The recess 840 is configured for housing an annular seal 842. Preferably, the seal 842 is a high pressure seal.

With reference to FIG. 25, each recess 840 comprises two sidewalls 844 joined by a base 846. The seal 842 is closely received within the recess 840. After a seal 842 is installed within a corresponding recess 840 within a bore 808, a suction plug 804 is installed within that bore.

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When a suction plug **804** is installed within a bore **808**, the seal **842** within the bore tightly engages the plug's sealing surface **826**. During operation, the seal **842** wears against the sealing surface **826** of the suction plug **804**. If the sealing surface **826** on one of the plugs **804** begins to erode, allowing fluid to leak around the plug **804**, that plug **804** is removed and replaced with a new plug. The seal **842** may also be removed and replaced with a new seal, if needed.

Continuing with FIG. 25, a small amount of clearance exists between the walls surrounding the bore **808** and the first, second, and third sections **828**, **830**, and **832** of the installed plug **804**. The clearance allows the suction plug **804** to rock back and forth on each side of its sealing surface **826**. The rocking motion helps to overcome friction between each of the plugs **804** and the walls surrounding its corresponding bore **808**. Thus, less force is required for installation or removal of one of the plugs **804** than is required for a traditional suction plug. Reduced forces mean fewer scrapes and scratches on the walls surrounding the bore, as compared to a traditional suction plug.

The suction plugs **804** may be installed and removed using a tool (not shown), which may be attached to a plug **804** at the threaded hole **820**, shown in FIG. 30. For example, a tool having an externally threaded end may mate with the internal threads formed in the threaded hole **820**. Once installed, an operator may rock the plug **804** back and forth using the tool while simultaneously pushing or pulling on the plug **804** with the tool.

Continuing with FIGS. 19 and 20, a component **806** installed within a discharge bore **810** may also be referred to as a discharge plug **806**. The discharge bore **810** may also be characterized as the first section of the vertical conduit. Turning to FIGS. 27-29, each of the discharge plugs **806** comprises a cylindrical body having opposed top and bottom surfaces **850** and **852**. The discharge plug **806** is substantially solid with the exception of two threaded holes. A first threaded hole **854** formed in its top surface **850** and a second threaded hole **856** formed in its bottom surface **852**. Each plug **806** includes an upper portion **858** joined to a lower portion **860** by a tapered portion **862**. The lower portion **860** includes a plurality of sections along its length; the sections have several different diameters. The section of the greatest diameter is situated midway along the length of the lower portion **860**, and presents an external sealing surface **864**. First and second sections **866** and **868** are formed on opposite sides of the sealing surface **864**. Each of the sections **866** and **868** has a reduced diameter relative to that of the sealing surface **864**. A third section **870** is formed below the second section **868** and has a reduced diameter relative to that of the second section **870**. The third section **870** includes a plurality of reduced diameter sections.

Each plug **806** further includes a connection portion **872**. The connection portion **872** extends between the third section **870** and the bottom surface **852**. The connection portion **872** has a reduced diameter relative to that of the lower portion **860**. The second threaded hole **856** extends within the connection portion **872**. The connection portion **872** is configured for connecting to, engaging, or aligning a spring **874** used with a discharge valve **876**, shown in FIGS. 19 and 20.

With reference to FIG. 22, a plurality of beveled corners **878** are formed in the fluid end body **836** at the intersection of the top surface **880** and the walls surrounding the opening of each discharge bore **810**. When a discharge plug **806** is installed within one of the discharge bores **810**, the tapered portion **862** of the plug **806** engages the beveled corners **878**. Such engagement prevents further axial movement of the

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plug **806** within the bore **810**. The upper portion **858** of the plug **806** projects from the top surface **880** of the fluid end body **836** when installed within one of the bores **810**. In alternative embodiments, the upper portion of the discharge plug may engage the top surface of the fluid end body. In further alternative embodiments, axial movement of the discharge plug within the bore may be prevented by engagement of the bottom surface of the plug with the walls surrounding the bore or protrusions from the walls surrounding the bore.

Turning back to FIGS. 27-29, the outer surface of the plug **806** includes no annular recess for housing a seal. Instead, an annular recess **882** is formed in the walls surrounding each of the discharge bores **810** adjacent the top surface **880** of the fluid end body **836**, as shown in FIG. 22. The recess **882** is configured for housing an annular seal **884**. Preferably, the seal **884** is a high pressure seal.

With reference to FIG. 26, each recess **882** comprises two sidewalls **886** joined by a base **888**. The seal **884** is closely received within the recess **882**. After a seal **884** is installed within a corresponding recess **882** within a bore **810**, a discharge plug **806** is installed within that bore.

When a discharge plug **806** is installed within a bore **810**, the seal **884** tightly engages the plug's sealing surface **864**. During operation, the seal **884** wears against the sealing surface **864** of the discharge plug **806**. If the sealing surface **864** on one of the plugs **806** begins to erode, allowing fluid to leak around the plug **806**, that plug **806** is removed and replaced with a new plug. The seal **884** may also be removed and replaced with a new seal, if needed.

Continuing with FIG. 26, a small amount of clearance exists between the walls surrounding the bore **810** and the first, second, and third sections **866**, **868** and **870** of the installed plug **806**. The clearance allows the discharge plug **806** to rock back and forth on each side of its sealing surface **864**. The rocking motion helps to overcome friction between each of the plugs **806** and the walls surrounding its corresponding bore **810**. The discharge plugs **806** may be installed and removed using a tool (not shown), which may be attached to a plug **806** at the threaded hole **854**, shown in FIG. 27.

With reference to FIGS. 19 and 20, when the fluid end **800** is operating, the bottom surfaces **818** and **852** of each of the plugs **804** and **806** will be exposed to the high fluid pressures within the interior of the fluid end **800**. The fluid pressure may be high enough to dislodge the suction and discharge plugs **804** and **806** from their respective bores **808** and **810**. To keep the plugs **804** and **806** within their respective bores **808** and **810**, a plurality of retainers **890** are attached to the fluid end body **836**. A retainer **890** is attached to the body **836** in front of each of the suction plugs **804** and above each of the discharge plugs **806**, as shown in FIG. 20.

With reference to FIG. 33, each retainer **890** has a cylindrical body having flat opposing top and bottom surfaces **892** and **894**. A threaded central passage **896** is formed in the center of each retainer **890**. The central passage **896** interconnects the top and bottom surfaces **892** and **894**. A plurality of peripheral passages **898** are formed in each retainer **890** and surround the central passage **896**. Each peripheral passage **898** interconnects the top and bottom surfaces **892** and **894** of each retainer **890**.

With reference to FIGS. 34 and 35, a retainer nut **900** is installed within the central passage **896** of each retainer **890**, as shown in FIGS. 19 and 20. A central passage **902** is formed in the retainer nut **900**. The central passage **902** interconnects the nut's top and bottom surfaces **904** and **906**. External threads are formed on the retainer nut **900** adjacent

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its bottom surface **906**. The external threads are matingly engageable with the internal threads formed in the retainer **890**, as shown in FIGS. **19** and **20**. The walls surrounding the central passage **902** adjacent the top surface **904** of the retainer nut **900** are shaped to closely receive a hex-shaped tool.

With reference to FIG. **20**, a plurality of peripheral openings **908** are formed in the fluid end body **836** around each opening of each suction and discharge bore **808** and **810**. The peripheral passages **908** formed in each retainer **890** are alignable with the peripheral openings **908** formed around each of the bores **808** and **810**, in a one-to-one relationship.

Each of the retainers **890** is secured to the fluid end body **836** using a fastening system **910**, as shown in FIGS. **17** and **18**. The fastening system **910** comprises a plurality of studs **912**, a plurality of washers **914**, and a plurality of nuts **916**. Each stud **912** is externally threaded adjacent its first end **918**, while each peripheral opening **908** formed in the fluid end body **836** has internal threads that mate with those of the stud **912**, as shown in FIGS. **23** and **24**. Studs **912** are threaded into place within each of the peripheral openings **908**.

Continuing with FIGS. **23** and **24**, once a first stud **912** has been installed in the fluid end body **836** at its first end **918**, its opposed second end **919** projects from the body's top or front surface **880** or **838**. Each peripheral passage **989** formed in each of the retainers **890** receives a corresponding one of the studs **912**. Each of the studs **912** receives a washer **914** and nut **916**, which hold the retainer **890** against the top and front surface **880** and **838** of the fluid end body **836**. Rather than applying a single large torque to a single retainer, the fastening system **910** contemplates distribution of smaller torques among a plurality of studs **912** and nuts **916**.

When a retainer **890** is attached to the fluid end body **836**, the central passage **896** surrounds the upper portion **822** or **858** of the plug **804** or **806**. The retainer nut **900** installed within the retainer **890** is torqued so that its bottom surface **906** tightly engages with the top surface **816** or **850** of the plug **804** or **806**. Such engagement maintains the plug **804** or **806** within its corresponding bore **808** or **810**. When the retainer nut **900** is engaged with the top surface **816** or **850** of the plug **804** or **806**, the threaded hole **820** or **854** formed in the plug **804** or **806** is exposed to the nut's central passage **902**.

During operation, an operator may need access to the inside of the fluid end **800** multiple times during a single fracking operation. For example, one of the plugs **804** or **806** may need to be replaced. Removing a retainer **890** to gain such access can be time-consuming, because of the need to remove multiple nuts **916** and washers **914**.

To avoid such delays, each retainer **890** includes a removable retainer nut **900**. Rather than remove all of the nuts **916** and washers **914**, the operator can simply remove the retainer nut **900**. When the retainer nut **900** is removed, the operator can access the interior of the fluid end body **836** through the central opening **896** of the retainer **890**. The retainer nut **900** may be removed using a hex-shaped tool that mates with the walls surrounding the central passage **902** of the retainer nut **900**.

While the fluid end **800** includes a plurality of threaded retainer nuts **900**, those retainer nuts **900** are not threaded into the walls surrounding the bores **808** and **810**. Thus, even if the threads on one of the retainer nuts **900** should crack, the fluid end body **836** remains intact. Only the retainer nut

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900 and/or its corresponding retainer **890** need be replaced. The high cost of repairing or replacing the fluid end body **836** is thereby avoided.

With reference to FIGS. **19**, **20**, **36** and **37**, a component **812** is installed within a plunger bore **814** of the fluid end **800**. The plunger bore **814** may be characterized as the third section of the horizontal conduit. The component **812** may be characterized as a stuffing box **812**. The stuffing box **812** has a cylindrical front portion **920** joined to a cylindrical rear portion **922**. The front portion **920** has a lesser diameter than that of the rear portion **922**. The stuffing box **812** is installed within the plunger bore **814** such that the front portion **920** is in front of or closer to the internal chamber **811** than the rear portion **922**.

In contrast to the stuffing box sleeve **158** shown in FIG. **3**, the stuffing box **812** does not include a tapered portion **164**. Instead, the front portion **920** is joined directly to a front surface **924** of the rear portion **922**. The stuffing box **812** is also more robust than the sleeve **158**. However, one of skill in the art will recognize that the stuffing box **812** and the sleeve **158** serve the same purpose and are functionally interchangeable with one another.

Continuing with FIGS. **36** and **37**, a central passage **926** extends through the stuffing box **812** and interconnects the box's rear and front surfaces **928** and **930**. An internal seat **932** is formed in the walls surrounding the central passage **926** adjacent the front surface **924** of the rear portion **922**, as shown in FIG. **19**. The internal seat **932** is formed as a result of the rear portion **922** having a larger inner diameter from that of the front portion **920**.

The rear portion **922** has a generally uniform outside diameter along its length. A plurality of peripheral passages **934** are formed in the rear portion **922** and surround the central passage **926**. The passages **934** interconnect the stuffing box's rear surface **928** and the front surface **924** of the rear portion **922**. A plurality of threaded openings **936** are formed in the rear surface **928** of the stuffing box **812**. The threaded openings **936** allow use of a tool for gripping the stuffing box **812** while it is being installed or removed.

Turning back to FIG. **20**, when the stuffing box **812** is installed within the fluid end **800**, a lubrication hole **938** is formed in the fluid end body **836** and faces the stuffing box **812**. Because of the alignment between the lubrication hole **938** and the stuffing box **812**, first, second, and third annular recesses **940**, **942**, and **944** are formed in an outer surface of the stuffing box **812**, as shown in FIGS. **36** and **37**. Each of the first and third recesses **940** and **944** are configured to house a seal **946**, as shown in FIG. **19**. Preferably, the seal **946** is an O-ring. The second recess **942** underlies the lubrication hole **938**, and is interconnected with the stuffing box's central passage **926** by a plurality of spaced passages **948**. Lubrication may be supplied to the stuffing box **812** by way of the lubrication hole **938**, the second recess **942**, and the passages **948**.

Turning back to FIGS. **36** and **37**, the outer surface of the stuffing box **812** includes no annular recess for housing a high pressure seal. Instead, an annular recess **950**, configured to house an annular seal **952**, is formed in the walls surrounding each plunger bore **814**, as shown in FIG. **19**. Preferably, the seal **952** is a high pressure seal.

Continuing with FIG. **19**, each recess **950** is characterized by two side walls **954** joined by a base **956**. The width of the base **956** is equal to at least one-third of the length of the front portion **920** of the stuffing box **812**. The recess **950** is sized so as to house a large seal **952**, much larger than an O-ring. The seal **952** is closely received within the recess **950** and may fill at least 95% of the volume of the recess

950. After a seal 952 is installed within a recess 950 formed within one of the plunger bores 814, a stuffing box 812 is installed within that bore 814.

When a stuffing box 812 is installed within a bore 814, the seal 952 tightly engages the outer surface of the stuffing box's front portion 920. During operation, the seal 952 wears against the front portion 920. If the outer surface of the front portion 920 begins to erode, allowing fluid to leak around the stuffing box 812, that stuffing box 812 can be removed and replaced with a new stuffing box. The seal 952 may also be removed and replaced with a new seal, if needed.

The seals 167 and 250 shown in FIGS. 3 and 6 may be identical to the seal 952. Likewise the seal 167 and 250 may be positioned in a recess that is identical to the recess 950.

Continuing with FIG. 20, a counterbore 958 is formed within the plunger bore 814. A plurality of openings 960 are formed in a base 962 of the counterbore 958. When a stuffing box 812 is installed within the plunger bore 814, the front surface 924 of the rear portion 922 engages the base 962 of the counterbore 958. Such engagement prevents further movement of the stuffing box 812 within the fluid end body 836. The stuffing box 812 is positioned within the plunger bore 814 such that its peripheral passages 934 and the openings 960 formed in the base 962 are aligned in a one-to-one relationship, as shown in FIG. 20.

With reference to FIGS. 38 and 39, a retainer 970 prevents the stuffing box 812 from being dislodged from the plunger bore 814. The retainer 970 comprises a cylindrical body having an internally threaded central passage 972. The central passage 972 interconnects the retainer's rear and front surfaces 974 and 976. A plurality of peripheral passages 978 surround the central passage 972 and interconnect the retainer's rear and front surfaces 974 and 976. A counterbore 980 is formed within each passage 978, adjacent the top surface 974 of the retainer 970.

With reference to FIG. 20, the retainer 970 is installed within the plunger bore 814 so that its front surface 976 engages the rear surface 928 of the stuffing box 812. The retainer 970 is installed over the stuffing box 812 such that the peripheral passages 934 and the peripheral passages 978 are aligned in a one-to-one relationship.

Each of the retainers 970 is secured to the fluid end body 836 using a fastening system 982 shown in FIGS. 17 and 18. The fastening system 982 comprises a plurality of studs 984 and a plurality of nuts 986. Each of the studs 984 is received within a corresponding one of the openings 960 formed in the base 962. From the base 962, each stud 984 extends through a corresponding one of the passages 934 in the stuffing box 812, and through a corresponding one of the passages 978 in the retainer 970.

A first end 988 of each stud 984 is positioned within one of the counterbores 980 formed in the retainer 970. A nut 986 is then placed on a second end 987 of each stud 984, and turned until it tightly engages the base of the counterbore 980. In alternative embodiments, the fastening system may comprise a plurality of screws instead of studs and nuts. The screws are preferably socket-headed cap screws. In further alternative embodiments, the fastening system may comprise a plurality of bolts, such as 12-point bolts.

Attaching the retainer 970 to the fluid end body 836 also helps ensure the stuffing box 812 remains tightly in place during operation. Because each of the retainers 970 is attached to the fluid end body 836 using the fastening system 982, no external threads are formed on the outer surface of each of the retainer 970. Likewise, no internal threads are formed within the walls of each plunger bore 814.

Continuing with FIGS. 19 and 20, a plunger packing 990 is installed within the central passage 926 of each stuffing box 812. When installed, the plunger packing 990 engages the stuffing box's internal seat 932. Thus, the plunger packing 990 is only positioned within the rear portion 922 of the stuffing box 812 and is spaced away from the front portion 920 and the seal 952. The plunger packing 990 prevents high pressure fluid from passing around a plunger 991 as the plunger reciprocates. Each plunger packing 990 comprises a plurality of annular seals compressed together and having aligned central passages. The outer seals 994 may be made of metal and compress the inner pressure seals 996, as shown in FIG. 19. The inner pressure seals 996 are preferably high pressure seals. The seal 952 is approximately the same width as one of the pressure seals 996.

Over time, the seals 994 and 996 wear against the inner surface of the stuffing box 812. If leakage occurs, the stuffing box 812 may be removed and replaced with a new stuffing box.

The plunger packing 990 is held within the stuffing box 812 by a packing nut 992. The packing nut 992 is generally identical to the packing nut 182 shown in FIGS. 2 and 3. However, the packing nut 992 may vary slightly in size from the packing nut 182 in order to properly fit within the retainer 970 and stuffing box 812. External threads formed on the outer surface of the packing nut 992 matingly engage the internal threads formed in the retainer 970. An O-ring may be installed within a groove 993 formed within the packing nut 992, as shown in FIG. 19.

When a packing nut 992 is installed within one of the retainers 970, a front surface 998 of the packing nut 992 engages the plunger packing 990. Such engagement compresses the plunger packing 990, creating a tight seal. After a packing nut 992 has been installed within a retainer 970, a central passage within that packing nut 992 will be aligned with a central passage in a plunger packing 990.

When assembling the fluid end 800, the stuffing box 812 is installed within the plunger bore 814 after installation of the seal 952 within the recess 950. After the stuffing box 812 is installed, the retainer 970 may be secured to the housing using the fastening system 982. After the retainer 970 is attached to the fluid end body 836, the plunger packing 990 may be installed within the rear portion 922 of the stuffing box 812. The plunger packing 990 is installed through the rear surface 928 of the stuffing box 812. Alternatively, the plunger packing 990 may be installed within the stuffing box 812 prior to attaching the retainer 970 to the fluid end body 836. During operation, the plunger packing 990 may be removed and replaced with a new plunger packing 990 without removing the stuffing box 812 from the fluid end 800.

Once a stuffing box 812, plunger packing 990, retainer 970, and packing nut 992 are installed within a plunger bore 814, a plunger 991 is next installed, as shown in FIG. 20. Alternatively, the plunger 991 may be installed prior to installing the packing nut 992. Once installed, the plunger 991 is surrounded by the other components within the plunger bores 814. During operation, the plunger 991 moves relative to the fluid end 800 and the components installed within the plunger bores 814.

The plunger 991 is identical to the plunger 188 shown in FIGS. 2 and 3. A clamp 1005 is attached to the end of each plunger 991. The clamp 1005 secures its plunger 991 to a pony rod 1007, shown in FIG. 40.

Continuing with FIG. 40, the fluid end 800 is shown attached to a power end 1009 via a plurality of stay rods 1011. The power end 1009 drives reciprocal motion of the

plungers **991** within the fluid end **800**. The fluid ends **100**, **200**, **300**, **400**, **500**, or **600** may be attached to the power end **1009** in place of the fluid end **800**.

Several kits are useful for assembling the fluid end **800**. A first kit comprises a plurality of the components **804**, a plurality of the components **806**, a plurality of the components **812**, a plurality of the retainers **890**, a plurality of the retainers **970**, and the fastening systems **910** and **982**. The kit may be assembled using the fluid end body **836**.

With reference to FIGS. **1-14** and **16-20**, a single fluid end body may use any combination of the kits described herein. The fluid end bodies, components, and retainer elements described herein are preferably made of high strength steel.

While the fluid end bodies **102**, **202**, **302**, **402**, **502** and **836** shown in FIGS. **1-13** and **16-20** are substantially rectangular in shape, the kits described herein may also be used with any shape of a fluid end body, such as that shown in FIG. **14**. Likewise, the retainer elements described herein may vary in shape and size, as desired. For example, the circular retainer elements described herein may be square or rectangular shaped.

The fastening systems **134**, **222**, **320**, **420**, **522** and **910** described herein each use eight studs around each bore opening. In alternative embodiments, more than eight studs or less than eight studs may be used to secure each retainer element over each bore opening. For example, FIG. **14** only shows six studs securing each retainer element **620** over each bore opening **610**. Likewise, fewer than 16 or more than 16 screws may be used with the fastening systems **178**, **260**, **352**, **444**, and **982**. The number of peripheral openings formed in each retainer element described herein may correspond with the number of openings formed around each bore opening in each fluid end body and the number of studs or screws being used.

The fastening systems described herein reduce the amount of torque required to secure each retainer element to the fluid end bodies. Rather than having to torque one large retaining nut, the torque is distributed throughout the plurality of studs, nuts, or screws. Decreasing the amount of torque required to seal the bores increases the safety of the assembly process.

Turning to FIG. **15**, a stud **700** is shown. The stud **700** may be used with the fastening systems **134**, **222**, **320**, **420**, **522**, **626**, and **910** shown in FIGS. **1**, **4**, **7**, **9**, **11**, **14**, **17** and **18**. For exemplary purposes, the stud **700** will be described with reference to fluid end **100**, shown in FIG. **1**.

The stud **700** has a first threaded section **702** and an opposite second threaded section **704**. The threaded sections **702** and **704** are joined by an elongate, cylindrical body **706**. The first threaded section **702** is configured for threading into one of the plurality of threaded openings **144** formed in the fluid end body **102**. The second threaded section **704** is configured for threading into the threaded opening formed in one of the nuts **152**.

The first section **702** may have fewer threads than that of the opening **144**. For example, if the opening **144** has 18 internal threads, the first section **702** of the stud **700** may only have 16 external threads. This configuration ensures that all of the threads formed on the first section **702** will be engaged and loaded when the first end **702** is threaded into the opening **144**. Engaging all of the threads helps to increase the fatigue life of the first end **702** of the stud **700**. Likewise, the second section **704** may have fewer external threads than there are internal threads formed in the nut **152**. The stud **700** may also be subjected to shot peening on its non-threaded sections prior to its use to help reduce the

possibility of fatigue cracks. The stud **700** may have a smooth outer surface prior to performing shot peening operations.

The body **706** of the stud **700** comprises a first section **708** and a second section **710**. The first section **708** has a smaller diameter than the second section **710**. The retainer element **132** is primarily held on the first section **708** of the body **706**. The diameter of the second section **710** is enlarged so that it may center the washer **150** on the stud **700**.

The diameter of the second section **710** is configured so that it is only slightly smaller than the diameter of the central opening of the washer **150**. This sizing allows the washer **150** to closely receive the second section **710** of the stud **700** when the washer **150** is positioned on the stud **700**. When the washer **150** is positioned on the second section **710**, the washer **150** is effectively centered on the stud **700**. The washer **150** is also effectively centered against the nut **152**, once the nut **152** is installed on the stud **700**.

Without placing the washer **150** on the second section **710**, the washer may have to be manually centered on the stud **700** prior to installing the nut **152**. If the washer **150** is not properly centered on the stud **700** or against the nut **152**, it may be difficult to effectively torque or un-torque the nut **152** from the stud **700**, depending on the type of washer used.

The plurality of washers used with each fastening system **134**, **222**, **320**, **420**, **522**, **626** and **910** shown in FIGS. **1**, **4**, **7**, **9**, **11**, **14**, **17** and **18** may be configured to allow a large amount of torque to be imposed on the nuts used with the washers without using a reaction arm. Instead, the washer itself may serve as the counterforce needed to torque a nut onto a stud. Not having to use a reaction arm increases the safety of the assembly process. The nuts used with the fastening systems **134**, **222**, **320**, **420**, **522**, and **626** may also comprise a hardened inner layer to help reduce galling between the threads of the nuts and studs during the assembly process. An example of the above described washers, nuts, and methods are described in Patent Cooperation Treaty Application Serial No. PCT/US2017/020548, authored by Junkers, et al, the entirety of which is incorporated herein by reference.

With reference to FIGS. **41-54**, an eighth embodiment of a fluid end **1000** is shown. The fluid end **1000** comprises a housing or fluid end body **1002**. The fluid end **1000** is identical to the fluid end **800**, but comprises an alternative embodiment of a stuffing box **1004** and a corresponding retainer **1006**. The stuffing box **1004** and retainer **1006** are each installed within a plunger bore **1008**. The plunger bore **1008** is identical to the plunger bore **814** formed in the fluid end **800** and shown in FIGS. **19** and **20**.

With reference to FIGS. **44-47**, the stuffing box **1004** is similar to the stuffing box **812** shown in FIGS. **36** and **37**. The stuffing box **1004** comprises a first portion **1010** joined to a second portion **1012**. The first and second portions **1010** and **1012** each have a cylindrical shape, such that the stuffing box **1004** may be considered primarily cylindrical. The first portion **1010** has an outer diameter, **D1**. The second portion **1012** has an outer diameter, **D2**. The diameter **D2** is greater than the diameter **D1**. The diameter **D2** is also greater than a maximum diameter of a groove **1014** formed in the housing **1002**, as shown in FIGS. **41-43**. Like the stuffing box **812**, the stuffing box **1004** is installed within the plunger bore **1008** such that the first portion **1010** is closely received within a first segment **1016**, and the second portion **1012** is closely received within a second segment **1018** of the plunger bore **1008**, as shown in FIG. **41**. The difference between the diameters **D1** and **D2** and the diameters of the

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plunger bore **1008** prevent further movement of the stuffing box **1004** into the housing **1002**, as shown in FIG. **41**.

Continuing with FIGS. **44-47**, the diameter **D1** is constant along at least a portion of the length of the first portion **1010** of the stuffing box **1004**. The diameter **D1** may be constant along the entire length of the first portion **1010**, with the exception of a tapered surface **1020** formed between the first portion **1010** and a first surface **1022** of the stuffing box **1004**. No grooves are formed in the outer surface of the first portion **1010** for housing a seal. Rather, the outer surface of the first portion **1010** has a sealing surface for a seal **1024**, as shown in FIGS. **41-43**. The diameter **D2** is constant along the entire length of the second portion **1012**. In alternative embodiments, the diameter **D2** may only be constant along a portion of the length of the second portion **1012**.

Continuing with FIGS. **44-47**, the first portion **1010** has an inner diameter, **D3**. The second portion **1012** has an inner diameter, **D4**. The diameter **D4** is greater than the diameter **D3**. The diameter **D3** may be constant along at least a portion of the length of the first portion **1010**, and the diameter **D4** may be constant along the length of the second portion **1012**.

Continuing with FIG. **47**, the inner surface of the first portion **1010** may have a slightly convex portion **1026** joined to a straight portion **1028**. The convex portion **1026** may extend between an internal seat **1030** and the straight portion **1028**. Because the first portion **1010** includes the convex portion **1026**, the first portion **1010** may also have an inner diameter, **D5**. The diameter **D3** is greater than the diameter **D5**. The convex portion **1026** helps increase the wall thickness of the first portion **1010**, which helps alleviate stress within the stuffing box **1004** during operation. In alternative embodiments, the inner surface of the first portion **1010** may be shaped like the stuffing box **812** shown in FIGS. **19** and **20**.

With reference to FIGS. **44-46**, like the stuffing box **812**, a plurality of passages **1032** are formed in the second portion **1012** and interconnect a first and a second surface **1034** and **1036** of the second portion **1012**. The passages **1032** are configured for receiving a plurality of fasteners **1038**, as shown in FIG. **42**. The fasteners **1038** may be threaded studs or socket-headed screws and are part of a fastening system, like the fastening system **982** shown in FIGS. **17** and **18**.

In contrast to the stuffing box **812**, a plurality of annular recesses are not formed in an outer surface of the second portion **1012**. Instead, as will be discussed in more detail below, such recesses are formed in the retainer **1006**. Because the recesses are formed in the retainer **1006**, the second portion **1012** of the stuffing box **1004** has a shorter length than that of the stuffing box **812**, shown in FIGS. **36** and **37**. Likewise, the retainer **1006** has a longer length than the retainer **970**, shown in FIGS. **38** and **39**.

With reference to FIGS. **48-54**, the retainer **1006** comprises a cylindrical body having a central passage **1040**. The central passage **1040** interconnects the retainer's first and second surfaces **1042** and **1044**. A plurality of peripheral passages **1046** surround the central passage **1040** and interconnect the retainer's first and second surfaces **1042** and **1044**. The passages **1046** are configured for receiving the fasteners **1038**, as shown in FIG. **42**. The retainer **1006** is attached to the housing **1002** using a fastening system in the same manner as the retainer **970**. The retainer **1006** is attached to the housing **1002** such that the first surface **1042** of the retainer **1006** abuts the second surface **1036** of the stuffing box **1004**.

Continuing with FIGS. **48-54**, a first, second, and third annular recess **1050**, **1052**, and **1054** are formed in an outer

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surface of the retainer **1006**. Each of the first and third recesses **1050** and **1054** are configured to house a seal (not shown). Preferably, the seal is an O-ring. A plurality of passages **1056** interconnect the central passage **1040** and the second recess **1052**. Each of the passages **1056** extends at a non-zero angle relative to a longitudinal axis extending through the central passage **1040**. The opening of each of the passages **1056** in the second recess **1052** is positioned closer to the second surface **1044** of the retainer **1006** than the opening of the passage **1056** within the central passage **1040**. In alternative embodiments, the passages **1056** may be formed at a different angle from that shown in FIGS. **41** and **51**. A portion of the central passage **1040** adjacent the second surface **1044** of the retainer **1006** is threaded and configured to engage a packing nut **1060**, as shown in FIG. **41**.

Turning back to FIG. **41**, when the stuffing box **1004** is installed within the housing **1002**, the second recess **1052** underlies a lubrication conduit **1062** formed in the housing **1002**. The lubrication conduit **1062** interconnects an external surface **1003** of the housing **1002** and the plunger bore **1008**. A plurality of lubrication conduits **1062** may be formed in the housing **1002**, each conduit corresponding with one of the plunger bores **1008**.

In operation, lubricant is pumped into the housing **1002** through the conduit **1062**. The lubricant enters the second recess **1052** and flows circumferentially around the retainer **1006** until it encounters one of the passages **1056**. Lubricant flows through one or more of the passages **1056** and into a plunger packing **1066** installed within the retainer **1006** and the stuffing box **1004**. The lubricant lubricates a plunger **1068** as it reciprocates. The angled nature of the passages **1056** help the flow of lubricant from the housing **1002** to the plunger packing **1066**.

With reference to FIGS. **43**, **46**, **47**, and **54**, a plurality of blind bores **1070** are formed in the second surface **1036** of the stuffing box **1004**. Each bore **1070** is configured for receiving a dowel pin **1072**, as shown in FIG. **43**. A plurality of corresponding bores **1074** are formed in the first surface **1042** of the retainer **1006** for also receiving the dowel pins **1072**. The dowel pins **1072** keep the retainer **1006** aligned with the stuffing box **1004** during assembly, and help carry any non-tensile loading imparted by the interface between the retainer **1006** and the stuffing box **1004**.

The bores **1074** formed in the retainer **1006** and shown in FIGS. **43** and **54** interconnect the first and second surfaces **1042** and **1044** of the retainer **1006**. In alternative embodiments, the bores **1074** may be blind bores that open on the first surface **1042** of the retainer **1006**. The bores **1070** and **1074** may include counter-bored areas configured for receiving the dowel pins **1072**, as shown in FIGS. **43**, **47**, and **54**.

The various features and alternative details of construction of the apparatuses described herein for the practice of the present technology will readily occur to the skilled artisan in view of the foregoing discussion, and it is to be understood that even though numerous characteristics and advantages of various embodiments of the present technology have been set forth in the foregoing description, together with details of the structure and function of various embodiments of the technology, this detailed description is illustrative only, and changes may be made in detail, especially in matters of structure and arrangements of parts within the principles of the present technology to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

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The invention claimed is:

1. An apparatus, comprising:

a plunger; and

a fluid end assembly, comprising:

a housing having an external surface with a plurality of 5 first openings and

a plurality of opposed plunger openings formed therein;

a plurality of bore pairs formed within the housing, each bore pair comprising:

a straight first bore and a straight second bore, the second 10 bore intersecting the first bore and interconnecting one of the first openings and one of the plunger openings;

a valve pair situated within a first bore of a selected bore pair;

a stuffing box sleeve installed within a second bore of the 15 selected bore pair; and

a first plate fastened to the external surface of the housing and covering the plurality of first openings; and

a second plate fastened to the external surface of the 20 housing, the second plate having a plurality of plate bores axially aligned with the plurality of plunger openings;

in which the stuffing box sleeve is retained within the second bore by the second plate; and in which the 25 second bore of the selected bore pair at least partially receives the plunger.

2. The apparatus of claim 1, further comprising:

a plurality of fasteners;

in which the first plate comprises a plurality of openings;

in which each of the plurality of fasteners extends through 30 one of the plurality of openings and into the housing to secure the first plate to the housing.

3. The apparatus of claim 1, further comprising a suction plug situated within the second bore of the selected bore pair; in which a top surface of the suction plug sits flush with 35 the external surface of the housing.

4. The apparatus of claim 3, in which the suction plug is a separate piece than the first plate.

5. The apparatus of claim 1, further comprising a discharge plug situated within the first bore of the selected bore 40 pair.

6. The apparatus of claim 1, in which the external surface of the housing includes at least one flat surface.

7. The apparatus of claim 1, in which the external surface of the housing comprises a plurality of threaded openings 45 formed in groups, each group positioned around one of the first openings.

8. The fluid end assembly of claim 7, further comprising a plurality of fasteners, the plurality of fasteners passing 50 through the first plate and into the plurality of threaded openings in a one-to-one relationship.

9. The fluid end assembly of claim 1, in which no threads are formed on the stuffing box sleeve.

10. A fluid end assembly, comprising:

a housing, comprising: 55

an external surface, comprising:

a first face having a plurality of first openings formed therein; and

a second face having a plurality of second openings 60 formed therein;

a plurality of first and second bores formed therein, each first bore intersecting a corresponding second bore, each second bore joining a first opening and a second opening;

a valve pair situated within one of the first bores; 65

a stuffing box sleeve situated within one of the second bores; and

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a retaining plate fastened to the second face of the housing; and

a second plate fastened to the external surface of the housing, the second plate having a plurality of plate bores axially aligned with the plurality of first openings;

in which the stuffing box sleeve is retained within the second bore by the second plate; and in which each first opening is configured to receive at least a portion of a plunger.

11. The fluid end assembly of claim 10, further comprising a seal positioned around the stuffing box sleeve, the seal configured to block fluid from leaking from the selected second bore.

12. The fluid end assembly of claim 10, further comprising a plurality of fasteners; in which the retaining plate comprises a plurality of passages; in which each of the plurality of fasteners extends through one of the plurality of passages and into the housing to secure the retaining plate to the housing; in which each of the plurality of fasteners is a threaded fastener; and in which each of the plurality of fasteners includes a threaded nut.

13. The fluid end assembly of claim 10, in which no internal threads are formed within the second openings.

14. The fluid end assembly of claim 10, in which no threads are formed on the stuffing box sleeve.

15. The fluid end assembly of claim 10, in which the housing is made of two sections joined together by a plurality of fasteners; in which each of the two sections receives at least a portion of the plunger.

16. The fluid end assembly of claim 10, in which the first and second faces are spaced and parallel.

17. An apparatus, comprising:

a housing, comprising:

an external surface that includes a plunger surface and a flat remote surface;

a first bore extending between different points on the external surface;

a plurality of second bores extending between the plunger surface and the remote surface and comprising:

a plurality of second bore openings formed in the plunger surface; and

a plurality of second bore openings formed in the remote surface;

in which a selected one of the second bores intersects the first bore;

a pair of valves installed within the first bore on opposite sides of the first bore's intersection with the selected second bore;

a plunger at least partially received within the selected second bore and extending through one of the second bore openings in the plunger surface; and

a flat plate disposed in overlaying engagement with the remote surface such that each second bore opening in the remote surface is covered; and

a second plate fastened to the plunger surface of the housing, the second plate having a plurality of plate bores axially aligned with the plurality of second bore openings;

in which the plunger extends through the second plate.

18. The apparatus of claim 17, in which the selected one of the second bores is straight.

19. The apparatus of claim 17, in which the plunger surface and the remote surfaces are spaced and parallel.

20. The apparatus of claim 17, further comprising a suction plug situated in the selected second bore.

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21. The apparatus of claim **20**, in which the suction plug does not extend past the remote surface of the external surface.

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